



MATH0057 notes

Probability and Statistics (University College London)



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LECTURE NOTES FOR

**MATH0057: PROBABILITY AND
STATISTICS
(15 CREDITS)**

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1 Introduction, and review of basic probability theory

There are many situations, in science and in life, where we have a question that cannot be answered precisely. Examples include:

1. Tossing a coin. Will it come down heads or tails?
2. Card games. Will I beat my opponent with the cards I currently hold (given that I can't see their cards)?
3. Who will win the next UK general election?
4. If a patient is ill, is it better to treat them with drug A or drug B?

Some uncertainty is involved in the answers to all of these questions. In everyday language, it is common to use the word 'random' to justify this uncertainty, for example to describe the result of tossing a coin. If we were asked to define 'random' in this context, we'd probably come up with something along the lines of 'the result is unpredictable'. However, in the coin tossing example it's not clear that the result *is* unpredictable. Presumably coins obey the laws of physics so that, if we had the time, inclination, measuring ability and necessary computer power, we could in fact determine the result.

The point of this is that 'uncertainty' and 'randomness' are not necessarily the same thing. If you like, you can argue about whether or not randomness exists. However the fact remains that uncertainty is an inescapable fact of life. So: in the presence of uncertainty, can we say anything at all useful in response to the questions above?

The answer is: yes, but in order to do so we need to use the calculus of probability to talk about uncertainty in a precise way. The kind of answers we might give to the questions above are:

1. If we tossed a coin a large number of times, we would expect roughly equal numbers of heads and tails.
2. In a card game, my opponent might be holding any of a large number of hands¹. If my own cards are such that only a small proportion of these can beat it, I will probably win.
3. At the time of writing (December 2018), the latest ICM² poll puts support for the Conservatives at 40%; Labour at 38%; UKIP at 5%; Liberal Democrats at 9%; and other parties at 7% (see http://www.ukpolitical.info/General_election_polls.htm). The 'margin of error' on these figures is around 3% for the main parties — this means that 95% of the time, ICM expect the 'true' proportions in the population to be within 3 percentage

¹A *hand* of cards is the collection of cards you have in your hand.

²I used to cite MORI here, because their polls are not commissioned by a newspaper with a specific — albeit under-represented — political bias. However, MORI have become a bit shy about publishing their opinion poll results since the 2015 election . . .

points of their quoted figures. Thus, if there were an election tomorrow, would we be able to say that “it is likely that the Conservative party would win”? Of course, the next election is *not* tomorrow and these figures will probably change over time.

4. The best treatment for an illness depends on what you mean by ‘best’ — do you mean ‘most effective in preventing the disease’ or ‘safest’ (amputation is a 100% effective, but somewhat excessive, remedy for athlete’s foot!) or a combination of the two? Ignoring this aspect, suppose that drugs A and B have been tested on different groups of people and that, on average, people on drug A recovered from their illness more quickly than those on drug B. However, time to recovery varies between individuals, so we cannot guarantee that the result will be reproduced if the experiment is repeated. If the average difference between the drugs is large relative to the variability between individuals’ responses, we can be fairly sure that drug A is better than drug B; otherwise, we may as well regard any difference between the drugs as due to chance alone.

In previous courses you have already encountered some basic ideas in probability, which enable you to be make statements in the spirit of those above, but in a rather more precise way. From an applied perspective, it is convenient to think of probability as the language of uncertainty³. In this course, we will build on the ideas you have encountered already to introduce a richer range of *probability models* that are applicable to a wider range of real-life situations.

The other part of the course — relating to *statistics* — is about applying ideas in probability to simplified real-life situations. Statistics is often perceived to be about the mindless and tedious business of collecting and tabulating data. Certainly, at the beginning of the 19th century when people first started to realise that you could find out useful things about society and the state of a nation by gathering data, this was pretty much what was involved⁴. However, statistics nowadays is rather more than that. Modern statistics is about *interpreting information* (the recommended textbook for the course gives, on pages 1–2, a variety of alternative definitions if you’re interested). Given that there is usually uncertainty associated with the interpretation of information, it is inevitable that probability finds its way, in some form or another, into most statistical methods.

The foundations of modern statistics, recognising the need to deal with uncertainty in interpreting data, were laid around the turn of the twentieth century. Most of the work was done at UCL by Karl Pearson (who left the Mathematics Department and founded a new department: the present departments of Statistical Science and Computer Science are descended from this, as well as the Genetics and Biometry group in Biology, and parts of Anthropology), Francis Galton (who was Charles Darwin’s cousin) and Walter Weldon (a zoologist, who was particularly interested in

³It is worth emphasizing that this *is* an applied perspective on probability. It is also possible to regard probability purely as a mathematical object satisfying certain properties. Many results in probability have been derived by pure mathematicians working on *measure theory*. This is a topic which is covered in some 3rd and 4th year Mathematics courses at UCL.

⁴In fact, the word *statistics* derives from ‘wishing to find out about the State’ (German *statistik* = ‘political science’, Latin *statisticus* = ‘person skilled in statecraft’, etc.).

discovering if it was possible to find evidence for Darwin's Theory of Evolution from observations of animal and plant populations). Most of the statistical techniques covered in the later part of this course were developed at this time. Perhaps surprisingly, the rigorous development of probability came later. It wasn't until 1933 that the Russian mathematician A.N. Kolmogorov came up with a clear, and generally acceptable, definition of what probability actually *is*. Kolmogorov's system is the one now generally accepted as providing a foundation for the rigorous study of probability.

In taking this course, we have the benefit of hindsight and will therefore cover the basics of probability *before* discussing how these may be applied, using statistical methods, to the interpretation of information.

All students taking the course should have studied a small amount of probability theory already during their degree: for example, as part of MATH0008 or during the post-examination short course. In this chapter we will summarise the fundamental results that you are expected to know.

1.1 Definitions and notation

Probability statements are made in the context of an EXPERIMENT which can result in a number of possible OUTCOMES. The symbol ω will be used in this course to denote an individual outcome. The set of all possible outcomes is called the SAMPLE SPACE for the experiment, and is usually denoted by Ω .

A collection of outcomes is called an EVENT. If E is an event then E is a subset of Ω . We say that E OCCURS if the outcome of the experiment is in E .

Set theoretic notation is used to talk about events and sample spaces. Table 1 summarises the main concepts and notation.

The following results are sometimes useful:

- **de Morgan's Laws:**

1. $(E \cap F)^c = E^c \cup F^c$ i.e. the complement of an intersection is a union of complements.
2. $(E \cup F)^c = E^c \cap F^c$ i.e. the complement of a union is an intersection of complements.

- **The Distributive Laws** for set operations:

1. $E \cap (F \cup G) = (E \cap F) \cup (E \cap G)$.
2. $E \cup (F \cap G) = (E \cup F) \cap (E \cup G)$.

Table 1: Set theory — *main concepts and notation.*

The COMPLEMENT of an event E is the set of outcomes that are not in E . In this course, E^c denotes the complement of E .

The UNION of E and F , $E \cup F$, is the set of outcomes that are in either E , F or both.

The INTERSECTION of E and F , $E \cap F$, is the set of outcomes that are in both E and F .

If all elements of E are also in F , we say that E is a subset of F , and write $E \subseteq F$. If $E \subseteq F$ and E and F are not identical, we write $E \subset F$.

$E \setminus F$ denotes the set of elements of E that are not in F i.e. $E \setminus F = E \cap F^c$.

The EMPTY SET, $\emptyset$, is the set containing no elements i.e. $\emptyset = \{ \}$. Note that $\emptyset = \Omega^c$.

If $E \cap F = \emptyset$ then there are no outcomes common to both E and F . The two events are said to be DISJOINT or MUTUALLY EXCLUSIVE (m.e.).

Events $E_1, \dots, E_n$ are MUTUALLY DISJOINT or PAIRWISE DISJOINT if all possible pairs of the events are disjoint. **Note:** This is a much stronger requirement than $\bigcap_{i=1}^n E_i = \emptyset$.

1.2 The axioms of probability

Let Ω be a sample space, and let $E \subseteq \Omega$ and $F \subseteq \Omega$ be any two events. To be absolutely rigorous we have to say that E and F are elements of an *event space* $\mathcal{F}$, which satisfies

1. $\Omega \in \mathcal{F}$.
2. If $E \in \mathcal{F}$ then $E^c \in \mathcal{F}$.
3. If $E_1 \in \mathcal{F}, \dots, E_n \in \mathcal{F}$ then $\bigcup_{i=1}^n E_i \in \mathcal{F}$.

An event space that satisfies these conditions is called a σ -ALGEBRA.

A PROBABILITY FUNCTION P is a mapping $P : \mathcal{F} \rightarrow \mathbb{R}$ such that

1. For all $E \in \mathcal{F}$, $P(E) \geq 0$.
2. $P(\Omega) = 1$.
3. For events $E, F \in \mathcal{F}$, if $E \cap F = \emptyset$ then $P(E \cup F) = P(E) + P(F)$.

These are KOLMOGOROV'S AXIOMS. For some purposes, axiom 3 must be replaced by that of COUNTABLE ADDITIVITY: this says that for any sequence $E_1, E_2, \dots$ of mutually disjoint sets in $\mathcal{F}$, $P(\bigcup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} P(E_i)$.

These axioms capture some aspects of our intuition as to how probabilities should behave. In many cases, this intuition comes from the RELATIVE FREQUENCY interpretation of probability as the long-run proportion of times that some event E occurs (i.e. the outcome is in E) in a large number of repetitions of an experiment. Unfortunately, the idea of relative frequency cannot be used to give a rigorous *definition* of probability. Moreover, the relative frequency interpretation is not universally accepted. The axioms above *are* universally accepted, however.

The axioms do not require us to define P in such a way that it relates usefully to any real-world situation. For example, in a coin-tossing experiment with sample space $\Omega = \{H, T\}$ (heads, tails), we could define a probability function as follows: $P(\emptyset) = 0, P(H) = 1, P(T) = 0, P(\Omega) = 1$. This satisfies all three axioms (**exercise: check this**), but is of little practical use unless your coin has heads on both sides. This is not a deficiency of the axioms, which merely give properties that probabilities ought to have. The allocation of useful probabilities in any particular situation is an exercise in mathematical modelling: the axioms merely provide constraints on the allocation.

If the sample space is countable, we can specify probabilities individually for each $\omega \in \Omega$, then deduce probabilities for more complicated events. Specifically, let the elements of Ω be $\omega_1, \omega_2, \dots$ and let $\{p_k : k = 1, 2, \dots\}$ be a set of 'weights' such that $p_k \geq 0, \sum_{k=1}^{\infty} p_k = 1$. Then, for any event $E \subseteq \Omega$, the function

$$P(E) = \sum_{\{k: \omega_k \in E\}} p_k$$

satisfies the axioms and hence is a probability function (**Exercise: prove this**). This means that we can calculate the probability of any event by summing the probabilities of the individual outcomes contained within it.

If the sample space is uncountable then things are a bit more difficult, but the axioms can still be used to deduce the probability of a 'complex' event from those of simpler events.

1.3 Deductions from the axioms

You should already be familiar with the following results, and should be able to prove them from the axioms.

1. $P(E^c) = 1 - P(E)$.
2. $P(E) \leq 1$ for all $E \subseteq \Omega$.
3. If $E \subseteq F$ then $P(F \setminus E) = P(F) - P(E)$, and $P(E) \leq P(F)$.
4. For any events E and F (not necessarily disjoint), $P(E \cup F) = P(E) + P(F) - P(E \cap F)$.
Note that in the disjoint case this reduces to axiom 3.

You may not have seen the following results, but they are also useful:

5. For any events $E_1, \dots, E_n$, $P(\bigcup_{i=1}^n E_i) \leq \sum_{i=1}^n P(E_i)$.

Proof: by induction. Suppose the result holds for some value of n , say r , and write $U_r = \bigcup_{i=1}^r E_i$. Then $P(U_r) \leq \sum_{i=1}^r P(E_i)$ by assumption. Now

$$P(U_{r+1}) = P(U_r \cup E_{r+1}) = P(U_r) + P(E_{r+1}) - P(U_r \cap E_{r+1}) ,$$

by result 4 above. Moreover, $P(U_r \cap E_{r+1}) \geq 0$ by axiom 1. Therefore

$$P(U_{r+1}) \leq P(U_r) + P(E_{r+1}) \leq \sum_{i=1}^r P(E_i) + P(E_{r+1}) = \sum_{i=1}^{r+1} P(E_i) .$$

This shows that if the result holds for $n = r$ then it also holds for $n = r + 1$.

The result holds trivially for $n = 1$, since $P(U_1) = P(E_1) = \sum_{i=1}^1 P(E_i)$. Therefore the result holds for $n = 2, 3, \dots$ and for all positive integers n . ■

6. For any events $E_1, \dots, E_n$,

$$\begin{aligned} P\left(\bigcup_{i=1}^n E_i\right) &= \sum_{i=1}^n P(E_i) - \sum_{i < j} P(E_i \cap E_j) \\ &\quad + \sum_{i < j < k} P(E_i \cap E_j \cap E_k) - \dots + (-1)^{n-1} P(E_1 \cap \dots \cap E_n). \end{aligned}$$

Proof: Again by induction, using result (4) above. This one's a bit tricky. We start as above, by assuming that the result holds for some value of n , say r , and writing $U_r = \bigcup_{i=1}^r E_i$. Note that, by assuming the result holds for $n = r$, we assume that it holds *for any collection of r events*. Now, from result (4) above, we have

$$P(U_{r+1}) = P(U_r \cup E_{r+1}) = P(U_r) + P(E_{r+1}) - P(U_r \cap E_{r+1}) .$$

From the first Distributive Law, we have

$$U_r \cap E_{r+1} = E_{r+1} \cap (E_1 \cup \dots \cup E_r) = (E_1 \cap E_{r+1}) \cup \dots \cup (E_r \cap E_{r+1}) = \bigcup_{i=1}^r I_i , \text{ say,}$$

where $I_i = E_i \cap E_{r+1}$. Now, since we are assuming the result holds for any union of r events, we are assuming that

$$\begin{aligned} P(U_r) &= \sum_{i=1}^r P(E_i) - \dots + (-1)^{r-1} P(E_1 \cap \dots \cap E_r) , \\ \text{and that } P\left(\bigcup_{i=1}^r I_i\right) &= \sum_{i=1}^r P(I_i) - \dots + (-1)^{r-1} P(I_1 \cap \dots \cap I_r) . \end{aligned}$$

Using these results, we find

$$\begin{aligned}
 P(U_{r+1}) &= \sum_{i=1}^r P(E_i) - \dots + (-1)^{r-1} P(E_1 \cap \dots \cap E_r) \\
 &\quad + P(E_{r+1}) - \left(\sum_{i=1}^r P(I_i) - \dots + (-1)^{r-1} P(I_1 \cap \dots \cap I_r) \right) \\
 &= \sum_{i=1}^{r+1} P(E_i) - \sum_{1 \leq i < j \leq r} P(E_i \cap E_j) + \dots + (-1)^{r-1} P(E_1 \cap \dots \cap E_r) \\
 &\quad - \sum_{i=1}^r P(I_i) + \dots + (-1)^r P(I_1 \cap \dots \cap I_r) .
 \end{aligned}$$

Notice now that $\sum_{i=1}^r P(I_i) = \sum_{i=1}^r P(E_i \cap E_{r+1})$, which can be combined with the term $\sum_{1 \leq i < j \leq r} P(E_i \cap E_j)$ to obtain

$$\begin{aligned}
 P(U_{r+1}) &= \sum_{i=1}^{r+1} P(E_i) - \sum_{1 \leq i < j \leq r+1} P(E_i \cap E_j) + \sum_{1 \leq i < j < k \leq r} P(E_i \cap E_j \cap E_k) - \dots \\
 &\quad + (-1)^{r-1} P(E_1 \cap \dots \cap E_r) + \sum_{1 \leq i < j \leq r} P(I_i \cap I_j) - \dots + (-1)^r P(I_1 \cap \dots \cap I_r) .
 \end{aligned}$$

The terms $\sum_{1 \leq i < j < k \leq r} P(E_i \cap E_j \cap E_k)$ and $\sum_{1 \leq i < j \leq r} P(I_i \cap I_j)$ can now be combined as in the previous step, and we can continue in this way to obtain (finally!)

$$\begin{aligned}
 P(U_{r+1}) &= \sum_{i=1}^{r+1} P(E_i) - \sum_{1 \leq i < j \leq r+1} P(E_i \cap E_j) + \sum_{1 \leq i < j < k \leq r+1} P(E_i \cap E_j \cap E_k) - \dots \\
 &\quad + (-1)^r P(E_1 \cap \dots \cap E_{r+1}) ,
 \end{aligned}$$

which demonstrates that if the result holds for $n = r$, it also holds for $n = r + 1$.

When $n = 1$, the result says that $P(E_1) = P(E_1)$ which, again, is trivially true. The case $n = 2$ is just result (5) above, which we used in the proof. Since the result holds for $n = 2$, it must hold for $n = 3, 4, \dots$ and hence for all positive integers n . ■

1.4 Conditional probability and independence

Often we need to study relationships between events within the framework of probability. The following definitions are fundamental:

- Events E and F are INDEPENDENT if $P(E \cap F) = P(E)P(F)$.
- The CONDITIONAL PROBABILITY of F given E is the probability that F occurs when E is known to have occurred. Denoted by $P(F|E)$, it is defined as

$$P(F|E) = \frac{P(E \cap F)}{P(E)},$$

when $P(E) > 0$.

- Events $E_1, \dots, E_n$ are MUTUALLY INDEPENDENT if, for *any collection* $E_{i_1}, \dots, E_{i_k}$ of these events,

$$P(E_{i_1} \cap \dots \cap E_{i_k}) = \prod_{j=1}^k P(E_{i_j}).$$

Notes:

1. The definition of independence captures the idea that events E and F are unrelated. In relative frequency terms, $P(E)$ is the long-run proportion of times that E occurs and $P(E \cap F)$ is the proportion of times that E and F both occur. If $P(E \cap F) = P(E)P(F)$ then, whenever E occurs, F also occurs a proportion $P(F)$ of the time, which is the same as the overall proportion of time that F occurs. Informally, we can think ‘ E does not affect F ’ (and vice versa).
2. Independence is *not* the same as being mutually exclusive. **Exercise:** is it possible for two events to not be m.e. but independent?
3. The definition of conditional probability works in effect by changing the sample space from Ω to E : to see this, note that $P(E|E) = 1$. The interpretation is that $P(F|E)$ is the probability of F *given that the outcome of the experiment is known to be in E* .
4. If E and F are independent and $P(E) \neq 0$ then

$$P(F|E) = \frac{P(E \cap F)}{P(E)} = \frac{P(E)P(F)}{P(E)} = P(F).$$

So if E and F are independent, the fact that E occurs tells us nothing about F . This is a more intuitive way of thinking about independence.

5. Mutual independence is more than just requiring $P(\bigcap_{i=1}^n E_i) = \prod_{i=1}^n P(E_i)$. For example, if $P(E_1) = 0$ we will have $P(\bigcap_{i=1}^n E_i) = \prod_{i=1}^n P(E_i) = 0$, regardless of the dependence among $E_2, \dots, E_n$.
6. Mutual independence is a stronger condition than PAIRWISE INDEPENDENCE, which is the condition that all pairs of events E_i and E_j are independent.

Example: Consider two tosses of a fair coin. Define E_1 to be the event ‘first toss is a head’; E_2 to be the event ‘second toss is a head’, and E_3 to be the event ‘the outcomes on the two tosses are different’. Show that E_1 , E_2 and E_3 are pairwise independent but not mutually independent.

■

1.5 Key results in conditional probability

This section reviews some of the most useful results in probability theory.

1. **Law of total probability** (also referred to as the ‘generalised addition law’): Let $\{E_i\}$ be a PARTITION of Ω i.e. a countable (but possibly infinite) collection of subsets of Ω such that each outcome belongs in exactly one of the E s. Then for any event F ,

$$P(F) = \sum_i P(F|E_i) P(E_i) .$$

Proof: Note that $F = F \cap \Omega = F \cap (\bigcup_i E_i) = \bigcup_i (E_i \cap F)$, by the distributive laws. Also note that the $\{E_i \cap F\}$ are mutually exclusive, since the $\{E_i\}$ are. Thus, by axiom 3, we must have $P(F) = \sum_i P(F \cap E_i)$, and $P(F \cap E_i) = P(F|E_i)P(E_i)$ from the definition of conditional probability. The result follows. ■

This result is especially useful for solving problems where you want to know the probability of F , and you think ‘if I knew whether or not E had occurred, I could do this’.

Example: A fair six-sided die is rolled twice. What is the probability that the numbers obtained add up to 9?

2.

Bayes’ Theorem: Let $\{E_i\}$ be a partition of Ω , and F be any event with $P(F) > 0$.

Then

$$P(E_i|F) = \frac{P(F|E_i) P(E_i)}{P(F)} = \frac{P(F|E_i) P(E_i)}{\sum_i P(F|E_i) P(E_i)}.$$

Proof: By definition, $P(E_i|F) = P(E_i \cap F) / P(F)$, and $P(E_i \cap F) = P(F|E_i) P(E_i)$. The expansion of the denominator for $P(F)$ is an application of the Law of Total Probability.

This result has a wide variety of applications. It is particularly important in problems where a situation could have arisen from a number of different causes, and it is required to determine which of the causes was probably responsible. In this case, F denotes the situation which has arisen and E_i denotes the event ‘the i th cause was present’. The result allows us to ‘invert’ the problem by expressing it in terms of $P(F|E_i)$, the probability that F would have

occurred if the i th cause was present. If you think this sounds abstract, it's about murder trials, forensic evidence and miscarriages of justice (what's the probability that a suspect is guilty given the evidence? Let E_1 be the event 'suspect is guilty', E_2 the event 'suspect is innocent', F the evidence — a body, DNA tests etc.).

Example: A doctor is called to see a sick child. The doctor knows (prior to the visit) that 90% of the sick children in the neighbourhood have the flu, while 10% have measles. No child is unfortunate enough to be suffering from both diseases.

A well-known symptom of measles is a rash. The probability of developing a rash if you have measles is 0.95. However, occasionally children with the flu also develop a rash — the probability of developing a rash if you have flu is 0.08.

Upon examining the child, the doctor finds a rash. What is the probability that the child has measles?

■

Generalised Multiplication Law: For arbitrary events $E_1, \dots, E_n$ such that $P(E_1 \cap \dots \cap E_{n-1}) > 0$,

$$3. \quad P\left(\bigcap_{i=1}^n E_i\right) = P(E_1) P(E_2|E_1) P(E_3|E_1 \cap E_2) \dots P(E_n|E_1 \cap \dots \cap E_{n-1}) .$$

Proof: First, note that all of the conditional probabilities here are defined. This is because result 3 from Section 1.3 guarantees that

$$P(E_1) \geq P(E_1 \cap E_2) \geq \dots \geq P(E_1 \cap \dots \cap E_{n-1}),$$

and the result requires $P(E_1 \cap \dots \cap E_{n-1}) > 0$.

A proof from left to right is by induction, and is left as an exercise. Here, we content ourselves by showing, working from right to left, that the equality holds. Using the definition of conditional probability throughout the RHS gives

$$P(E_1) \frac{P(E_1 \cap E_2)}{P(E_1)} \frac{P(E_1 \cap E_2 \cap E_3)}{P(E_1 \cap E_2)} \dots \frac{P(E_1 \cap \dots \cap E_n)}{P(E_1 \cap \dots \cap E_{n-1})}.$$

The numerator of each term cancels with the denominator of the next, and we are left with $P(E_1 \cap \dots \cap E_n)$, which is the LHS of the result. ■

This one is useful for analysing problems where an event of interest can be regarded as arising from a sequence of contributing events. For example, to find the probability of a student getting a degree we could define the event E_i to be ‘student passes i th year of course’ ($i = 1, 2, 3$ and possibly 4). For 3-year degrees, you get a degree if you pass each year of your course, which corresponds to the event $E_1 \cap E_2 \cap E_3$. This result allows you to decompose the event into manageable chunks e.g. $P(E_2|E_1)$ is the probability that you pass your second year given that you passed your first year (which is hopefully higher than $P(E_2)$, in which case E_1 and E_2 are *not* independent).

1.6 Probability distributions

We often need to calculate probabilities in situations where the outcome of an experiment is a number (or where it can be converted to one). This can be done by specifying *probability distributions* for the associated *random variables*. You should already be familiar with the binomial, Poisson and normal distributions, and with the notions of *expectation* (or mean value) and variance. However, you probably have not studied random variables formally. This needs to be done before we tackle some of the more advanced material in the course.

2 Discrete random variables

2.1 Random variables and distributions

Definition: Suppose Ω is a countable sample space. Then a function

$$X : \Omega \rightarrow \mathbb{R}$$

is called a DISCRETE RANDOM VARIABLE (discrete r.v.).

In effect, this definition says that a discrete random variable is a rule for allocating numbers to outcomes of an experiment.

Example 2.1: When tossing a coin twice, we could study the number of heads obtained as a discrete random variable. This is a mapping from the sample space $\{HH, HT, TH, TT\}$ as follows:

Outcome		Value
HH	→	2
HT	→	1
TH	→	1
TT	→	0

■

Example 2.2: If Ω is a set of n individuals, we could choose to study their heights by defining a random variable $X(\cdot)$ such that $X(\omega)$ is the height of individual ω .

■

Example 2.3: Suppose a six-sided fair die is thrown twice. The sample space for this experiment is $\Omega = \{(i, j) : i, j = 1, \dots, 6\}$ (NB the elements of the sample space are not numbers, they're *pairs* of numbers). We could define a random variable X as the total score on the two throws i.e. $X((i, j)) = i + j$.

■

A probability function P on Ω induces a PROBABILITY DISTRIBUTION on X . This is obtained by considering events of the form ' $X(\omega) = x$ ', i.e. collections of outcomes which are allocated to x by X . In mathematical notation, the event is written as $\{\omega : X(\omega) = x\}$. Then $P(X(\omega) = x) = P(\{\omega : X(\omega) = x\})$. We abbreviate this to $P(X = x)$ for convenience, although this is a slight abuse of notation. Bear in mind when you write things like this that X is actually a *function* (or mapping) but x is a *number*. By convention, we use *UPPER CASE LETTERS* to denote random variables, and *lower case letters* for the values they take.

Example 2.3 continued: $X((i, j)) = 8$ iff $(i, j) \in \{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\}$. We would therefore write $P(X = 8) = 5/36$.

■

We are now in a position to introduce the following:

Definition: Let X be a discrete random variable taking values in the set $\{x_i\}$ (which must be countable, since Ω is). The function $p_X(x) = P(X = x)$ is called the PROBABILITY MASS FUNCTION (pmf) of X .

A discrete probability distribution is specified by the set of values $\{x_i\}$ together with the corresponding probabilities $\{p_X(x_i)\}$. If there is no ambiguity regarding which random variable is being referred to, we can drop the X subscript from the pmf and write $p(x)$ instead. Often a discrete random variable is integer-valued, in which case you may see p_n used to denote $P(X = n)$.

For a collection $\{p(x_i)\}$ to be a pmf, it must have the following properties in order that the axioms of probability are satisfied:

1. $p(x_i) \geq 0 \forall i$ (since the p s are supposed to be probabilities of events).
2. $\sum_i p(x_i) = 1$. This must hold since $\Omega = \bigcup_i \{\omega : X(\omega) = x_i\}$, which is a union of disjoint events. Hence $P(\bigcup_i \{\omega : X(\omega) = x_i\}) = \sum_i P(\{\omega : X(\omega) = x_i\}) = P(\Omega) = 1$, by axiom (3).

This shows that the collection $\{x_i\}$ can be regarded as a sample space in its own right, with probability function $P(\{x_i\}) = p(x_i)$. Intuitively, we are often only interested in the values taken by X , in which case this may be an appropriate form in which to express the sample space.

As well as studying the probability that X takes any particular value, we can consider the probability that the value lies within an interval e.g.

$$P(a \leq X \leq b) = P(\{\omega : a \leq X(\omega) \leq b\}) = \sum_{i: a \leq x_i \leq b} p_X(x_i) .$$

In particular, we can consider the probability that the value lies *below* some threshold, and this leads to the following:

Definition: for any random variable X , the function

$$F_X(x) = P(X \leq x) = \sum_{i: x_i \leq x} p_X(x_i)$$

is called the (CUMULATIVE) DISTRIBUTION FUNCTION (cdf) of X .

Note: if we refer to the *distribution function* of a random variable, we are talking about F and nothing else!

The properties of a cdf, again derivable from the axioms, are as follows:

1. $F(x) \rightarrow 0$ as $x \rightarrow -\infty$. This follows immediately from the definition above.
2. $F(x) \rightarrow 1$ as $x \rightarrow \infty$.

3. $F(x)$ is monotonic increasing. **Proof: exercise.**
4. For a discrete random variable, $F(x)$ is a step function with discontinuities at the ATOMS $\{x_i\}$ of the distribution, the magnitude of the jump at x_i being $p(x_i)$.

From the definition of the distribution function, it is clear that we can derive the cdf from the pmf. The converse is also true. To see this, note that by definition, F is right continuous (i.e. the value of the function at each ‘step’ is the higher of the two possible values). Thus

$$p(x) = P(X = x) = P(X \leq x) - P(X < x) = \lim_{\delta \rightarrow 0+} [F(x) - F(x - \delta)] .$$

In fact, this is a special case of a more general result, and we skipped over something that needs to be derived a bit more rigorously. The general result is:

Result: If $a < b$ then $P(a < X \leq b) = F(b) - F(a)$.

Proof:



Now we know all about discrete random variables and how we can allocate probabilities to them. Often, it is useful to be able to summarize a probability distribution by giving a single ‘typical’ value for a random variable. One way of doing this is to take the value in the ‘middle’ of the distribution:

Definition: The MEDIAN of a distribution is the value of x such that $F(x) = 1/2$.

So if we repeat an experiment over and over again, observing the value of a random variable each time, in the long run we expect to observe equal proportions of results above and below the median.

Of course, this idea of summarising a probability distribution by a single a value ‘in the middle’ is related in some way to our idea of ‘the average’. We now explore this further.

2.2 Expectation

We are all familiar with the notion of an ‘average’. In everyday usage, the term is usually used to denote what you get by adding together a load of numbers and dividing the result by how many there are. The intuitive justification for this is something like ‘if all the values had been the same, this is what they would have been’. So the operation of ‘averaging’ provides us with a convenient single-number summary of an entire dataset.

Can we extend this idea of a ‘convenient single-number summary’ to random variables? Recall the relative frequency interpretation of probability in terms of the notional repetition of experiments. Suppose X is a discrete random variable defined on a sample space Ω , taking values $\{x_i\}$ with probabilities $\{p(x_i)\}$. If we were to repeat the experiment a large number (n , say) of times

then we could observe a large sequence of realised values of X and, in our ‘relative frequency’ interpretation, we would expect the value x_i to come up roughly $n \times p(x_i)$ times. If we calculated the ‘average’ of all the observed x s, we would therefore expect the result to come out as roughly

$$\frac{1}{n} \sum_i [x_i \times np(x_i)] = \sum_i x_i p(x_i) .$$

This motivates the following:

Definition: The EXPECTED VALUE, or EXPECTATION, of a discrete random variable X , denoted by $E(X)$, is defined as

$$E(X) = \sum_i x_i P(X = x_i) = \sum_i x_i p(x_i) ,$$

where $p(\cdot)$ is the probability mass function of X . The definition is valid providing the sum is well-defined.

The sum will always be well-defined if the sample space is finite. In the countably-infinite case, the sum is not guaranteed to converge. We say that $E(X)$ EXISTS if the sum converges absolutely i.e. if $\sum_i |x_i| P(X = x_i) < \infty$.

$E(X)$ represents an idealised long-run average for the values of X . It is also called the (POPULATION) MEAN of X . Note that it need not necessarily be a value that X can take.

Note that $p(x_i) = P\{\omega : X(\omega) = x_i\}$, so that

$$E(X) = \sum_i x_i P\{\omega : X(\omega) = x_i\} = \sum_{\omega \in \Omega} X(\omega) P(\{\omega\}) .$$

This says that we can perform the average *either* on the values taken by X *or* directly on the sample space.

Expectation of a function: If X is a random variable and $\phi : \mathbb{R} \rightarrow \mathbb{R}$, then $\phi(X)$ is also a random variable since $\phi(X) : \Omega \rightarrow \mathbb{R}$. Thus its expectation is

$$E(\phi(X)) = \sum_{\omega \in \Omega} \phi(X(\omega)) P(\{\omega\})$$

with respect to the original sample space. Gathering together all the outcomes that result in the same value for X , we get

$$E(\phi(X)) = \sum_i \phi(x_i) P(\{\omega : X(\omega) = x_i\}) = \sum_i \phi(x_i) p_X(x_i) .$$

Properties of Expectation. The following results can save time when calculating expectations.

1. If X is a constant c (by which we mean $P(X = c) = 1$), then $E(X) = c$.

Proof: trivial!

2. If a and b are constants and $Y = aX + b$, then $E(Y) = aE(X) + b$.

Proof: From the definition of expectation of a function, $E(Y) = \sum_i (ax_i + b)p(x_i) = a \sum_i x_i p(x_i) + b \sum_i p(x_i)$. But $\sum_i x_i p(x_i) = E(X)$ by definition, and $\sum_i p(x_i) = 1$. The result follows. ■

Symmetry: A particularly useful result, which can save you a whole load of calculation, is that if X is a random variable with a symmetric pmf, and if $E(X)$ exists, then $E(X)$ is just the central value (i.e. point of symmetry of the pmf). This might be obvious to you because you have an intuitive idea of what ‘expectation’ really is. However, all we’ve really done so far is to define expectation in mathematical terms and derive some properties. We’ll now use these to show that the symmetry result does indeed hold.

Suppose first of all that the pmf is symmetric about zero. What this means is that X takes values $0, \pm x_1, \pm x_2, \dots$ with probabilities $p_0, p_1, p_2, \dots$ say (if X cannot take the value 0 we just set $p_0 = 0$). Then, from the definition, we find that where $E(X)$ exists, it is $(0 \times p_0) + (x_1 \times p_1) + (-x_1 \times p_1) + (x_2 \times p_2) + (-x_2 \times p_2) + \dots$, which is clearly zero.

If the pmf of X is symmetric about some other value, μ say, which is not equal to zero, then it takes values $\mu, \mu \pm x_1, \mu \pm x_2, \dots$ with probabilities $p_0, p_1, p_2, \dots$. Define $Y = X - \mu$. Then Y has a pmf which is symmetric about zero. Hence $E(Y) = 0$. But $E(Y) = E(X) - \mu$ by result 2 above; hence $E(X) = \mu$.

Note unless you *know* that $E(X)$ exists, you can’t use the symmetry argument. $E(X)$ will always exist if the set of values taken by X is finite.

Example 2.3 continued: Let X be the total score on two successive throws of a fair die. Write down the pmf of X , and find $E(X)$ ($= \mu$, say). Also, find $E(X - \mu)^2$.



We have seen that, if the pmf of X is symmetric about μ then $E(X) = \mu$ if it exists. Another way of thinking of this is that the pmf ‘balances’ at μ (i.e. if you had a load of physical weights $\{p_j\}$, stuck on a beam at positions $\{x_j\}$, the whole lot would balance at position μ). This is related to the idea of *moments* in physics, which govern where things balance. This is the reason for the nomenclature in the following:

Definition: The r TH MOMENT OF X ABOUT α is given by

$$E((X - \alpha)^r) ,$$

when it exists.

Don’t panic, it’s just a definition. You know how to calculate it, since it’s just the expectation of a function of X (r and α are numbers). We see this most frequently with $\alpha = 0$ or $\alpha = \mu$. For example, $E(X)$ is the first moment of X about 0.

The most important moment apart from $E(X)$ is the second moment about μ :

Definition: Let X be a random variable with mean μ (remember: ‘mean’ is another word for ‘expectation’). The VARIANCE of X is defined by

$$\text{Var}(X) = E((X - \mu)^2),$$

where it exists.

The following alternative forms are often useful for calculating variance in practice ($\mu = E(X)$ throughout):

1. $\text{Var}(X) = E(X^2) - \mu^2.$
2. $\text{Var}(X) = E(X(X - 1)) - \mu(\mu - 1).$

To see the first of these, note that $E((X - \mu)^2) = E(X^2 - 2\mu X + \mu^2) = E(X^2) - 2\mu E(X) + \mu^2$, using properties 1 and 2 of expectation as derived previously. The second term here is just $-2\mu E(X) = -2\mu \times \mu = -2\mu^2$, and the expression simplifies to the desired result.

The derivation of the second alternative is left for you as an exercise.

■

Properties of variance: the variance of a random variable is, as we shall see, an extremely useful quantity. As with expectation, there are some fundamental properties that are helpful in manipulating variances:

1. $\text{Var}(X) \geq 0$ wherever it exists.

Proof: Variance is defined as

$$\text{Var}(X) = E(X - \mu)^2 = \sum_i (x_i - \mu)^2 p(x_i).$$

This is a sum of terms which are all non-negative.

■

2. $\text{Var}(X) = 0$ if and only if X is a constant i.e. $P(X = c) = 1$.

Proof: Suppose $P(X = c) = 1$. Then $E(X) = c$ (from properties of expectation), and $\text{Var}(X) = 0$ from the definition of variance.

Now suppose that X is not a constant i.e. X takes at least two values with non-zero probability. In this case, at least one of the terms in the sum above must be non-zero, so that $\text{Var}(X) > 0$. So the only way we can have $\text{Var}(X) = 0$ is if X is a constant. ■

3. If a and b are constants and $Y = aX + b$, then $\text{Var}(Y) = a^2 \text{Var}(X)$.

Proof: Let μ_X and μ_Y be the expected values of X and Y respectively. Then $\text{Var}(Y) = E(Y - \mu_Y)^2$. Now $\mu_Y = E(Y) = aE(X) + b$ (properties of expectation), so $Y - \mu_Y = a(X - \mu_X)$. Hence

$$\text{Var}(Y) = E([a(X - \mu_X)]^2) = a^2 E(X - \mu_X)^2 = a^2 \text{Var}(X),$$

as required. ■

Often we will work with the square root of the variance, rather than the variance itself:

Definition: The STANDARD DEVIATION of X is

$$\text{sd}(X) = \sqrt{\text{Var}(X)},$$

where it exists.

The motivation for this is that, if X has a physical interpretation in terms of some units of measurement (e.g. when X is the height of a person in centimetres), standard deviation is measured in the same units as X and hence has a direct interpretation. We will see this illustrated in the Statistics part of the course. For the moment, just accept the definition!

Notation: We have already seen the symbol μ used to denote the expected value of a random variable. This is a standard convention, as is the use of σ^2 to denote a variance and σ to denote a standard deviation. Where necessary, we write $\mu_X, \mu_Y, \sigma_X^2, \sigma_X$ etc. to avoid ambiguity.

Interpretation: The expected value tells us where, on the number line, the pmf of a random variable ‘balances’. It is therefore a MEASURE OF LOCATION (since it ‘locates’ the pmf on the number line). If you examine the definition of variance, you see that it is the expected squared distance between a random variable and its mean — so it tells us something about how ‘spread out’ a pmf is about its mean. In particular, we have seen that if X is a constant, $\text{Var}(X) = 0$. For this reason, variance and standard deviation are called MEASURES OF SPREAD or DISPERSION.

Example 2.3 continued: Let X be the total score on two successive throws of a fair die. Find $\text{Var}(X)$ and $\text{sd}(X)$.



The standard deviation is expressed in the same units of measurement as the mean. We sometimes wish to work with a dimensionless measure of dispersion, and therefore introduce:

Definition: Let X be a random variable taking non-negative values, with mean μ and standard deviation σ . The COEFFICIENT OF VARIATION of X is defined as the ratio σ/μ .

A further expectation-related quantity, which is useful for more advanced study, is introduced in the following:

Definition: Let X be a random variable taking values $0, 1, 2, \dots$. The PROBABILITY GENERATING FUNCTION (pgf) of X is defined as

$$\Pi_X(z) = E(z^X) = \sum_{k=0}^{\infty} z^k p(k),$$

for all values of z at which the expectation is defined.

Note that the pgf is a function of z . Therefore we can do things like differentiate it and evaluate it for different values of z . Where there is no ambiguity we drop the subscript and write $\Pi(z)$.

The pgf is useful for a variety of reasons, as follows:

1. If we know the functional form of the pgf for a random variable, and can expand this as a power series in z , then we can find the probability that X takes any value by picking out the appropriate term in the power series (look at the definition above). This explains why the definition is restricted to random variables taking non-negative integer values — otherwise we could not pick out the probabilities in any useful way. Note that many random variables of practical interest *do* take non-negative values. Typically they represent counts e.g. numbers

of people in a queue, number of hurricanes hitting the USA in a given year etc.

You might think that this isn't a very useful feature, since in order to find the pgf you have to know the probabilities to start with! However, in advanced applications it is, believe it or not, often easier to derive the pgf than to derive the probabilities directly. We will see this later in the course.

2. The behaviour of the function at the value $z = 1$ is particular important. We have:

(a) $\Pi(1) = 1$, always.

Proof: trivial from the definition. ■

(b) $\Pi'(1) = E(X)$, where it exists (by $\Pi'(1)$, we mean the gradient $d\Pi(z)/dz$ evaluated at $z = 1$).

Proof: From the definition, we have

$$\Pi'(z) = \frac{d\Pi(z)}{dz} = \frac{d}{dz} \left(\sum_{k=0}^{\infty} z^k p(k) \right) = \sum_{k=0}^{\infty} k z^{k-1} p(k).$$

Putting $z = 1$ into this gives $\sum_{k=0}^{\infty} k p(k)$, which is the definition of $E(X)$ (since X takes values $0, 1, 2, \dots$ with probabilities $p(0), p(1), p(2), \dots$). ■

(c) $\Pi''(1) = E(X(X-1))$ where it exists (recall that $E(X(X-1))$ is used in one of the alternative formulae for variance).

Proof: as in the previous case. We have

$$\begin{aligned} \Pi''(z) &= \frac{d\Pi'(z)}{dz} = \frac{d}{dz} \left(\sum_{k=0}^{\infty} k z^{k-1} p(k) \right) = \frac{d}{dz} \left(0 + \sum_{k=1}^{\infty} k z^{k-1} p(k) \right) \\ &= \sum_{k=1}^{\infty} k(k-1) z^{k-2} p(k). \end{aligned}$$

When $z = 1$, this gives $\sum_{k=1}^{\infty} k(k-1)p(k)$. But this is just the definition of $E(X(X-1))$, since $E(X(X-1)) = \sum_{k=0}^{\infty} k(k-1)p(k) = 0 + \sum_{k=1}^{\infty} k(k-1)p(k)$. ■

In general, it can be shown that the k th derivative of $\Pi(z)$, evaluated at $z = 1$, gives $E(X(X-1)\dots(X-k+1))$. This quantity is called the k TH FACTORIAL MOMENT of X .

Example 2.4: Suppose that X is a random variable having pgf $\Pi(z) = pz \left[(1 - (1-p)z)^{-1} \right]$, where $p \in (0, 1)$. Find $P(X = k)$ for $k = 0, 1, 2, \dots$; also find $E(X)$ and $\text{Var}(X)$.



2.3 Standard discrete distributions

The examples of random variables encountered so far have been simple, and have been dealt with on a case-by-case basis. In practice, we often find ourselves in situations which are similar to those we have encountered before, and it is useful to assemble a ‘cookbook’ of formulae which are applicable in any given situation without having to start from scratch every time. Some common situations are as follows:

1. X takes the value 1 if the outcome of an experiment is a ‘success’, 0 if it is a ‘failure’ (NB success and failure are used here as generic terms for two outcomes which form a partition of the sample space Ω — it could just as easily be heads and tails, alive and dead etc.). Such an experiment is called a BINARY EXPERIMENT or a BERNOULLI TRIAL, and we describe the random variable X as having a BERNOULLI DISTRIBUTION.
2. X is the number of ‘successes’ recorded during a sequence of n (≥ 1) independent Bernoulli trials, each with the same probability of success. In this case, X has a BINOMIAL DISTRIBUTION.
3. X is the number of independent Bernoulli trials required before a ‘success’ is observed. X has a GEOMETRIC DISTRIBUTION.

4. X is the number of independent Bernoulli trials required before r (≥ 1) ‘successes’ have been observed. X has a NEGATIVE BINOMIAL DISTRIBUTION.
5. X is the number of objects having a particular attribute obtained when we draw n objects at random from a population of N , of which M have the attribute of interest. X has a HYPERGEOMETRIC DISTRIBUTION.
6. X is the number of ‘accidents’ occurring during a time period of fixed duration (again, ‘accident’ is a generic term, taken to mean simply an instantaneous occurrence of something). X has a POISSON DISTRIBUTION.

In all of these situations, the basic setup is the same but there will be features that differ from one case to another. For example, in example (1), the probability of a ‘success’ (p , say) will be different in different situations; in example (2), both p and n may differ. These varying factors are referred to as PARAMETERS of the distribution. So really if we talk about ‘a binomial distribution’ we are talking about a whole *family* of distributions which are all effectively the same except for the values of their parameters.

A discrete random variable is completely specified by its probability mass function: from there, we can derive any property of the random variable we wish (such as mean and variance). You should already be familiar with the binomial and Poisson distributions; we now derive pmfs, and other properties, for the other distributions described above.

2.3.1 The Bernoulli distribution (parameter $p \in (0, 1)$):

Suppose an experiment has only 2 possible outcomes, labelled ‘success’ and ‘failure’, and $P(\text{success}) = p$. Let

$$X = \begin{cases} 1 & \text{if outcome is ‘success’} \\ 0 & \text{if outcome is ‘failure’} \end{cases}$$

Then $P(X = 1) = p$, $P(X = 0) = 1 - p$. X is said to have a BERNOULLI DISTRIBUTION WITH PARAMETER p .

$$\begin{aligned} E(X) &= (1 \times p) + (0 \times (1 - p)) = p \\ E(X^2) &= (1^2 \times p) + (0^2 \times (1 - p)) = p \\ \text{So } Var(X) &= E(X^2) - (E(X))^2 = p - p^2 = p(1 - p). \end{aligned}$$

2.3.2 The Binomial distribution (parameters $n \in \mathbb{N}, p \in (0, 1)$)

Consider an experiment consisting of n independent Bernoulli trials, each with success probability p , and let X be the total number of ‘successes’. X takes values in $\{0, 1, \dots, n\}$, and its pmf can be derived (see your previous probability courses) as

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k} \quad (k = 0, 1, \dots, n).$$

We say that X follows a BINOMIAL DISTRIBUTION WITH PARAMETERS n AND p , and write $X \sim \text{Bin}(n, p)$. n is sometimes referred to as the INDEX of the distribution.

Result: If $X \sim \text{Bin}(n, p)$ then $E(X) = np$, $\text{Var}(X) = np(1 - p)$.

You have seen the proof of this result in your previous probability courses. We will give an alternative and simpler proof in Chapter 5.

Example 2.5: The guidance system for a rocket operates correctly with probability 0.85 when called upon. Independent but identical backup systems are installed in the rocket. Let n be the number of systems. Find n if the probability that none of the systems work is to be at most 0.01.

If $n = 5$, find the probability that at least 2 systems work.



2.3.3 The Geometric distribution (parameter: $p \in (0, 1)$)

Consider a sequence of independent Bernoulli trials, each with success probability p . Let X be the number of trials up to and including the first success. X is a discrete random variable taking values in $\{1, 2, 3, \dots\} = \mathbb{N}$.

To derive the pmf of X , we note that

$$\begin{aligned} P(X = k) &= P(\text{first } k-1 \text{ trials fail, } k^{\text{th}} \text{ succeeds}) \\ &= (1-p)^{k-1}p = q^{k-1}p \text{ where } q = 1-p. \end{aligned}$$

This result holds for all $k \in \mathbb{N}$. X is said to have a GEOMETRIC DISTRIBUTION WITH PARAMETER p , and we write $X \sim \text{Geo}(p)$.

To check that this defines a pmf, we need to verify that $P(X = k) \geq 0$ for all k (trivial) and that $\sum_{k=1}^{\infty} P(X = k) = 1$. To show the latter we use the expansion

$$(1-x)^{-1} = 1 + x + x^2 + \dots = \sum_{k=0}^{\infty} x^k \text{ for } |x| < 1.$$

We have

$$\begin{aligned} \sum_{k=1}^{\infty} P(X = k) &= \sum_{k=1}^{\infty} q^{k-1}p = p \sum_{k=1}^{\infty} q^{k-1} = p(1 + q + q^2 + \dots) \\ &= p[(1-q)^{-1}] \text{ (if } q < 1) = p/p = 1. \end{aligned}$$

Therefore we have a valid pmf, as required. ■

Result: If $X \sim \text{Geo}(p)$ then $E(X) = 1/p$, $\text{Var}(X) = q/p^2$.

Proof: We've already done most of the work for this, in fact, in Example 2.4. The probability generating function for a geometric distribution is

$$\Pi(z) = E(z^X) = \sum_{k=1}^{\infty} z^k p q^{k-1} = pz \sum_{k=1}^{\infty} (zq)^{k-1} = pz(1 - zq)^{-1},$$

providing $|zq| < 1$. We're interested in the behaviour at $z = 1$, which is OK if $|q| < 1$.

If you compare this expression with the pgf given in Example 2.4, you'll see it's exactly the same, and that example proves the result.

The result can also be derived directly from the definitions of expectation and variance, using $E(X) = \sum_{k=1}^{\infty} k p q^{k-1}$ and $E(X(X-1)) = \sum_{k=1}^{\infty} k(k-1) p q^{k-1} = 0 + \sum_{k=2}^{\infty} k(k-1) p q^{k-1}$. In order to do it this way however, you need to use the identities

$$(1-x)^{-2} = 1 + 2x + 3x^2 + \dots = \sum_{k=1}^{\infty} k x^{k-1} \quad (|x| < 1)$$

for $E(X)$, and

$$(1-x)^{-3} = 1 + 3x + \frac{(4 \times 3)}{2}x^2 + \frac{(5 \times 4)}{2}x^3 \dots = \frac{1}{2} \sum_{k=2}^{\infty} k(k-1)x^{k-2} \quad (|x| < 1)$$

for $E(X(X-1))$. This is left as an exercise for those of you who are interested. ■

Note that in general for a geometric distribution the event $X > k$ corresponds to the event ‘first k trials fail’ and has probability q^k . This is a favourite trick to watch out for on exam papers — if X is geometric, and you get asked for $P(X > 10)$, *don’t* calculate $P(X = 1) + P(X = 2) + \dots + P(X = 10)$. If you do, you deserve to lose the 15 or so minutes it’ll take!

2.3.4 The Negative Binomial distribution (parameters $r \in \mathbb{N}, p \in (0, 1)$)

The geometric distribution is the number of trials up to and including the first success. We now generalise this. Let X be the number of independent Bernoulli trials, each with success probability p , up to and including the r^{th} success. X is a discrete random variable taking values in $\{r, r+1, \dots\}$. We have

$$\begin{aligned} P(X = k) &= P(r-1 \text{ successes in first } k-1 \text{ throws, 1 success on } k^{\text{th}} \text{ throw}) \\ &= \binom{k-1}{r-1} p^{r-1} q^{k-r} \times p = \binom{k-1}{r-1} p^r q^{k-r} \text{ for } k = r, r+1, \dots \end{aligned}$$

The random variable X is said to have a **NEGATIVE BINOMIAL DISTRIBUTION WITH PARAMETERS r AND p** ; we write $X \sim NB(r, p)$.

As usual, it is trivial to show that $P(X = k) \geq 0$ for all k . To show that the probabilities sum to 1, we go

$$\begin{aligned} \sum_{k=r}^{\infty} P(X = k) &= \sum_{k=r}^{\infty} \binom{k-1}{r-1} p^r (1-p)^{k-r} = p^r \left(1 + r(1-p) + \frac{r(r+1)}{2}(1-p)^2 + \dots \right) \\ &= p^r (1 - (1-p))^{-r} = 1. \end{aligned}$$

The last line here is, again, an application of the binomial theorem. The name of the distribution derives from the fact that the probabilities are terms in the ‘negative binomial expansion’ of $(1-q)^{-r}$.

Result: If $X \sim NB(r, p)$ then $E(X) = r/p$, $\text{Var}(X) = rq/p^2$.

This result will be proved in Chapter 5. To remember it, just note that the mean and variance are both r times the corresponding results for the geometric distribution.

If $X \sim NB(1, p)$ then X is the number of trials up to and including the first success i.e. $NB(1, p)$ is the same as a geometric distribution with success probability p .

Example 2.6: A large stockpile of used pumps contains 20% that are unusable and need repair. There are 3 repair kits. Pumps are selected for testing 1 at a time, at random. If one needs repairing, then one of the repair kits is used to repair it. It takes 10 minutes to test a pump if it works; 30 minutes to test and repair a pump that does not work. Find the expected value and variance of the total time it takes to use up the three kits.

■

As for the Geometric distribution, watch out for questions of the form ‘ X has a negative binomial distribution; find $P(X > c)$ ’. The event ‘ $X > c$ ’ is equivalent to ‘less than r successes in c trials’ and, if r is small, it may be easier to calculate this using a Binomial distribution for the number of successes in the first c trials.

Some authors define the negative binomial as the distribution of the number of *failures* before the r th success. If Y is the number of failures and X is the number of trials, then clearly $X = Y + r$, so we can derive all the properties of Y from those of X .

2.3.5 The Hypergeometric distribution (parameters $n, M, N \in \mathbb{N} : n, M \leq N$)

Consider a population of finite size N , containing M individuals of interest (we will refer to these as ‘Type 1’ individuals); and consider an experiment which consists of choosing a sample of n objects from this population *at random* (which, in this context, means that all possible subsets of n objects have the same probability). Let X be the number of type 1 objects in the sample.

The distribution of X depends on the sampling strategy:

SAMPLING WITH REPLACEMENT: here, objects are ‘replaced’ after being selected so that, each time an object is selected, the probability of its being Type 1 is M/N . In this case, clearly $X \sim \text{Bin}(n, M/N)$.

SAMPLING WITHOUT REPLACEMENT: in this case, once an object has been selected it cannot be selected again. The binomial distribution is not appropriate in this case, since the probability of drawing a type 1 object changes each time an object is selected.

In the case of sampling without replacement, X takes values in $\{0, 1, \dots, \min(n, M)\}$. We can derive the distribution of X by counting outcomes. There are $\binom{N}{n}$ possible collections of n objects chosen from N . How many of these contain exactly k objects of type 1? Well, if we wanted to obtain a sample with exactly k objects of type 1, we could split the population up, choose k objects of type 1 and then, to make up the numbers, choose $n - k$ objects of type 2. There are $\binom{M}{k}$ possible collections of k type 1 objects, and $\binom{N - M}{n - k}$ collections of $n - k$ type 2 objects. Therefore the total number of ways in which we could obtain a sample with exactly k objects of type 1 is $\binom{M}{k} \binom{N - M}{n - k}$, and we have

$$P(X = k) = \frac{\binom{M}{k} \binom{N - M}{n - k}}{\binom{N}{n}} \text{ for } k = 0, 1, \dots, \min(n, M).$$

We say that X has a **HYPERGEOMETRIC DISTRIBUTION WITH PARAMETERS n, M AND N** , and write $X \sim H(n, M, N)$.

Clearly, all the quantities here are non-negative. Unfortunately, that’s about the only thing that’s easy in terms of derivations for this distribution! To prove that the probabilities sum to 1, we have to show that

$$\sum_{k=0}^{\min(n, M)} \binom{M}{k} \binom{N - M}{n - k} = \binom{N}{n}.$$

It is worth noting that if $M < n$, $\binom{M}{k} = 0$ for $k > M$ so that, without loss of generality, we can show that

$$\sum_{k=0}^n \binom{M}{k} \binom{N - M}{n - k} = \binom{N}{n}.$$

This is done by recognising that $\binom{N}{n}$ is the coefficient of x^n in the series expansion of

$(1+x)^N$ (by the binomial theorem). But

$$(1+x)^N = (1+x)^M (1+x)^{N-M} = \sum_{r=0}^M \binom{M}{r} x^r \times \sum_{s=0}^{N-M} \binom{N-M}{s} x^s.$$

The coefficient of x^n in the right-hand side here is

$$\left[\binom{M}{0} \times \binom{N-M}{n} \right] + \left[\binom{M}{1} \times \binom{N-M}{n-1} \right] + \dots + \left[\binom{M}{n} \times \binom{N-M}{0} \right],$$

which is exactly the required quantity (NB again, if $n > N-M$ it doesn't matter since the first few terms will just be zero in this case). ■

Result: If $X \sim H(n, M, N)$ then

$$E(X) = \frac{nM}{N}, \quad \text{and} \quad \text{Var}(X) = \frac{nM(N-M)(N-n)}{N^2(N-1)}.$$

Proof: For the mean, we have

$$\begin{aligned} E(X) &= \sum_{k=0}^{\min(n,M)} k P(X=k) = \sum_{k=0}^{\min(n,M)} k \frac{\binom{M}{k} \binom{N-M}{n-k}}{\binom{N}{n}} \\ &= 0 + \sum_{k=1}^{\min(n,M)} k \frac{\binom{M}{k} \binom{N-M}{n-k}}{\binom{N}{n}} \\ &= \sum_{k=1}^{\min(n,M)} M \frac{\binom{M-1}{k-1} \binom{N-M}{n-k}}{\binom{N}{n}} \\ &= M \sum_{k=1}^{\min(n,M)} \frac{\binom{M-1}{k-1} \binom{N-M}{n-k}}{\frac{N}{n} \binom{N-1}{n-1}} \\ &= (\text{putting } \ell = k-1) \\ &\quad \frac{nM}{N} \sum_{\ell=0}^{\min(n-1, M-1)} \frac{\binom{M-1}{\ell} \binom{N-M}{n-1-\ell}}{\binom{N-1}{n-1}}. \end{aligned}$$

But the sum here is just the sum of probabilities for a Hypergeometric distribution with parameters $(n-1, M-1, N-1)$ and is therefore 1 (as already demonstrated). Hence $E(X) = nM/N$, as required.

The variance is derived similarly. First, we have

$$\begin{aligned}
 E(X(X-1)) &= \sum_{k=0}^{\min(n,M)} k(k-1) \frac{\binom{M}{k} \binom{N-M}{n-k}}{\binom{N}{n}} \\
 &= 0 + 0 + \sum_{k=2}^{\min(n,M)} k(k-1) \frac{\binom{M}{k} \binom{N-M}{n-k}}{\binom{N}{n}} \\
 &= \sum_{k=2}^{\min(n,M)} M(M-1) \frac{\binom{M-2}{k-2} \binom{N-M}{n-k}}{\binom{N}{n}} \\
 &= M(M-1) \sum_{k=2}^{\min(n,M)} \frac{\binom{M-2}{k-2} \binom{N-M}{n-k}}{\frac{N(N-1)}{n(n-1)} \binom{N-2}{n-2}} \\
 &= (\text{putting } \ell = k-2) \\
 &\quad \frac{n(n-1)M(M-1)}{N(N-1)} \sum_{\ell=0}^{\min(n-2, M-2)} \frac{\binom{M-2}{\ell} \binom{N-M}{n-2-\ell}}{\binom{N-2}{n-2}}.
 \end{aligned}$$

Again, the sum is equal to 1 so we have found $E(X(X-1))$. Now use $\text{Var}(X) = E(X(X-1)) - [E(X)][E(X) - 1]$ to obtain the required result. ■

Example 2.7: A purchaser of electrical components buys them in lots of size 10. It is his policy to inspect 3 components at random from a lot and to accept the lot only if none of them are defective. If 30% of the lots have 4 defective components and 70% have only 1, what proportion of lots does the purchaser reject?



Approximating the Hypergeometric with the binomial: Intuitively, it seems reasonable to think that if N (the size of the population) is very large compared to n (the number of items we draw from it) then the probability of drawing an item of type 1 at each stage will hardly vary, and hence the experiment will be very similar to a sequence of independent Bernoulli trials with success probability $p = M/N$ at each trial. This in turn suggests that, when the ratio n/N is small, the hypergeometric (n, M, N) distribution may be approximated by the binomial (n, p) distribution with $p = M/N$.

We demonstrate that this is indeed the case, by considering the limit of $P(X = k)$ as $N \rightarrow \infty$ with n fixed and the ratio M/N fixed at p . First we expand $P(X = k)$:

$$\begin{aligned} \frac{\binom{M}{k} \binom{N-M}{n-k}}{\binom{N}{n}} &= \frac{M!}{k!(M-k)!} \frac{(N-M)!}{(n-k)!(N-M-n+k)!} \frac{n!(N-n)!}{N!} \\ &= \left[\frac{n!}{k!(n-k)!} \right] \times \left[\frac{M!}{(M-k)!} \frac{(N-M)!}{(N-M-n+k)!} \right] \div \left[\frac{N!}{(N-n)!} \right] \end{aligned}$$

$$= \binom{n}{k} \frac{\overbrace{M(M-1)\dots(M-k+1)}^{k \text{ terms}} \times \overbrace{(N-M)(N-M-1)\dots(N-M-n+k+1)}^{(n-k) \text{ terms}}}{\underbrace{N(N-1)\dots(N-n+1)}_{n \text{ terms}}}.$$

Both numerator and denominator here contain n terms; if we divide each term by N , then the net effect is to divide both top and bottom by N^n . We get

$$P(X = k) = \binom{n}{k} \frac{\frac{M}{N} \left(\frac{M}{N} - \frac{1}{N}\right) \dots \left(\frac{M}{N} - \frac{k-1}{N}\right) \times \left(\frac{N-M}{N}\right) \left(\frac{N-M}{N} - \frac{1}{N}\right) \dots \left(\frac{N-M}{N} - \frac{n-k-1}{N}\right)}{1 \left(1 - \frac{1}{N}\right) \dots \left(1 - \frac{n-1}{N}\right)}.$$

Now, as $N \rightarrow \infty$ with $M/N = p$, n and k fixed, it is clear that this expression tends to

$$\binom{n}{k} \left(\frac{M}{N}\right)^k \left(\frac{N-M}{N}\right)^{n-k} = \binom{n}{k} p^k (1-p)^{n-k},$$

which is the required Binomial probability. ■

As a quick check we can examine the limiting behaviour of $E(X)$ and $Var(X)$, to ensure that these tend to the corresponding Binomial expressions. This is left as an exercise:

■

2.3.6 The Poisson distribution (parameter $\mu \in \mathbb{R}^+$)

You have encountered the Poisson distribution in your previous probability courses. The pmf of a POISSON DISTRIBUTION WITH PARAMETER μ is

$$P(X = k) = \frac{\mu^k e^{-\mu}}{k!} \quad (k = 0, 1, 2, \dots).$$

The mean and variance are both equal to μ .

The Poisson distribution is often used in the following situations:

- As an approximation to the binomial(n, p) distribution when n is large and p is small.
- As the distribution of the number of ‘accidents’ occurring in some fixed period of time or region of space. The term ‘accident’ here is a generic term, like ‘success’ and ‘failure’ in the context of Bernoulli trials.

The reason for using the Poisson distribution in the second situation above is that the Poisson distribution can be shown to be the distribution of the number of accidents in a POISSON PROCESS. Specifically, suppose that we are interested in the number of accidents occurring in a time interval $[0, t]$. Suppose in addition that:

1. The probability of one accident during some small time interval of length δ time units is $\lambda\delta + o(\delta)$ (where λ is a constant).
2. The probability of no accidents in the same time interval is $1 - \lambda\delta + o(\delta)$.
3. The number of accidents in any time interval is independent of the number in any other interval, providing the two intervals do not overlap.

The first two assumptions mean that the probability of more than accident in a time period of length δ is approximately zero, which is reasonable providing δ is small enough. In effect, by requiring δ to be ‘small’, we are prohibiting exactly simultaneous accidents.

To find the distribution of $X(t)$, we split the interval $[0, t]$ into n very small intervals each of length t/n , and set $\delta = t/n$:

We can set up a Bernoulli trial in each interval, where we count a ‘success’ if an accident occurs in that interval and a ‘failure’ otherwise. In this case, the probability of a success in any interval is approximately $\lambda\delta$ and so, because of the assumed independence of non- overlapping

time intervals, the distribution of $X(t)$ can be approximated by $\text{Bin}(n, \lambda\delta)$. The distribution is not exact, because of the possibility of more than 1 call arriving in an interval, but the approximation clearly becomes better as δ gets smaller. So, as $n \rightarrow \infty$, the distribution of $X(t)$ approaches $\text{Bin}(n, \lambda\delta) = \text{Bin}(n, \lambda t/n)$ (from the definition of δ). But this is a binomial with “large n and small p ”, which is Poisson in the limit. The mean of this Poisson distribution is the same as that of the approximating binomial: $\mu = n\lambda t/n = \lambda t$.

Definition: A process in which ‘accidents’ occur in such a way as to satisfy assumptions 1–3 above is called a POISSON PROCESS. The parameter λ is called the RATE of the process.

To see why we call λ the rate of the process, note that in a Poisson process with rate λ ,

$$E(X(t)) = \lambda t \quad \Rightarrow \quad E(X(1)) = \lambda$$

i.e. λ is the mean number of accidents in a time interval of length 1 time unit. λ is measured in units of $(\text{time})^{-1}$ (e.g. ‘events per minute’).

Following the same argument as in the derivation above, if we define $N(a, b)$ to be the number of accidents occurring between times a and b ($b > a$), then $N(a, b)$ is a random variable following a Poisson distribution with mean $\lambda(b - a)$. This shows that in a Poisson process, the number of accidents in an interval will depend on the length of the interval but not on its position.

Example 2.8: Patients arrive at a doctor’s surgery in a Poisson process of rate 0.5 per minute. What is the probability that no patients arrive in a given 10 minute interval?

■

3 Continuous Random Variables

In the previous section we studied the properties of discrete random variables, which could be regarded as mappings from the sample space of an experiment to the real numbers. The key feature of discrete random variables is that they take values in a countable set. This feature allows us to define concepts, such as the probability mass function, expectation etc., which make sense intuitively.

What happens if the sample space is uncountable? For example, suppose we throw a dart at a dartboard (randomly), and consider the outcome to be the point of contact. There is an uncountable infinity of such points, and it is not clear as to how we should allocate probabilities to them. However, it is clear that, if we divide the board into two halves, then approximately 50% of the time we will hit one half, and 50% the other. This suggests that we might be able to make progress with uncountable sample spaces by considering *regions* of the sample space, rather than individual points. In particular, if X is a random variable defined on an uncountable sample space, we can define regions which correspond to events of the form ' $X < c$ ', and study their probability structure.

3.1 Probability density functions and properties

Firstly, we generalise the definition of a discrete r.v.:

Definition: A RANDOM VARIABLE is a function from a sample space Ω to $\mathbb{R}$.

Suppose the sample space Ω has associated event space $\mathcal{F}$ and probability function P . Let X be a random variable defined upon this sample space. For any real numbers a and b , the event $a \leq X \leq b = \{\omega : a \leq X(\omega) \leq b\}$ is an element of the event space $\mathcal{F}$. In particular, we can define the cumulative distribution function $F_X(x) = P(X \leq x)$ as before. In the previous section, when X took values in a countable set, $F_X(\cdot)$ was seen to be a step function with a countable set of jumps. This may be the case for r.v.s defined on uncountable sample spaces as well (in the dartboard example, we could let X be the score obtained). However, when the sample space is uncountable we also have the possibility that F_X is continuous and differentiable.

Definition: If F_X is continuous and differentiable, with derivative f_X , then X is a CONTINUOUS RANDOM VARIABLE. f_X is the PROBABILITY DENSITY FUNCTION (pdf) of X .

As usual, we drop the ' X ' subscript where there is no ambiguity.

Since differentiation and integration are inverse operations and $F(-\infty) = 0$, we have

$$F(x) = \int_{-\infty}^x f(u)du \quad \text{and} \quad P(a < X \leq b) = F(b) - F(a) = \int_a^b f(x)dx \quad (a \leq b) .$$

This gives us an interpretation for the pdf of a continuous random variable: the probability that X lies in an interval $(a, b]$ is given by the area under the graph of the pdf between a and b :

In particular, note that $F(\infty) = P(X \leq \infty) = \int_{\mathbb{R}} f(x)dx = 1$. Also note that a continuous r.v. can be described completely by knowing *either* F or f , since each can be obtained from the other.

Can we find another interpretation for the probability density function? Well, from the definition of differentiation we have

$$f(x) = \lim_{\delta x \rightarrow 0} \frac{F(x + \delta x) - F(x)}{\delta x} = \lim_{\delta x \rightarrow 0} \frac{P(x < X < x + \delta x)}{\delta x} .$$

This shows that $f(x)$ must be non-negative everywhere. Also, providing δx is small, we see that

$$P(x < X < x + \delta x) \approx f(x)\delta x .$$

This says that we could choose to approximate a continuous random variable by a discrete one which takes values on a very finely spaced grid of points (separated by intervals of length δx). In this case, the probability associated with an interval starting at point x would be $f(x)\delta x$. Note that $f(x)$ itself is *not* a probability; in particular, it does not have to be less than 1. In fact, the only two conditions required for a function f to be a pdf are:

1. $f(x) \geq 0 \quad \forall x \in \mathbb{R} .$

2. $\int_{-\infty}^{\infty} f(x)dx = 1 .$

Can we say anything about $P(X = x)$ when X is continuous? We have

$$P(X = x) = \lim_{\epsilon \rightarrow 0} P(x - \epsilon < X \leq x) = \lim_{\epsilon \rightarrow 0} F(x) - F(x - \epsilon) = 0 ,$$

since F is continuous. This might seem counter-intuitive — in particular, you might think that it can't be right because the sample space is made up of a union of individual outcomes, and the probabilities have got to add up to 1. This reasoning, however, makes use of Axiom 3. If you look back, you'll see that the axiom only applies to *countable* unions, so it is inapplicable here.

It is possible to imagine situations where F is neither a step function nor everywhere differentiable. For example, it may be a mixture of the two (differentiable except at a countable set of x values). This is straightforward to handle, and has some practical application (the distribution of daily rainfall amounts, for example, has a step at the origin corresponding to 'dry' days, and is continuous elsewhere). Alternatively, it may be discontinuous on an uncountable set, or everywhere continuous but non-differentiable. This is unlikely to arise in many (if any) applications however!

We have seen that a continuous r.v. can be approximated by a discrete r.v. taking values on a fine grid of resolution δx , with probabilities given by $f(x)\delta x$. The expectation of this approximating discrete variable is therefore $\sum x \times f(x)\delta x$, where the sum is over all the x -values in the grid. As the grid becomes finer and finer (i.e. as $\delta x \rightarrow 0$), this sum tends to an integral and we are led to the following:

Definition: Let X be a continuous random variable with density $f(\cdot)$. The EXPECTED VALUE of X is defined as

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx ,$$

if the integral converges absolutely i.e. $\int_{\mathbb{R}} |x| f(x) dx < \infty$.

This is the same as the definition for discrete random variables, but the sum has been replaced by an integral and ' $p(x)$ ' has been replaced by ' $f(x)dx$ '. This is generally true for all formulae relating to continuous random variables. For example, the random variable $\phi(X)$ has expected value $\int_{\mathbb{R}} \phi(x) f(x) dx$. The variance is defined in the same way as in the discrete case:

$$\text{Var}(X) = E(X - \mu)^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx ,$$

where $\mu = E(X)$. Expectation and variance have the same properties for continuous random variables as in the discrete case.

3.1.1 The Moment Generating Function

For discrete random variables taking the form of counts, we defined the probability generating function $\Pi(s) = E(s^X)$. A similar idea, applicable both to continuous and discrete r.v.s but which we will mainly find useful in the continuous case, is

Definition: The MOMENT GENERATING FUNCTION (mgf) of a random variable X is defined as $M_X(t) = E(e^{tX})$, for those values of t where the expectation exists.

The motivation for this is the following: for any real number m , we have

$$e^m = 1 + m + \frac{m^2}{2!} + \frac{m^3}{3!} + \dots$$

so that $M(t) = E(e^{tX}) = E\left(1 + tX + \frac{(tX)^2}{2!} + \frac{(tX)^3}{3!} + \dots\right)$.

If we put $t = 0$ we get $M(0) = E(1 + 0 + 0 + \dots) = 1$.

Suppose we differentiate $M(t)$ with respect to t :

$$\frac{dM}{dt} = E\left(X + tX^2 + \frac{t^2X^3}{2!} + \frac{t^3X^4}{3!} + \dots\right)$$

If we put $t = 0$ we get $\frac{dM}{dt}|_{t=0} = E(X + 0 + 0 + \dots) = E(X)$.

Differentiate again:

$$\frac{d^2M}{dt^2} = E\left(X^2 + tX^3 + \frac{t^2X^4}{2!} + \frac{t^3X^5}{3!} + \dots\right)$$

and putting $t = 0$ we get $\frac{d^2M}{dt^2}|_{t=0} = E(X^2 + 0 + 0 + \dots) = E(X^2)$.

In general, $\frac{d^kM}{dt^k}|_{t=0} = E(X^k)$, the k^{th} moment of X . This explains why $M(\cdot)$ is called the moment generating function. MGFs often provide a very quick way of obtaining means and variances. For discrete r.v.s where a pgf exists, we have the relation $M(t) = \Pi(e^t)$.

3.1.2 Other measures of location

For discrete random variables, we defined the median of a distribution to be the solution of the equation $F(x) = 0.5$. This definition also holds for continuous random variables. In addition, we define a further measure of location:

Definition: If X is a continuous random variable, the MODE of X is the value of x at which $f(x)$ is a maximum. If X is discrete, the mode is the value with highest probability.

3.2 Some particular continuous distributions

3.2.1 The Uniform (or rectangular) distribution (parameters $a, b \in \mathbb{R} : a < b$)

A continuous random variable X has a UNIFORM DISTRIBUTION if it has pdf

$$f(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise.} \end{cases}$$

We write $X \sim U(a, b)$.

For $f(x)$ to be a valid pdf we must check (i) $f(x) \geq 0$ everywhere (trivial) (ii) $\int_{\mathbb{R}} f(x)dx = 1$ (also trivial, since the integral is the area of a rectangle).

For some interval $(c, c + d) \in (a, b)$, we have

$$P(c < X \leq c + d) = \int_c^{c+d} \frac{1}{b-a} dx = \frac{1}{b-a} [x]_c^{c+d} = \frac{d}{b-a},$$

so that $P(X \in (c, c + d))$ depends only on the length of the interval, not its location.

For the cdf, we have

$$F(x) = \int_{-\infty}^x f(u)du = \begin{cases} 0 & x < a \\ \int_a^x \frac{1}{b-a} du = \frac{x-a}{b-a} & a \leq x \leq b \\ 1 & x \geq b. \end{cases}$$

Result: If $X \sim U(a, b)$ then $E(X) = (a + b)/2$ and $\text{Var}(X) = (b - a)^2/12$.

Proof: $E(X) = (a + b)/2$ by symmetry. Exercise: derive this from the definition:

For the variance, we use $Var(X) = E(X^2) - (E(X))^2$. Now

$$\begin{aligned}
 E(X^2) &= \int_{\mathbb{R}} x^2 f(x) dx = \int_a^b \frac{x^2}{b-a} dx = \frac{1}{b-a} \left[\frac{x^3}{3} \right]_a^b = \frac{b^3 - a^3}{3(b-a)} \\
 &= \frac{(b-a)(b^2 + ab + a^2)}{3(b-a)} = \frac{1}{3} (b^2 + ab + a^2), \text{ whence} \\
 Var(X) &= \frac{1}{3} (b^2 + ab + a^2) - \frac{(a+b)^2}{4} = \frac{4(b^2 + ab + a^2) - 3(a^2 + 2ab + b^2)}{12} \\
 &= \frac{b^2 - 2ab + a^2}{12} = \frac{(b-a)^2}{12},
 \end{aligned}$$

as required. ■

Uses of the uniform distribution: The uniform distribution is seldom of interest in its own right. Its most important use is in the simulation of random numbers (on a computer or hand calculator, for example), for it turns out that if we can generate a random number from the distribution $U(0, 1)$ then we can quite easily transform this into a random number from almost any other distribution.

3.2.2 The Exponential distribution (parameter $\lambda > 0$)

Suppose accidents occur in a Poisson process of rate λ . Starting at time 0, let X be the time until the first accident occurs. What is the distribution of X ?

We know that if $N(t)$ is the number of accidents occurring in the time interval $(0, t)$, $N(t) \sim \text{Poi}(\lambda t)$ for any $t > 0$. Thus we can make the following statement about X , for any $x > 0$:

$$\begin{aligned} P(X \leq x) &= P(\text{first accident occurs before time } x) = P(\text{at least 1 accident in } (0, x]) \\ &= P(N(x) \geq 1) = 1 - P(N(x) = 0) = 1 - \frac{(\lambda x)^0 e^{-\lambda x}}{0!} = 1 - e^{-\lambda x}. \end{aligned}$$

But by definition, $P(X \leq x) = F(x)$, the cdf of X evaluated at x . So the pdf is

$$f(x) = \frac{dF(x)}{dx} = \begin{cases} \lambda e^{-\lambda x} & x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

We say that X has a (NEGATIVE) EXPONENTIAL DISTRIBUTION WITH PARAMETER λ , and write $X \sim \text{Exp}(\lambda)$. $f(\cdot)$ defines a valid pdf, for $f(x) \geq 0$ everywhere, and

$$\int_{\mathbb{R}} f(x) dx = \int_0^{\infty} \lambda e^{-\lambda x} dx = [-e^{-\lambda x}]_0^{\infty} = 0 - (-1) = 1.$$

We have derived the exponential distribution as the time between consecutive accidents in a Poisson process of rate λ . For this reason, the distribution is often used as a model for the times to failure of electrical components, lifetimes of individuals etc.

Result: If $X \sim \text{Exp}(\lambda)$ then $E(X) = 1/\lambda$ and $\text{Var}(X) = 1/\lambda^2$.

Proof: For the mean we have

$$E(X) = \int_{\mathbb{R}} x f(x) dx = \lambda \int_0^{\infty} x e^{-\lambda x} dx.$$

Use integration by parts (put $u = x$, $dv/dx = e^{-\lambda x}$):

$$\lambda \int_0^{\infty} x e^{-\lambda x} dx = \lambda \left(\left[\frac{x e^{-\lambda x}}{-\lambda} \right]_0^{\infty} + \frac{1}{\lambda} \int_0^{\infty} e^{-\lambda x} dx \right) = 0 + \left[\frac{e^{-\lambda x}}{-\lambda} \right]_0^{\infty} = 0 + \left(0 - \left(-\frac{1}{\lambda} \right) \right) = \frac{1}{\lambda}.$$

For the variance, use $Var(X) = E(X^2) - (E(X))^2$, and evaluate $E(X^2)$ using integration by parts in a very similar manner to that shown above. This derivation is left as an exercise:

More generally, however, we can derive all moments of the exponential distribution via its moment generating function, very easily. Recall that the mgf of X is defined as $M(t) = E(e^{tX})$, and is a function of t . When $X \sim Exp(\lambda)$ we have

$$M(t) = E(e^{tX}) = \int_{\mathbb{R}} e^{tx} f(x) dx = \lambda \int_0^{\infty} e^{tx} e^{-\lambda x} dx = \lambda \int_0^{\infty} e^{(t-\lambda)x} dx = \lambda \left[\frac{e^{(t-\lambda)x}}{t-\lambda} \right]_0^{\infty}.$$

At this point we have to be a little bit careful, as this expression will only be finite if $t < \lambda$. However, λ is fixed, and is > 0 , so we can always find $t > 0$ for which the expression is finite, and for which therefore

$$M(t) = \lambda \left(0 - \frac{1}{t-\lambda} \right) = \frac{\lambda}{\lambda-t} = \lambda(\lambda-t)^{-1}.$$

Now we obtain the moments of X by differentiating with respect to t . For example,

$$\begin{aligned} \frac{dM}{dt} &= \lambda(\lambda-t)^{-2} \Rightarrow \text{set } t=0 \text{ to get } E(X) = \frac{1}{\lambda}. \\ \frac{d^2 M}{dt^2} &= 2\lambda(\lambda-t)^{-3} \Rightarrow \text{set } t=0 \text{ to get } E(X^2) = \frac{2}{\lambda^2}. \\ \frac{d^3 M}{dt^3} &= 3 \times 2\lambda(\lambda-t)^{-4} \Rightarrow \text{set } t=0 \text{ to get } E(X^3) = \frac{3!}{\lambda^3}. \end{aligned}$$

In general,

$$\frac{d^r M}{dt^r} = r! \lambda(\lambda-t)^{-(r+1)} \Rightarrow \text{set } t=0 \text{ to get } E(X^r) = \frac{r!}{\lambda^r}.$$

So the mgf gives all the moments of X very quickly. (Note that the same result was obtained for $E(X)$ as previously). Using these results, then, yields

$$Var(X) = E(X^2) - (E(X))^2 = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda} \right)^2 = \frac{1}{\lambda^2}.$$



Example 3.1: Find the median of an exponential distribution with parameter λ . Is it possible for an exponential distribution to have equal mean and median?



Example 3.2: Particles are emitted from a radioactive material in a Poisson Process of rate λ per unit time. What is the mean time between emissions? If the last particle was emitted L time units ago, what is the distribution of the time till the next emission (starting now)?

In the same way that the Poisson distribution can be regarded as the limit of a Binomial as $n \rightarrow \infty$ with np fixed, so the exponential distribution can be regarded as the limit of a geometric, using the same arguments as were used in the Poisson process construction (recall that the geometric distribution arises as the number of independent Bernoulli trials up to and including the first success). ■

3.2.3 The Gamma distribution (parameters $\alpha > 0, \lambda > 0$)

The exponential distribution was derived from the time until the first accident in a Poisson process, and as such could be regarded in some sense as the limit of a geometric distribution. We could equally be interested in knowing about the time until the k th accident (just as we extended the geometric distribution to the negative binomial by considering the number of trials required before k successes were obtained).

The derivation proceeds in the same way as for the exponential distribution: let X be the time until the k th accident in a Poisson process. Then X takes values in $\mathbb{R}^+$, and its distribution function is

$$\begin{aligned} F(x) = 1 - P(X > x) &= 1 - P(\text{at most } k-1 \text{ accidents in } (0, x]) \\ &= 1 - \sum_{n=0}^{k-1} \frac{e^{-\lambda x} (\lambda x)^n}{n!} \quad (x > 0). \end{aligned}$$

To obtain the pdf we differentiate:

$$\begin{aligned}
 f(x) = \frac{dF}{dx} &= -\sum_{n=0}^{k-1} \frac{1}{n!} \frac{d}{dx} [e^{-\lambda x} (\lambda x)^n] = -\sum_{n=0}^{k-1} \frac{1}{n!} \left[(\lambda x)^n \frac{d}{dx} (e^{-\lambda x}) + e^{-\lambda x} \frac{d}{dx} (\lambda x)^n \right] \\
 &= \sum_{n=0}^{k-1} \frac{1}{n!} [\lambda (\lambda x)^n e^{-\lambda x}] - \sum_{n=1}^{k-1} \frac{1}{n!} [\lambda n e^{-\lambda x} (\lambda x)^{n-1}] \\
 &= \text{(putting } \ell = n - 1 \text{ in the second sum)} \\
 &\quad \sum_{n=0}^{k-1} \frac{1}{n!} [\lambda (\lambda x)^n e^{-\lambda x}] - \sum_{\ell=0}^{k-2} \frac{1}{\ell!} [\lambda e^{-\lambda x} (\lambda x)^\ell] = \frac{1}{(k-1)!} \lambda^k x^{k-1} e^{-\lambda x} .
 \end{aligned}$$

The distribution of X in this case is called a GAMMA DISTRIBUTION WITH INDEX k AND SCALE PARAMETER λ . The name comes from a generalisation of the distribution to the case when k is not an integer. Obviously, if we generalise in this way then we lose the interpretation in terms of time before an accident; however, the increased flexibility that results is very useful in many practical applications.

Definition: The GAMMA FUNCTION is defined as

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$$

for any $\alpha > 0$.

What does this have to do with anything? Let's explore some of its properties: firstly, for $\alpha = 1$ we have

$$\Gamma(1) = \int_0^\infty e^{-x} dx = \left[\frac{e^{-x}}{-1} \right]_0^\infty = 0 - (-1) = 1 .$$

For most values of α the integral in the Gamma function cannot be evaluated analytically. However, for any $\alpha > 1$ we can use integration by parts to obtain a recurrence relation, as follows: set $u = x^{\alpha-1}$ and $dv/dx = e^{-x}$. Then $du/dx = (\alpha-1)x^{\alpha-2}$ and $v = \int e^{-x} dx = -e^{-x}$. So, for $\alpha > 1$, we have

$$\begin{aligned}
 \Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx &= [-x^{\alpha-1} e^{-x}]_0^\infty + \int_0^\infty (\alpha-1)x^{\alpha-2} e^{-x} dx \\
 &= 0 + (\alpha-1) \int_0^\infty x^{\alpha-2} e^{-x} dx = (\alpha-1)\Gamma(\alpha-1) .
 \end{aligned}$$

So for $\alpha > 1$, $\Gamma(\alpha) = (\alpha-1)\Gamma(\alpha-1)$. Equivalently, for $\alpha > 0$, $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$. This property holds regardless of whether or not α is an integer. Integer values are of particular interest,

however. We have already seen that $\Gamma(1) = 1$. Now we can use the recurrence relation to obtain

$$\begin{aligned}\Gamma(2) &= 1 \times \Gamma(1) = 1 \times 1 = 1; \\ \Gamma(3) &= 2 \times \Gamma(2) = 2 \times 1 = 2; \\ \Gamma(4) &= 3 \times \Gamma(3) = 3 \times 2 \times 1 = 6; \\ \Gamma(5) &= 4 \times \Gamma(4) = 4 \times 3 \times 2 \times 1 = 24;\end{aligned}$$

and, in general, $\Gamma(\alpha) = (\alpha - 1)!$ for $\alpha \in \mathbb{N}$. For this reason, and because $\Gamma(\alpha)$ is also defined for non-integer values of α , the gamma function is sometimes called the *generalised factorial*. Note that it provides a motivation for defining $0! = 1$, for $0! = \Gamma(1) = 1$. It is also sometimes useful to know that $\Gamma(1/2) = \sqrt{\pi}$.

The Gamma function is frequently encountered in probability and statistics. In particular, it allows us to generalise distributions containing factorials, such as the one we are currently studying. The density of the time to the k th accident in a Poisson process has $(k - 1)!$ in the denominator. Noting that this is equal to $\Gamma(k)$ for integer k , this gives us a clue as to how we might generalise the distribution:

Definition: A continuous random variable X has a GAMMA DISTRIBUTION WITH PARAMETERS $\alpha > 0$ AND $\lambda > 0$ if it has pdf

$$f(x) = \begin{cases} \lambda^\alpha x^{\alpha-1} e^{-\lambda x} / \Gamma(\alpha) & x \geq 0 \\ 0 & \text{otherwise} \end{cases}.$$

We write $X \sim \Gamma(\alpha, \lambda)$.

NB do not confuse the notation for the gamma *distribution* with that for the function.

Note that the pdf of the $\Gamma(1, \lambda)$ distribution is, for $x > 0$,

$$f(x) = \lambda e^{-\lambda x}$$

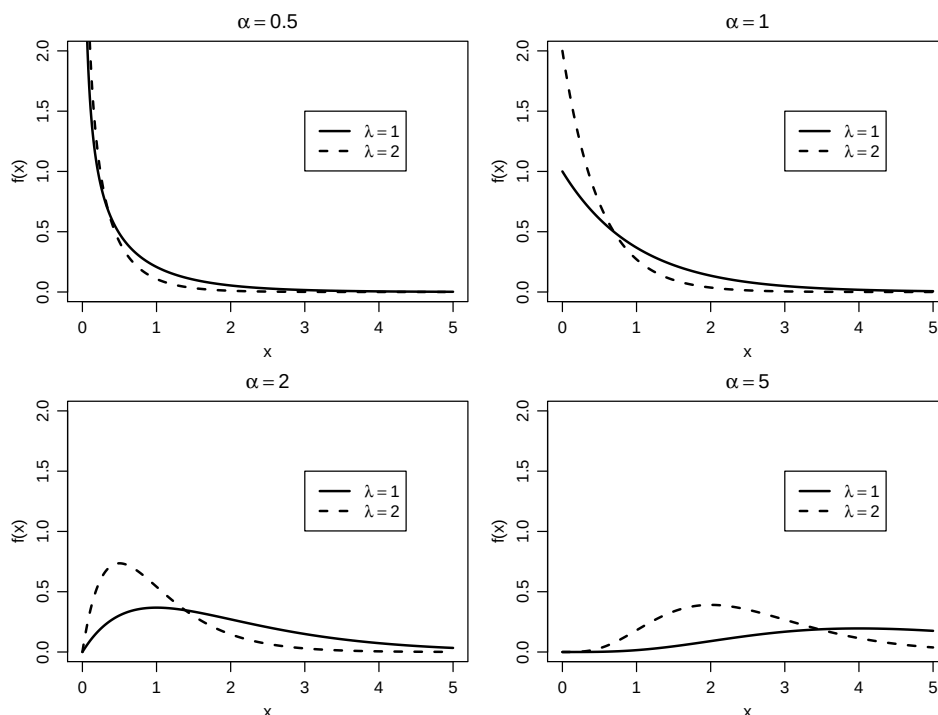
which is the same as that for $Exp(\lambda)$. So the $\Gamma(1, \lambda)$ distribution is the same as the $Exp(\lambda)$ distribution. This is to be expected, since $\Gamma(1, \lambda)$ is the distribution of the time until the first accident in a Poisson process of rate λ .

We can check that $f(x)$ is a valid pdf; it is non-negative everywhere, and

$$\begin{aligned}\int_{\mathbb{R}} f(x) dx &= \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{\alpha-1} e^{-\lambda x} dx \\ &= (\text{making the substitution } u = \lambda x \text{ so that } dx = du/\lambda) \\ &= \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_{u=0}^\infty \left(\frac{u}{\lambda}\right)^{\alpha-1} e^{-u} \frac{du}{\lambda} = \frac{1}{\Gamma(\alpha)} \int_0^\infty u^{\alpha-1} e^{-u} du = \frac{1}{\Gamma(\alpha)} \Gamma(\alpha) = 1,\end{aligned}$$

the last step following from the definition of the Gamma function.

Examples of gamma densities:



α may be called the *index* or *shape parameter*, and λ the *scale parameter* (or sometimes, confusingly, just the *parameter*) of the distribution. The gamma family is often used to study random variables which take values in $\mathbb{R}^+$, because of the wide variety of pdf shapes which can be generated — it is not just restricted to ‘time until accident’ problems.

Result: If $X \sim \Gamma(\alpha, \lambda)$ then $E(X) = \alpha/\lambda$ and $\text{Var}(X) = \alpha/\lambda^2$.

Proof: The ‘obvious’ approach is to do the integrals. This is tedious. It involves lots of integration by parts, substitution and recognition of Gamma functions when they appear.

We will prove the result using the mgf of the gamma distribution. We have

$$\begin{aligned} M(t) = E(e^{tX}) &= \int_0^\infty e^{tx} \frac{\lambda^\alpha x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)} dx = \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{\alpha-1} e^{-(\lambda-t)x} dx \\ &= \frac{\lambda^\alpha}{(\lambda-t)^\alpha} \left(\frac{(\lambda-t)^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{\alpha-1} e^{-(\lambda-t)x} dx \right). \end{aligned}$$

If $t < \lambda$, the expression in brackets is the integral of the pdf of a gamma distribution with parameters α and $(\lambda - t)$, and hence evaluates to 1. Since $\lambda > 0$, we can always find such a t . So in this case we have

$$M(t) = \frac{\lambda^\alpha}{(\lambda-t)^\alpha} = \lambda^\alpha (\lambda-t)^{-\alpha}.$$

Differentiating, we get

$$\frac{dM}{dt} = \lambda^\alpha \alpha (\lambda-t)^{-(\alpha+1)} \Rightarrow \quad \text{set } t = 0 \text{ to get } E(X) = \frac{\alpha}{\lambda}.$$

$$\frac{d^2 M}{dt^2} = \lambda^\alpha \alpha(\alpha + 1)(\lambda - t)^{-(\alpha+2)} \Rightarrow \text{set } t = 0 \text{ to get } E(X^2) = \frac{\alpha(\alpha + 1)}{\lambda^2}.$$

Hence

$$\text{Var}(X) = E(X^2) - (E(X))^2 = \frac{\alpha(\alpha + 1)}{\lambda^2} - \frac{\alpha^2}{\lambda^2} = \frac{\alpha}{\lambda^2}.$$

■

Example 3.3: The response times (in tenths of a second) of an online terminal are found to have a gamma distribution with expectation 4 and variance 8. Write down the pdf for these response times, and sketch it.

■

Summary of ‘time till occurrence’ distributions: We have now seen several distributions which deal with situations where we are concerned with the time (discrete or continuous) until an ‘accident’, or ‘success’ is observed. There is a duality between the distributions which occur in discrete, and those which occur in continuous time, as shown in the following table:

Random variable	Distribution	
	Discrete time	Continuous time
No. of accidents/successes	Binomial	Poisson
Time till first accident/success	Geometric	Exponential
Time till r th accident/success	Negative binomial	Gamma

3.2.4 The Beta distribution (parameters $\alpha > 0, \beta > 0$)

Sometimes we wish to study a random variable that represents a proportion, measured on some continuous scale. For example, in pollution studies we might be interested in the proportion of contaminants in river water. Other examples include proportions of land affected by flooding, and the proportion of time that a machine is under repair.

We could study such situations by considering a sequence of Binomial (n, p) variables $\{X_n : n = 1, 2, \dots\}$ and studying the distribution of X_n/n as $n \rightarrow \infty$. However, this is not particularly straightforward. Instead, what we shall do is to simply define an extremely flexible class of distributions. It can be shown that the ‘limiting Binomial’ proportions arise as a special case of this class.

As with the gamma distribution, we need to introduce another special function here:

Definition: The BETA FUNCTION is defined as

$$B(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx$$

for any $\alpha, \beta > 0$.

It can be shown that

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}.$$

The derivation of this result will not be given here (it’s done by evaluating $\Gamma(\alpha)\Gamma(\beta)$ as a double integral, making various substitutions and using polar co-ordinates in a way which is very non-obvious!). However, we can use the result to see that when α and β are both integers > 1 ,

$$B(\alpha, \beta) = \frac{(\alpha-1)!(\beta-1)!}{(\alpha+\beta-1)!}$$

We are now in a position to give the following:

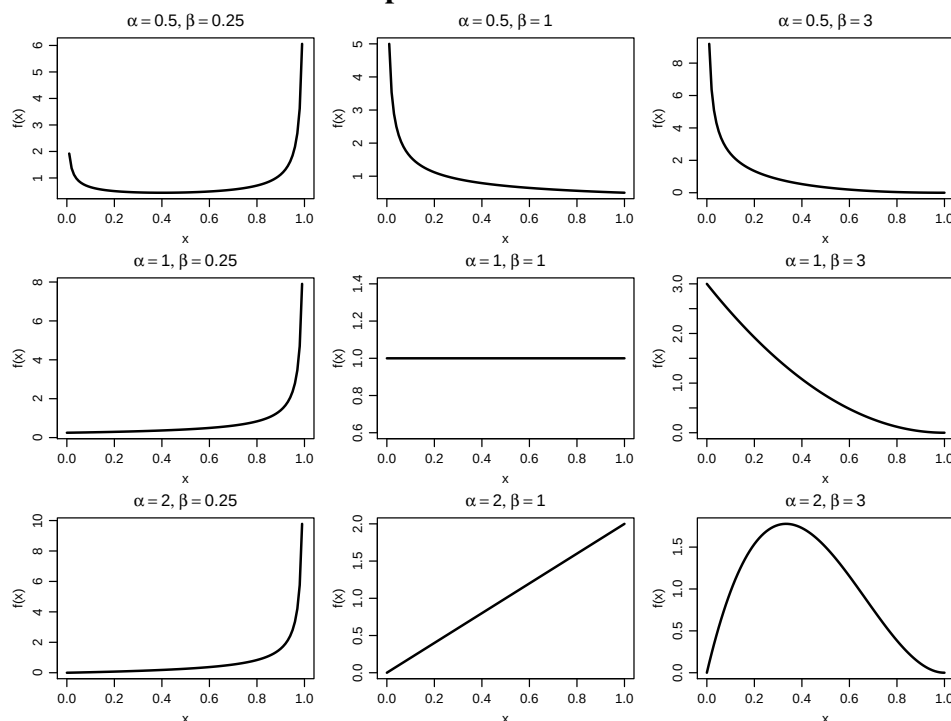
Definition: A continuous random variable X has a BETA DISTRIBUTION WITH PARAMETERS $\alpha > 0$ AND $\beta > 0$ if it has pdf

$$f(x) = \begin{cases} x^{\alpha-1}(1-x)^{\beta-1}/B(\alpha, \beta) & x \in (0, 1) \\ 0 & \text{otherwise} \end{cases}.$$

We write $X \sim B(\alpha, \beta)$.

This function is everywhere non-negative, and integrates to 1 by the definition of the beta function. Hence it defines a valid pdf. ■

Examples of beta densities:



These examples show that the beta family is extremely flexible. Although it has not been derived using any arguments about an ‘underlying process’ therefore, it might be expected to provide a good model in practice for the distribution of any random variable which arises as a proportion.

Result: If $X \sim B(\alpha, \beta)$ then

$$E(X) = \frac{\alpha}{\alpha + \beta} \quad \text{and} \quad \text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

Proof: For the mean, we have

$$E(X) = \frac{1}{B(\alpha, \beta)} \int_0^1 x^\alpha (1-x)^{\beta-1} dx = \frac{B(\alpha+1, \beta)}{B(\alpha, \beta)} = \frac{\Gamma(\alpha+1)\Gamma(\beta)\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)\Gamma(\alpha+\beta+1)} = \frac{\alpha}{\alpha + \beta},$$

since we know that $\Gamma(n+1) = n\Gamma(n)$ for $n > 0$.

For the variance, we find $E(X^2)$ in a similar manner and use $\text{Var}(X) = E(X^2) - [E(X)]^2$. The derivation is left as an exercise.



Example 3.4: The proportion of sky covered by a cloud in a certain location has a beta distribution with mean $2/3$ and coefficient of variation $1/2$. Find the parameters of the distribution, and sketch its pdf.

Note that the $B(1, 1)$ distribution is the same as $U(0, 1)$. ■

3.2.5 The Normal (or Gaussian) distribution (parameters $\mu \in \mathbb{R}, \sigma^2 > 0$)

The gamma family of distributions can be used for studying positive-valued random variables, and the beta family can be used for r.v.s taking values in a finite range. You should already be familiar with the normal family of distributions, for random variables taking all real values.

Definition: A continuous random variable X has a NORMAL DISTRIBUTION WITH PARAMETERS μ AND σ^2 if its density is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(x - \mu)^2}{2\sigma^2} \right], x \in \mathbb{R}.$$

We write $X \sim N(\mu, \sigma^2)$.

It is straightforward to verify that $f(x) \geq 0$ everywhere. Checking that $\int_{\mathbb{R}} f(x)dx = 1$ is difficult, and this will not be proved here (the derivation involves making the substitution $z = (x - \mu)/\sigma$, and evaluating the *square* of the integral as a double integral via polar co-ordinates).

The density is symmetric about $x = \mu$, and is unimodal; it is often described as ‘bell-shaped’.

Result: If $X \sim N(\mu, \sigma^2)$ then $E(X) = \mu$ and $\text{Var}(X) = \sigma^2$.

Proof: We only do the variance here. The proof for the mean is very similar, and is left as an exercise (note that, because of the symmetry, we must have $E(X) = \mu$ if it exists).

To find $\text{Var}(X)$, it is easiest to work directly:

$$\begin{aligned} \text{Var}(X) = E[(X - \mu)^2] &= \int_{-\infty}^{\infty} (x - \mu)^2 \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left\{-\frac{(x - \mu)^2}{2\sigma^2}\right\} dx \\ &= \text{(putting } t = (x - \mu)/\sigma \text{ so that } dx = \sigma dt) \\ &\quad \int_{-\infty}^{\infty} t^2 \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt. \end{aligned}$$

We now use integration by parts: put $u = t$, $dv/dt = t \exp(-\frac{t^2}{2})$ so that $du/dt = 1$ and $v = -\exp(-\frac{t^2}{2})$. We obtain

$$\int_{-\infty}^{\infty} t^2 \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt = \frac{\sigma^2}{\sqrt{2\pi}} \left[-t \exp\left(-\frac{t^2}{2}\right) \right]_{-\infty}^{\infty} + \sigma^2 \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt.$$

The first term here is zero; and the integrand in the second is the pdf of a $N(0, 1)$ density; hence the integral is 1 and we obtain $\text{Var}(X) = \sigma^2$. ■

The $N(0, 1)$ distribution is called the STANDARD NORMAL distribution. We write $\phi(x)$ for its density and $\Phi(x)$ for its distribution function, so that

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \quad \Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}u^2} du.$$

Frequently we need to calculate probabilities like $P(a < X \leq b) = F_X(b) - F_X(a)$, when X is a normally distributed random variable. This is done by STANDARDISING the distribution to take advantage of the following result, which will be proved as an example later in the course:

Result: If $X \sim N(\mu, \sigma^2)$ and $Z = (X - \mu)/\sigma$, then $Z \sim N(0, 1)$.

This result allows us to convert probability statements about any normal random variable into statements about a standard normal random variable, whence we can use tables to find any required probabilities. For example, if $X \sim N(\mu, \sigma^2)$ then

$$P(a < X < b) = P\left(\frac{a - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{b - \mu}{\sigma}\right) = P\left(\frac{a - \mu}{\sigma} < Z < \frac{b - \mu}{\sigma}\right),$$

where $Z \sim N(0, 1)$. The cdf of Z is $\Phi(\cdot)$ (by definition) and so

$$P\left(\frac{a - \mu}{\sigma} < Z < \frac{b - \mu}{\sigma}\right) = \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right).$$

These probabilities can be read from tables. When using most tables however, it is useful to remember that $\Phi(0) = 0.5$ and that, by symmetry, $\Phi(-x) = 1 - \Phi(x)$ for all x .

Example 3.5: A machine automatically fills bottles with juice. The amount of juice dispensed into a bottle has a normal distribution with standard deviation 2cm^3 . The expected value μ can be altered by a dial on the machine. Find μ such that a 1 litre bottle will overflow only 5% of the time.

For any normal distribution, approximately 68% of observations will fall within 1 standard deviation of the mean, and approximately 95% within 2 standard deviations. This provides an interpretation for σ . ■

Uses of the normal distribution The normal distribution is perhaps the most important model for continuous random variables, and much statistical theory is based upon it. The reasons for its importance are:

1. Empirically, many ‘real-world’ phenomena such as heights, weights etc. often have a distribution which is approximately normal. Even if not, a transformation can often be found (e.g. by taking logs or cube roots) that produces an approximately normal distribution.
2. The distribution often provides a good model for random measurement errors in experiments.
3. There are useful normal approximations to other distributions, such as the binomial, Poisson and gamma distributions.
4. If X is a normal random variable, and $Y = aX + b$ for some constants a and b , then Y is also a normal random variable (we have already shown this, for the special case $a = 1/\sigma, b = -\mu/\sigma$). Moreover, sums of normal random variables are themselves normally distributed. We will prove this later in the course.
5. The distribution of the sum of a large number of independent random variables from *any* distribution, none of which predominate, tends in the limit to a normal distribution. This result is known as the *Central Limit Theorem*, and underlies many statistical techniques in

common use. For example, the sample mean of a data set is obtained by summing all the values and then dividing by the number of points; so the mean of a large sample has an approximately normal distribution.

Results governing sums of random variables are so important that we will shortly devote an entire chapter to their study.

3.3 Functions of random variables

We sometimes wish to study random variables of the form $Y = g(X)$, where $g : \mathbb{R} \rightarrow \mathbb{R}$ and X has a known density $f_X(\cdot)$. The question is: can we find the density of Y ? The general procedure for doing this is to consider events of the form $Y \leq y$, express these in terms of X to obtain the distribution function of Y , and differentiate to obtain the density. We will illustrate the procedure for the case when g is a strictly increasing function (i.e. if $x_1 < x_2$ then $g(x_1) < g(x_2)$). In this case, the distribution function of Y is

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

(the requirement that g is strictly increasing ensures the existence of the inverse function $g^{-1}(\cdot)$). Differentiating, we get

$$f_Y(y) = \frac{dF_Y}{dy} = \frac{d}{dy} F_X(g^{-1}(y)) = f_X(g^{-1}(y)) \frac{d}{dy} [g^{-1}(y)] = f_X(g^{-1}(y)) \frac{dx}{dy}.$$

If g is strictly *decreasing*, the event $Y \leq y$ corresponds to the event $X \geq g^{-1}(y)$, and the argument follows through in the same way except for the incorporation of a minus sign. The general result is

Result: If $Y = g(X)$, where $g(\cdot)$ is a strictly monotone function (increasing or decreasing) and X has density $f_X(\cdot)$, the density of Y is

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} (g^{-1}(y)) \right| = f_X(x) \left| \frac{dx}{dy} \right|.$$

The final form here can be remembered as “ $f_Y(y)|dy| = f_X(x)|dx|$ ”.

Example 3.6: $X \sim \text{Exp}(\lambda)$ and $Y = \sqrt{X}$. Find the density of Y .



Example 3.7: $X \sim N(\mu, \sigma^2)$. Show that $Z = (X - \mu)/\sigma$ has a standard normal distribution.



Example 3.8: $Z \sim N(0, 1)$ and $Y = Z^2$. What is the distribution of Y ?



4 Joint probability distributions

We have seen that a random variable is a mapping from a sample space to the real line, for example $X : \Omega \Rightarrow \mathbb{R}$. Obviously, for any given experiment we could define many different random variables on the same sample space: these are just different mappings.

Definition: Suppose $X_1, \dots, X_n$ are random variables defined on the same sample space. The JOINT DISTRIBUTION FUNCTION of $X_1, \dots, X_n$ is the function

$$F(x_1, \dots, x_n) = P(X_1 \leq x_1, \dots, X_n \leq x_n),$$

where the notation “ $X_1 \leq x_1, \dots, X_n \leq x_n$ ” is used as a shorthand for the event

$$\{\omega : X_1(\omega) \leq x_1\} \cap \dots \cap \{\omega : X_n(\omega) \leq x_n\}.$$

Note that if $n = 1$, this definition reduces to the (hopefully) familiar definition for a single random variable.

The distribution function of any individual random variable can be recovered from the joint distribution function. To see this, consider the distribution function of X_1 :

$$F_1(x) = P(X_1 \leq x), \text{ say.}$$

Now

$$P(X_1 \leq x) = P(X_1 \leq x, X_2 \leq \infty, \dots, X_n \leq \infty) = F(x_1, \infty, \dots, \infty).$$

A similar approach can be used to find the distribution function of any of the other random variables taken individually.

In this context the distribution function of a single random variable is called its MARGINAL DISTRIBUTION FUNCTION. The reason for this will soon become clear.

Joint distribution functions have similar properties to their univariate counterparts:

- If any of its arguments is $-\infty$, $F(x_1, \dots, x_n) = 0$.
- $F(\infty, \infty, \dots, \infty) = 1$
- $F(x_1, \dots, x_n)$ is a nondecreasing function of each of its arguments.

In this course, we will consider the case when the marginal distribution functions are either all step functions so that the individual random variables are all discrete, or are all continuous and differentiable so that the individual random variables are all continuous (recall the definition at the start of Chapter 3).

4.1 The discrete case

If the random variables $X_1, \dots, X_n$ are discrete then we can define the following analogue of the probability mass function for a single random variable:

Definition: If $X_1, \dots, X_n$ are discrete random variables, their JOINT PROBABILITY MASS FUNCTION is the collection

$$p(x_1, \dots, x_n) = P(X_1 = x_1, \dots, X_n = x_n) .$$

Example 4.1: Suppose we roll a fair die twice. Let X_1 denote the number of sixes obtained and X_2 the number of threes. Find the joint pmf of X_1 and X_2 .

■

As in this example, it is often convenient to represent a discrete BIVARIATE distribution (i.e. a joint distribution of two variables) in tabular form. Notice also that the probabilities sum to 1.

For a bivariate discrete distribution, notice that we can write (for example)

$$P(X_1 = x_1) = P(\{X_1 = x_1\} \cap \left(\bigcup_{x_2} \{X_2 = x_2\} \right) ,$$

where the union is over all of the values taken by X_2 — this forms a partition of the sample space

(see Section 1.5). Therefore we have

$$\begin{aligned}
 P(X_1 = x_1) &= P\left(\{X_1 = x_1\} \cap \left(\bigcup_{x_2} \{X_2 = x_2\}\right)\right) \\
 &= P\left(\bigcup_{x_2} \{X_1 = x_1 \cap X_2 = x_2\}\right) \quad (\text{distributive laws}) \\
 &= \sum_{x_2} P(X_1 = x_1, X_2 = x_2) = \sum_{x_2} p(x_1, x_2) .
 \end{aligned}$$

But this last sum is just the row total for the x_1 -row of the table: thus the row totals give the marginal pmf of X_1 . Similarly, the column totals give the marginal pmf of X_2 . The term ‘marginal’ comes from the fact that these totals can be written in the margins of the table.

Example 4.1 (continued): Find the marginal pmfs for the variables X_1 and X_2 , and verify that these match the appropriate binomial probabilities.

■

4.1.1 Expectation and its properties

The definition of expectation generalises immediately to functions of several variables, because any function $\phi(X_1, \dots, X_n)$ is itself a mapping from the sample space Ω to the real numbers. The

argument involves summing over the outcomes in the sample space in exactly the same way as for the expectation of a function on page 20, to yield

$$E[\phi(X_1, \dots, X_n)] = \sum_{x_1} \dots \sum_{x_n} \phi(x_1, \dots, x_n) p(x_1, \dots, x_n).$$

Note that this definition is identical to that for the univariate case in the previous chapter: the only thing that has changed is that we now substitute the joint pmf for the marginal pmf, and sum over the values of all the variables. It can be verified that if $\phi(X_1, \dots, X_n) = X_1$ (say) then the expression above simplifies to $\sum_{x_1} x_1 p_{X_1}(x_1)$ in agreement with the definition in the previous chapter (**Exercise**: show this).

Example 4.1 (continued): Let μ_1 and μ_2 be the marginal expectations of X_1 and X_2 respectively. Find $E[(X_1 - \mu_1)(X_2 - \mu_2)]$.

■

In general, if X_1 and X_2 are two random variables with expectations μ_1 and μ_2 then the quantity $E[(X_1 - \mu_1)(X_2 - \mu_2)]$ is called the COVARIANCE between X_1 and X_2 , denoted $\text{Cov}(X_1, X_2)$. For calculation purposes, it is often easier to use the equivalent formula $\text{Cov}(X_1, X_2) = E(X_1 X_2) - \mu_1 \mu_2$ (**Exercise**: show that the formulae are equivalent).

A useful application of the definition of expectation is to sums of random variables:

Result: $E(\sum_{i=1}^n X_i) = \sum_{i=1}^n E(X_i).$

Proof: $E(\sum_{i=1}^n X_i) = E(X_1 + \dots + X_n)$. By definition, this is

$$\sum_{x_1} \dots \sum_{x_n} (x_1 + \dots + x_n) p(x_1, \dots, x_n)$$

$$\begin{aligned}
&= \left[\sum_{x_1} \dots \sum_{x_n} x_1 p(x_1, \dots, x_n) \right] + \dots + \left[\sum_{x_1} \dots \sum_{x_n} x_n p(x_1, \dots, x_n) \right] \\
&= E(X_1) + \dots + E(X_n), \text{ as required.}
\end{aligned}$$

■

4.1.2 Independence of discrete random variables

In the context of events and sample spaces, we know that events E and F are independent if $P(E \cap F) = P(E)P(F)$, and that this can be interpreted as saying that knowledge of E tells us nothing about F . We would like to extend this to the study of random variables i.e. to find a way of expressing the idea that if we observe the value of X_1 , this tells us nothing about X_2 . We have

Definition: Discrete random variables X_1 and X_2 are INDEPENDENT RANDOM VARIABLES if, for *all* possible pairs (x_1, x_2) ,

$$P(X_1 = x_1 \text{ and } X_2 = x_2) = P(X_1 = x_1)P(X_2 = x_2)$$

i.e. if $p(x_1, x_2) = p_1(x_1)p_2(x_2)$, where $p_1(\cdot)$ and $p_2(\cdot)$ are the marginal pmfs of X_1 and X_2 respectively.

NB: it follows from this definition that, for any sets A, B of real values, $P(X_1 \in A \text{ and } X_2 \in B) = P(X_1 \in A)P(X_2 \in B)$ (**Exercise:** prove this). In particular,

$$P(X_1 \leq x_1 \text{ and } X_2 \leq x_2) = P(X_1 \leq x_1)P(X_2 \leq x_2) = F_1(x_1)F_2(x_2),$$

where $F_1(\cdot)$ and $F_2(\cdot)$ are the marginal distribution functions of X_1 and X_2 respectively.

If two random variables are independent then their joint probabilities are the products of the marginal probabilities. In Example 4.1 we see that X and Y are *not* independent.

If two random variables are independent then the covariance between them is zero (**Exercise:** prove this). The converse is not necessarily true, however.

Example 4.2: The joint pmf of discrete random variables X_1 and X_2 is as follows:

		x_2		
		0	1	2
x_1	0	0.25	0	0.25
	1	0	0.5	0

Show that the covariance between X_1 and X_2 is zero but that they are not independent.

■

The definition of independence extends to higher-dimensional distributions: a collection of random variables is mutually independent if all joint probabilities are the products of the corresponding marginal probabilities.

The following is a useful property of independent random variables:

Result: If X_1 and X_2 are independent random variables and their expectations exist, then $E(X_1 X_2) = E(X_1)E(X_2)$.

Proof: $E(X_1 X_2) = \sum_{x_1} \sum_{x_2} x_1 x_2 p(x_1, x_2)$ by definition, where $p(\cdot, \cdot)$ is the joint pmf of X_1 and X_2 . But if X_1 and X_2 are independent then $p(x_1, x_2) = p(x_1)p(x_2)$. Thus

$$E(X_1 X_2) = \sum_{x_1} \sum_{x_2} x_1 x_2 p(x_1)p(x_2) = \sum_{x_1} x_1 p(x_1) \sum_{x_2} x_2 p(x_2) = E(X_1)E(X_2),$$

as required. ■

This result generalises to products of more than two independent variables: if $X_1, \dots, X_n$ are mutually independent then $E(\prod_{i=1}^n X_i) = \prod_{i=1}^n E(X_i)$.

The most important consequence of this result is the following:

Result: If $X_1, \dots, X_n$ are mutually independent random variables, then

$$\text{Var} \left(\sum_{i=1}^n X_i \right) = \sum_{i=1}^n \text{Var}(X_i).$$

Proof: For each $i = 1, \dots, n$, let $\mu_i = E(X_i)$. From the properties of expectation, we have

$E(\sum_{i=1}^n X_i) = \sum_{i=1}^n E(X_i) = \sum_{i=1}^n \mu_i$. Hence, using one of the alternative definitions of variance, we have

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = E\left(\left[\sum_{i=1}^n X_i\right]^2\right) - \left(\sum_{i=1}^n \mu_i\right)^2.$$

Expanding the brackets and applying the properties of expectation, we get

$$\begin{aligned} \text{Var}\left(\sum_{i=1}^n X_i\right) &= E\left(\sum_{i=1}^n \left[X_i^2 + \sum_{j \neq i} X_i X_j\right]\right) - \sum_{i=1}^n \left[\mu_i^2 + \sum_{j \neq i} \mu_i \mu_j\right] \\ &= E\left(\sum_{i=1}^n [X_i^2 - \mu_i^2]\right) + E\left(\sum_{j \neq i} [X_i X_j - \mu_i \mu_j]\right) \\ &= \sum_{i=1}^n [E(X_i^2) - \mu_i^2] + \sum_{j \neq i} [E(X_i X_j) - \mu_i \mu_j] \\ &= \sum_{i=1}^n \text{Var}(X_i) + \sum_{j \neq i} [E(X_i X_j) - \mu_i \mu_j]. \end{aligned}$$

The previous result tells us that for $i \neq j$, $E(X_i X_j) = \mu_i \mu_j$, so the second term here is zero. The result is proved. ■

Note that this result is not true in general for *any* collection of random variables $X_1, \dots, X_n$ — although we can relax the restriction of independence to the slightly weaker restriction that the covariance between each pair of variables is zero, and the result still holds.

This result for the variance of a sum of independent random variables, coupled with the corresponding result for the expectation of a sum, provides a very easy way to find the mean and variance of the binomial and negative binomial distributions.

Binomial distribution: recall that the $\text{Binomial}(n, p)$ distribution arises as the distribution of the number of ‘successes’ in n independent Bernoulli trials, each with success probability p . It follows that if $X \sim \text{Bin}(n, p)$ then X can be regarded as the sum of n independent Bernoulli random variables, each with parameter p . Specifically, let $Y_1, \dots, Y_n$ be these Bernoulli random variables, and define $X = \sum_{i=1}^n Y_i$. X is just the number of 1s — or successes — out of the n Y s, and because the Y s are independent it must therefore have a $\text{Binomial}(n, p)$ distribution. In this case, we can use the results for the Bernoulli distribution to obtain $E(X) = E(\sum_{i=1}^n Y_i) = \sum_{i=1}^n E(Y_i) = \sum_{i=1}^n p = np$, and $\text{Var}(X) = \text{Var}(\sum_{i=1}^n Y_i) = \sum_{i=1}^n \text{Var}(Y_i) = \sum_{i=1}^n p(1-p) = np(1-p) = npq$, as required (note that we couldn’t do this to find the variance of X unless all the Y s were independent). ■

Negative binomial distribution: If $X \sim \text{NB}(r, p)$ then it can be regarded as the the number of trials until the r th success, in a sequence of independent Bernoulli trials with success probability

p . X can therefore be written as the sum of r independent *geometric* random variables:

$$X = Y_1 + Y_2 + \dots + Y_r$$

say, where the $\{Y_i\}$ are independent random variables each having a geometric distribution with parameter p . To see this, note that Y_i is simply the number of trials after the $(i-1)^{th}$ success up to and including the i^{th} success: the independence of the Y s follows from the independence of trials. Thus each Y has mean $1/p$ and variance q/p^2 (recall the properties of the geometric distribution). Thus we have $E(X) = E(Y_1 + \dots + Y_r) = E(Y_1) + \dots + E(Y_r) = r \times 1/p = r/p$ and $\text{Var}(X) = \text{Var}(Y_1 + \dots + Y_r) = \text{Var}(Y_1) + \dots + \text{Var}(Y_r) = r \times q/p^2 = rq/p^2$, as claimed in Section 2.3.4. ■

4.1.3 Conditional probability distributions

We can also calculate conditional probabilities from joint distributions. For example:

$$P(X_1 = x_1 \mid X_2 = x_2) = \frac{P(X = x_1, X_2 = x_2)}{P(X_2 = x_2)} = \frac{p(x_1, x_2)}{p_2(x_2)}.$$

Example 4.1 (continued): Find $P(X_1 = x_1 \mid X_2 = 0)$ for $x_1 = 0, 1, 2$. Repeat the calculations, conditioning on $X_2 = 1$ and on $X_2 = 2$. ■

In this example, for each value of x_2 the conditional probabilities for X_1 sum to 1 and therefore

form a valid pmf in their own right. This is generally true:

$$\sum_{x_1} P(X_1 = x_1 | X_2 = x_2) = \sum_{x_1} \frac{p(x_1, x_2)}{p_2(x_2)} = \frac{1}{p_2(x_2)} \sum_{x_1} p(x_1, x_2) = \frac{p_2(x_2)}{p_2(x_2)} = 1,$$

the penultimate step following from the fact that summing the joint probabilities over the values of x_1 gives the marginal probability $P(X_2 = x_2)$.

Definition: the CONDITIONAL PROBABILITY DISTRIBUTION of X_1 given $X_2 = x_2$ is defined by the probability mass function

$$p_{X_1|X_2}(x_1|x_2) = \frac{p(x_1, x_2)}{p_2(x_2)}$$

This conditional distribution gives the probabilities for observing $X_1 = x_1$ given that we already know $X_2 = x_2$.

Note that we can rearrange the above definition to obtain the multiplication rule for pmfs:

$$p(x_1, x_2) = p_{X_1|X_2}(x_1|x_2)p_2(x_2).$$

This can be used to find a bivariate pmf when we know one marginal distribution and one conditional distribution. The result extends in the obvious way when more than two variables are involved (this is the analogue of the Generalized Multiplication Law — see Section 1.5).

Note that if X_1 and X_2 are independent then $p(x_1, x_2) = p_1(x_1)p_2(x_2)$ so that $p_{X_1|X_2}(x_1|x_2) = p_1(x_1)$ i.e. the conditional and marginal distributions are the same.

4.1.4 Conditional expectation

Since $p_{X_1|X_2}(x_1|x_2)$ is a probability distribution, it has an expected value:

$$E(X_1|X_2 = x_2) = \sum_{x_1} x_1 p_{X_1|X_2}(x_1|x_2)$$

which represents the average value of X_1 among outcomes ω for which $X_2(\omega) = x_2$. This may also be written (say) $E_{X_1|X_2}(X_1|X_2 = x_2)$ to emphasise the distribution over which expectations are being taken.

Example 4.1 (continued): Find $E_{X_1|X_2}(X_1|X_2 = x_2)$ for $x_2 = 0, 1, 2$. Describe the relationship between x_2 and $E_{X_1|X_2}(X_1|X_2 = x_2)$.



Notice here that $E_{X_1|X_2}(X_1|X_2 = x_2)$ is a random variable: it takes values $E_{X_1|X_2}(X_1|X_2 = 0)$, $E_{X_1|X_2}(X_1|X_2 = 1)$ and $E_{X_1|X_2}(X_1|X_2 = 2)$ with probabilities $p_2(0)$, $p_2(1)$ and $p_2(2)$ respectively. These probabilities are from the marginal distribution of X_2 .

Example 4.1 (continued): Calculate the expectation $E_{X_2} [E_{X_1|X_2}(X_1|X_2)]$. Also calculate the marginal expectation $E(X_1)$. Comment on the result.



The result in this example is not a coincidence. It is an illustration of

Iterated expectation law: for any two random variables X_1 and X_2 ,

$$E(X_1) = E_{X_2} [E_{X_1|X_2} (X_1|X_2)] .$$

Proof:

$$\begin{aligned}
 E_{X_2} [E_{X_1|X_2} (X_1|X_2)] &= \sum_{x_2} p_{X_2}(x_2) \left[\sum_{x_1} x_1 p_{X_1|X_2}(x_1 | x_2) \right] \\
 &= \sum_{x_1} \sum_{x_2} x_1 p_{X_2}(x_2) \cdot \frac{p(x_1, x_2)}{p_{X_2}(x_2)} \quad \text{by definition of } p_{X_1|X_2} \\
 &= \sum_{x_1} x_1 \sum_{x_2} p(x_1, x_2) \\
 &= \sum_{x_1} x_1 p_{X_1}(x_1) \quad (\text{summing over } x_2 \text{ gives marginal pmf of } X_1) \\
 &= E(X_1) .
 \end{aligned}$$

■

This result is most useful when the conditional distribution of X given $Y = y$ is known and is easier to handle than the joint distribution.

Example 4.3: A fair die is rolled once, and the number showing is recorded as n . A fair coin is then tossed n times. What is the expected number of heads obtained?

■

There is an analogous version of the iterated expectation law for variances. This will be covered in later probability courses.

4.1.5 The multinomial distribution

Example 4.1 has some features in common with the binomial distribution: in fact, the marginal distributions of both X_1 and X_2 in that example are binomial. However, in the example there are more than just two outcomes labelled ‘success’ and ‘failure’: the outcomes of interest are ‘three’, ‘six’ and ‘neither’, occurring on each trial with probabilities $1/6$, $1/6$ and $2/3$ respectively.

The example can be generalised further, in an obvious way. Suppose that the outcome of an experiment falls in one of $m + 1$ categories, and that the probability of falling in category i is p_i with $\sum_{i=1}^{m+1} p_i = 1$. Suppose in addition that n repetitions of the experiment are carried out and, for $i = 1, \dots, m+1$, let X_i denote the number of times that the outcome is in category i . The collection $X_1, \dots, X_m$ is said to have a MULTINOMIAL DISTRIBUTION WITH INDEX n AND PARAMETERS $p_1, \dots, p_m$ (p_{m+1} is then determined from all of the other parameters as $p_{m+1} = 1 - \sum_{i=1}^m p_i$).

To find the joint pmf of the multinomial distribution, we need to find $P(X_1 = x_1, \dots, X_m = x_m)$ for all collections of non-negative integers $x_1, \dots, x_m$ with $\sum_{i=1}^m x_i \leq n$ (note that $X_{m+1} = n - \sum_{i=1}^m X_i$ so that we don’t have to consider X_{m+1} explicitly in the joint pmf). Now the probability of any sequence of n outcomes containing x_1 in category 1, x_2 in category 2, and so on, is $p_1^{x_1} p_2^{x_2} \dots p_{m+1}^{x_{m+1}}$. And there are $n! / x_1!(n - x_1)!$ sequences containing exactly x_1 outcomes in category 1; for each of these, in the remaining $n - x_1$ positions there are $(n - x_1)! / x_2!(n - x_1 - x_2)!$ possible locations for the x_2 outcomes in category 2, and so forth. The total number of sequences with x_i outcomes in category i ($i = 1, \dots, m$) is therefore

$$\begin{aligned} & \frac{n!}{x_1!(n - x_1)!} \frac{(n - x_1)!}{x_2!(n - x_1 - x_2)!} \dots \frac{(n - x_1 - x_2 - \dots - x_{m-1})!}{x_m!(n - x_1 - x_2 - \dots - x_{m-1} - x_m)!} \\ &= \frac{n!}{x_1! x_2! \dots x_m! x_{m+1}!}, \end{aligned}$$

and the joint pmf of $X_1, \dots, X_m$ is given by

$$p(x_1, \dots, x_m) = \begin{cases} \frac{n!}{x_1! \dots x_{m+1}!} p_1^{x_1} \dots p_{m+1}^{x_{m+1}} & \sum_{i=1}^m x_i \leq n \\ 0 & \text{otherwise.} \end{cases}$$

It can be shown that this does indeed define a valid joint pmf.

Example 4.1 (continued): Verify that the multinomial pmf yields the correct joint probabilities for this example.



Some useful properties of the multinomial distribution are as follows:

- For each $i = 1, \dots, m + 1$, the marginal distribution of X_i is $\text{Bin}(n, p_i)$ (this is obvious from the definition); thus, the marginal means and variances can be derived directly from the binomial distribution.
- Given $X_j = x_j$, the conditional distribution of X_i is $\text{Bin}(n - x_j, p_i / (1 - p_j))$. Again, with a bit of thought this should be obvious from the definition. **Exercise:** verify the result using the definition of conditional probability together with the joint pmf of X_i and X_j .

4.2 The continuous case

If a joint distribution function $F(\cdot, \dots, \cdot)$ is everywhere continuous and differentiable then the X s are all continuous random variables.⁵ Then, by analogy with the univariate case, we can consider events of the form

$$\{\omega : X_1(\omega) \in (x_1, x_1 + \delta x_1], \dots, X_n(\omega) \in (x_n, x_n + \delta x_n]\}$$

and define the JOINT PROBABILITY DENSITY FUNCTION OF $X_1, \dots, X_n$ as

$$f(x_1, \dots, x_n) = \lim_{\delta x_j \rightarrow 0 \forall j} \frac{P(X_1 \in (x_1, x_1 + \delta x_1], \dots, X_n \in (x_n, x_n + \delta x_n])}{\delta x_1 \dots \delta x_n}.$$

In the previous chapter we saw that formulae and definitions for discrete random variables could be transferred directly to continuous random variables, just by replacing sums by integrals and probabilities by probability density functions including the ‘ dx ’ part — see page 42. The same is true for joint distributions. For example:

⁵Perhaps surprisingly, the converse is not necessarily true — more detail in subsequent courses.

- To be a joint probability density function, a function $f(x_1, \dots, x_n)$ must be non-negative everywhere and must integrate to 1:

$$\int_{x_1 \in \mathbb{R}} \cdots \int_{x_n \in \mathbb{R}} f(x_1, \dots, x_n) dx_1 \dots dx_n = 1.$$

- If X_1 and X_2 are continuous random variables with joint density $f(x_1, x_2)$ then the marginal density of X_1 is obtained by integrating over x_2 :

$$f_{X_1}(x_1) = \int_{x_2 \in \mathbb{R}} f(x_1, x_2) dx_2.$$

Similarly,⁶ the marginal density of X_2 is obtained by integrating over x_1 .

- The expected value of $\phi(X_1, \dots, X_n)$ is

$$E[\phi(X_1, \dots, X_n)] = \int_{x_1 \in \mathbb{R}} \cdots \int_{x_n \in \mathbb{R}} \phi(x_1, \dots, x_n) f(x_1, \dots, x_n) dx_1 \dots dx_n.$$

Moreover, the probability that the random variables simultaneously fall in some set A is obtained by integrating the joint density over that set (this is the equivalent of finding the area under the curve of the pdf for a single variable).

Example 4.4: Two continuous random variables X_1 and X_2 have joint density

$$f(x_1, x_2) = \begin{cases} Kx_1e^{-x_1(3+x_2)} & x_1, x_2 > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Find the value of K . Find the marginal distributions of X_1 and X_2 , and name them if possible. Find $P(X_1 > 1)$. Find $P(X_1 > 1, X_2 < 1)$.

⁶Strictly speaking, if we apply the ‘replace the pmf with $f(x)dx$ rule’ then we should have $f_{X_1}(x_1)dx_1$ on the left-hand side of the preceding expression; and $f(x_1, x_2)dx_1dx_2$ on the right-hand side (NB the dx_1 doesn’t do anything here because the integration is only over x_2). However, the dx_1 s ‘cancel’.



4.2.1 Independence for continuous random variables

We have defined discrete random variables X_1 and X_2 to be independent if $P(X_1 = x_1 \text{ and } X_2 = x_2) = P(X_1 = x_1)P(X_2 = x_2)$ for all x_1 and x_2 . This is a way of saying that knowing the value of one of them tells us nothing about the other. This definition cannot usefully be carried straight over to the continuous case however, since if X_1 and X_2 are continuous then $P(X_1 = x_1) = P(X_2 = x_2) = 0$, which is equal to $P(X_1 = x_1)P(X_2 = x_2)$ regardless of any dependence between them! We can deal with the problem by considering sets of outcomes:

Definition: X_1 and X_2 are INDEPENDENT if, for *all* $A, B \subseteq \mathbb{R}$,

$$P(X_1 \in A \text{ and } X_2 \in B) = P(X_1 \in A)P(X_2 \in B) .$$

This definition covers both discrete and continuous cases. Note in particular that if we take $A = (x_1, x_1 + \delta x_1)$, $B = (x_2, x_2 + \delta x_2)$ and X_1 and X_2 are independent, then $P(x_1 < X_1 < x_1 + \delta x_1, x_2 < X_2 < x_2 + \delta x_2) = P(x_1 < X_1 < x_1 + \delta x_1)P(x_2 < X_2 < x_2 + \delta x_2) \approx f_{X_1}(x_1)f_{X_2}(x_2)\delta x_1\delta x_2$. Also, $P(X_1 < x_1, X_2 < x_2) = F_{X_1}(x_1)F_{X_2}(x_2)$. Both of these are in fact equivalent to the definition above, for continuous random variables. Indeed, a consequence of the former is usually used as the operational definition of independence for continuous random

variables:

Operational definition (independence of continuous random variables): Continuous random variables X_1 and X_2 are independent if their joint density factorises as

$$f(x_1, x_2) = f_{X_1}(x_1)f_{X_2}(x_2).$$

It can be shown that for continuous random variables, this definition is equivalent to the previous one (**Proof:** exercise).

Example 4.4 (continued): Are X_1 and X_2 independent?

■

4.2.2 Conditional distributions

As with independence, we must be careful when defining conditional distributions for continuous random variables: we cannot condition on $X_2 = x_2$ for example, because this is an event with probability zero. However, we can consider $P(x_1 < X_1 \leq x_1 + \delta x_1 | x_2 < X_2 \leq x_2 + \delta x_2)$

$$= \frac{P(x_1 < X_1 \leq x_1 + \delta x_1, x_2 < X_2 \leq x_2 + \delta x_2)}{P(x_2 < X_2 \leq x_2 + \delta x_2)} \simeq \frac{f(x_1, x_2)\delta x_1\delta x_2}{f_{X_2}(x_2)\delta x_2} = \frac{f(x_1, x_2)\delta x_1}{f_{X_2}(x_2)}.$$

This motivates the following

Definition: The CONDITIONAL DENSITY of X_1 given $X_2 = x_2$ is defined as

$$f_{X_1|X_2}(x_1|x_2) = \frac{f(x_1, x_2)}{f_{X_2}(x_2)}.$$

Note that $f_{X_1|X_2}(\cdot|\cdot)$ is a valid pdf: it is non-negative and

$$\int_{x_1 \in \mathbb{R}} f_{X_1|X_2}(x_1|x_2) dx_1 = \int_{x_1 \in \mathbb{R}} \frac{f(x_1, x_2)}{f_{X_2}(x_2)} dx_1 = \frac{1}{f_{X_2}(x_2)} \underbrace{\int_{x_1 \in \mathbb{R}} f(x_1, x_2) dx_1}_{f_{X_2}(x_2)} = 1.$$

If X_1 and X_2 are independent then, as before, the conditional density of X_1 given $X_2 = x_2$ is just the marginal density of X_1 .

The iterated expectation law carries over directly to continuous random variables, as do the other properties of expectation that were studied in the discrete case.

Example 4.4 (continued): Give the conditional distributions of X_1 when $X_2 = x_2$, and of X_2 when $X_1 = x_1$. Name these distributions if possible, and find their expectations. Find $E(X_2)$.

■

4.2.3 The bivariate normal distribution

There aren't many 'standard' multivariate distributions for continuous random variables. The most common one is the multivariate generalisation of the normal distribution. In this course we will only consider the bivariate version though:

Definition: continuous random variables X_1 and X_2 are said to have a BIVARIATE NORMAL DISTRIBUTION if they have joint probability density function

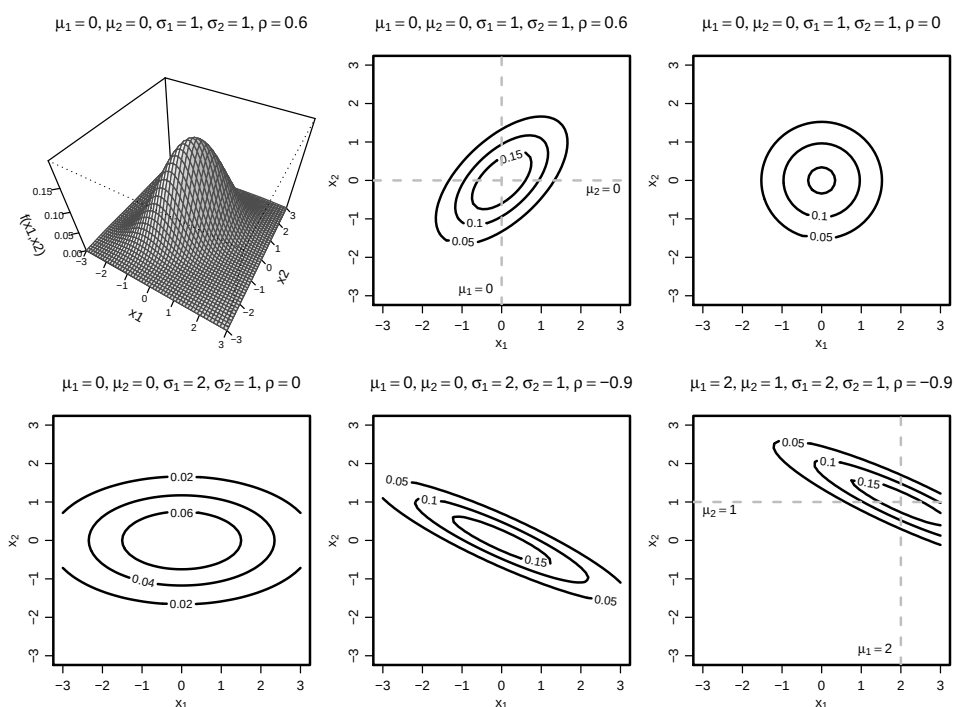
$$f_{X_1, X_2}(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp \left[-\frac{1}{2(1-\rho^2)} (z_1^2 - 2\rho z_1 z_2 + z_2^2) \right]$$

$$(x_1, x_2 \in \mathbb{R}; \mu_1, \mu_2 \in \mathbb{R}; \sigma_1, \sigma_2 > 0; -1 < \rho < 1).$$

where $z_1 = (x_1 - \mu_1)/\sigma_1$ and $z_2 = (x_2 - \mu_2)/\sigma_2$. The parameters of this distribution are $\mu_1, \mu_2, \sigma_1^2, \sigma_2^2$, and ρ .

It is helpful to think of the bivariate normal density as a surface with elliptical contours:

Examples of bivariate normal densities:



Marginal densities: as you might expect from the notation, the marginal densities of X_1 and X_2 are $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$ respectively. To show this for X_1 , remember that the marginal density is given by $f_{X_1}(x_1) = \int_{x_2 \in \mathbb{R}} f(x_1, x_2) dx_2$. Writing $z_1 = (x_1 - \mu_1)/\sigma_1$ as above, and making a formal change of variables in the integration: $z_2 = (x_2 - \mu_2)/\sigma_2$ so that $dx_2 = \sigma_2 dz_2$, we have

$$f_{X_1}(x_1) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \int_{z_2 \in \mathbb{R}} \exp \left[-\frac{1}{2(1-\rho^2)} (z_1^2 - 2\rho z_1 z_2 + z_2^2) \right] \sigma_2 dz_2.$$

Now, if the bracketed term in the exponent was $z_2^2 - 2\rho z_1 z_2 + \rho^2 z_1^2$ then it could be written as $(z_2 - \rho z_1)^2$; in this case the integrand would start to look like the density of a normal distribution with mean ρz_1 and variance $\sqrt{1 - \rho^2}$. To get the bracketed term into this form, just write

$$z_1^2 - 2\rho z_1 z_2 + z_2^2 = (z_2 - \rho z_1)^2 + z_1^2(1 - \rho^2)$$

so that the exponent itself can be written as

$$\exp \left[-\frac{(z_2 - \rho z_1)^2 + z_1^2(1 - \rho^2)}{2(1 - \rho^2)} \right] = \exp \left[-\frac{(z_2 - \rho z_1)^2}{2(1 - \rho^2)} \right] \exp \left[-\frac{z_1^2}{2} \right]$$

(this procedure is known as *completing the square*). We can now take the final term outside the integral, since it doesn't depend on z_2 :

$$f_{X_1}(x_1) = \frac{1}{2\pi\sigma_1\sqrt{1 - \rho^2}} \exp \left[-\frac{z_1^2}{2} \right] \int_{z_2 \in \mathbb{R}} \exp \left[-\frac{(z_2 - \rho z_1)^2}{2(1 - \rho^2)} \right] dz_2.$$

If the integrand was preceded by a factor $1/\sqrt{2\pi(1 - \rho^2)}$ then it would be exactly the density of $N(\rho z_1, 1 - \rho^2)$ and hence the integral would be 1. This is easy to arrange:

$$f_{X_1}(x_1) = \frac{1}{\sigma_1\sqrt{2\pi}} \exp \left[-\frac{z_1^2}{2} \right] \int_{z_2 \in \mathbb{R}} \frac{1}{\sqrt{2\pi(1 - \rho^2)}} \exp \left[-\frac{(z_2 - \rho z_1)^2}{2(1 - \rho^2)} \right] dz_2 = \frac{1}{\sigma_1\sqrt{2\pi}} \exp \left[-\frac{z_1^2}{2} \right].$$

Finally, remember that we have been using z_1 as shorthand for $(x_1 - \mu_1)/\sigma_1$:

$$f_{X_1}(x_1) = \frac{1}{\sigma_1\sqrt{2\pi}} \exp \left[-\frac{(x_1 - \mu_1)^2}{2\sigma_1^2} \right],$$

which is the density of $N(\mu_1, \sigma_1^2)$ as required. Obviously, the corresponding result for X_2 follows in exactly the same way (just swap all the subscripts around).

Conditional distributions. The conditional distribution of X_1 given $X_2 = x_2$ can be found from the definition $f_{X_1|X_2}(x_1|x_2) = f(x_1, x_2)/f_{X_2}(x_2)$, by substituting the definition of the bivariate normal density into the numerator and the density of $N(\mu_2, \sigma_2^2)$ into the denominator. The algebra is straightforward, but lengthy and not particularly instructive so we just give the result:

Result: If the joint distribution of X_1 and X_2 is bivariate normal then, conditional on $X_2 = x_2$, the distribution of X_1 is $N(\mu_1 + \rho\frac{\sigma_1}{\sigma_2}(x_2 - \mu_2), \sigma_1^2(1 - \rho^2))$.

Note the following features of this result:

- The conditional mean $E(X_1|X_2 = x_2)$ is a linear function of x_2 : this suggests that in some sense X_1 and X_2 are linearly related (which makes sense given the elliptical contours of bivariate normal densities).

- If $x_2 > \mu_2$, then the conditional mean $E(X_1|X_2 = x_2)$ is greater than the marginal mean $E(X_1) = \mu_1$ when $\rho > 0$: this says that if X_2 is greater than average then we expect X_1 to be greater than average as well. Conversely, if $\rho < 0$ then we expect X_1 to be *less* than average whenever X_2 is greater than average. So ρ controls the *direction of association* between X_1 and X_2 .
- The conditional variance of X_1 is $\sigma_1^2(1 - \rho^2)$. Since $-1 < \rho < 1$, this conditional variance is never greater than σ_1^2 (in fact, this is true for any conditional distribution, not just in the bivariate normal case). Moreover, as $\rho^2 \rightarrow 1$ the conditional variance tends to zero: in this case the value of X_1 is completely determined by the value of X_2 (and vice versa).

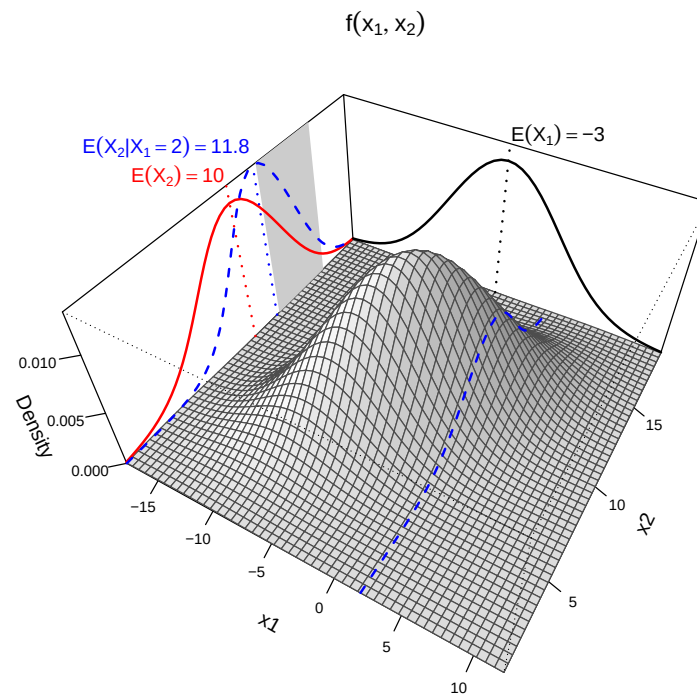
These features all suggest that ρ can be thought of as a measure of the *strength of linear association* between X_1 and X_2 . In fact, it can be shown that $\rho = \text{Cov}(X_1, X_2)/\sigma_1\sigma_2$ where $\text{Cov}(X_1, X_2)$ is the covariance between X_1 and X_2 . This quantity may be defined for other joint distributions as well: it is called the *correlation coefficient* between X_1 and X_2 , and it always lies between -1 and +1.

Example 4.5: X_1 and X_2 have a bivariate normal distribution with $E(X_1) = -3$, $E(X_2) = 10$, $\text{Var}(X_1) = 25$, $\text{Var}(X_2) = 9$ and $\text{Corr}(X_1, X_2) = 0.6$. Calculate $P(12 < X_2 < 16)$ and $P(12 < X_2 < 16|X_1 = 2)$. Compare and comment on the results.

■

To visualise the preceding example, and to get a feel for how marginal and conditional distri-

butions relate to the joint distributions from which they are derived, the following figure may be helpful. It shows the surface of the bivariate density, together with the marginal densities of X_1 and X_2 , the conditional density of X_2 given $X_1 = 2$ and the region $12 < X_2 < 16$:



5 Sums of independent variables

Often in statistics we need to know the the distribution of sums, of the form $S_n = X_1 + \dots + X_n$ where the X s are independent. The results in the previous section show that the expectation and variance of S_n are $\sum_{i=1}^n E(X_i)$ and $\sum_{i=1}^n \text{Var}(X_i)$ respectively. In this section we consider how to make more detailed probability statements about S_n .

In general, if S_n is a discrete random variable,

$$P(S_n = k) = \sum P(X_1 = i_1, \dots, X_n = i_n),$$

where the sum is over all $i_1, \dots, i_n$ such that $i_1 + \dots + i_n = k$. Although this is conceptually straightforward however, evaluation of the n -fold sum to find the distribution of S_n is generally difficult.

5.1 Distribution of S_n

The easiest way to find the distribution of S_n is often via its probability or moment generating function. Suppose the X s are independent discrete random variables taking non-negative integer values. Then S_n also takes non-negative integer values, so it has a PGF which is defined by

$$\Pi_{S_n}(z) = E(z^{S_n}) = E(z^{X_1} z^{X_2} \dots z^{X_n}) = E(z^{X_1}) E(z^{X_2}) \dots E(z^{X_n}),$$

since if the X s are independent r.v.s the z^X s must be also, and we know that the expectation of a product of independent r.v.s is the product of their expectations. Now the terms in this last expression are just the PGFs of the individual X s. Thus we have demonstrated the following

Result: If $X_1, \dots, X_n$ are independent discrete random variables taking non-negative integer values, the PGF of their sum is the product of their PGFs.

If all of the X s have the same distribution with PGF $\Pi_X(\cdot)$, then the PGF of S_n is $[\Pi_X(z)]^n$.

The PGF is only defined for non-negative integer-valued random variables. We have, however, encountered the Moment Generating Function as an alternative which works for discrete and continuous random variables: we have $M_X(t) = E(e^{tX})$. If we study the MGF for a sum of discrete random variables, the above argument is unchanged (effectively we just replace z by e^t), and we obtain

Result: If $X_1, \dots, X_n$ are independent random variables, the MGF of their sum is the product of their MGFs.

This result holds for continuous random variables as well.

Example 5.1: Suppose $X_1, \dots, X_n$ are independent Poisson random variables with means $\mu_1, \dots, \mu_n$ respectively. What is the distribution of $X_1 + \dots + X_n$?



If you consider the Poisson process, this result is obvious (**Exercise:** explain why!). However, the example illustrates the power of the PGF to solve problems which would otherwise be difficult.

Sums of variables arise in practice when we compute averages of data values. Regarding these as realised values of random variables, we are now in a position to study the behaviour of such averages. Specifically, suppose $X_1, \dots, X_n$ are independent with the same distribution (we say they are INDEPENDENT AND IDENTICALLY DISTRIBUTED (IID)), and define $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i = S_n/n$. If the X s have mean μ and variance σ^2 then

$$\begin{aligned} E(\bar{X}_n) &= E\left(\frac{1}{n}S_n\right) = \frac{1}{n}E(S_n) = \frac{1}{n}(n\mu) = \mu, \\ \text{and } \text{Var}(\bar{X}_n) &= \text{Var}\left(\frac{1}{n}S_n\right) = \frac{1}{n^2}\text{Var}(S_n) = \frac{1}{n^2}(n\sigma^2) = \frac{\sigma^2}{n}. \end{aligned}$$

We see that, as $n \rightarrow \infty$, the variance of $\bar{X}_n$ tends to zero. We can interpret this as saying that its distribution becomes increasingly concentrated about μ .

Definition: $\bar{X}_n$ is called the SAMPLE MEAN (it is the average of all the observations in a sample). Its distribution is called the SAMPLING DISTRIBUTION OF THE MEAN.

Because the variance of the sample mean tends to zero as $n \rightarrow \infty$, we suspect that the sample mean of a large number of observations will tend to be close to μ . This idea is encapsulated in the following result, which we will not prove here:

Result (the STRONG LAW OF LARGE NUMBERS):

$$P\left(\lim_{n \rightarrow \infty} \bar{X}_n = \mu\right) = 1.$$

An informal interpretation of this result is this: *almost every possible sequence of sample means tends strictly to μ as $n \rightarrow \infty$* i.e. you can get as close to μ as you like just by collecting a large enough sample. This appears to justify our definition of expectation, as an idealised long-term average.

Another consequence of the Strong Law is obtained by considering the case when the X s are independent Bernoulli random variables, each with probability p . Then $\bar{X}_n$ is the relative frequency of successes in the first n trials and $\mu = p$, so the relative frequency apparently does tend to the probability p .

This all seems to justify our interpretation of probability as an idealised long-term relative frequency. However, we have defined probability and expectation as purely mathematical abstractions. In particular, the allocation of probabilities to events in any sample space is arbitrary. So how can the Strong Law be right? There must be a catch somewhere!



5.2 Sums of normal random variables

We have seen that the average of a large number of iid random variables converges in probability to their common mean, so that by taking a larger and larger sample, we can be more and more certain that the average will be ‘close’ to the underlying mean. However, it is unlikely that the average will ever be *exactly* equal to the underlying mean. Can we say anything about the distribution of this average (or, equivalently, of the sum of a large number of random variables)?

Before we do this in the general case, it is instructive to examine the behaviour of sums of *normal* random variables. Earlier in the course, we said that sums of normal random variables are

themselves normally distributed. We now prove this result. The proof uses moment generating functions, whence we need the following:

Result: If $X \sim N(\mu, \sigma^2)$ then its MGF is

$$M(t) = \exp \left[\mu t + \frac{\sigma^2 t^2}{2} \right] .$$

Proof: First consider $Z \sim N(0, 1)$. Then the MGF of Z is

$$\begin{aligned} M_Z(t) = E(e^{tZ}) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tz} e^{-\frac{1}{2}z^2} dz = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp \left[tz - \frac{1}{2}z^2 \right] dz \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp \left[-\frac{1}{2}(z^2 - 2tz + t^2) + \frac{1}{2}t^2 \right] dz \\ &= \exp \left[\frac{1}{2}t^2 \right] \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{1}{2}(z - t)^2 \right] dz = e^{t^2/2} , \end{aligned}$$

since the integrand on the last line is the pdf of a normal distribution with mean t and variance 1, so the integral is 1.

Now in the last chapter, we saw that if $Z \sim N(0, 1)$ and $X = \mu + \sigma Z$, then $X \sim N(\mu, \sigma^2)$. The MGF of X is

$$M_X(t) = E(e^{tX}) = E(e^{t(\mu + \sigma Z)}) = e^{t\mu} E(e^{t\sigma Z}) = e^{t\mu} M_Z(t\sigma) = e^{t\mu} e^{(t\sigma)^2/2} ,$$

which is the required result. ■

We can now show that a sum of independent normally distributed random variables is itself normally distributed. Specifically, let $X_1, \dots, X_n$ be independent with $X_i \sim N(\mu_i, \sigma_i^2)$, and define $S_n = \sum_{i=1}^n X_i$. Now we know from the above result that the MGF of X_i is

$$M_{X_i}(t) = \exp \left[\mu_i t + \frac{\sigma_i^2 t^2}{2} \right] ,$$

and we know that the MGF of a sum of independent random variables is the product of their MGFs. Hence the MGF of S_n is

$$M_{S_n}(t) = \prod_{i=1}^n \exp \left[\mu_i t + \frac{\sigma_i^2 t^2}{2} \right] = \exp \sum_{i=1}^n \left[\mu_i t + \frac{\sigma_i^2 t^2}{2} \right] = \exp \left[t \sum_{i=1}^n \mu_i + \frac{t^2}{2} \sum_{i=1}^n \sigma_i^2 \right] .$$

This has the same form as the MGF of a normal distribution with mean $\sum_{i=1}^n \mu_i$ and variance $\sum_{i=1}^n \sigma_i^2$; the result is shown. Note that the mean and variance agree with the results we have already derived for sums of independent variables. The important thing in this particular example is that the shape of the distribution (i.e. the normality) is preserved.

The main reason for deriving this result here is to show the power of the moment generating function approach to deriving distributional results. The approach is now used to derive one of the most important results in statistics.

5.3 The Central Limit Theorem

Suppose that $X_1, \dots, X_n$ are independent variables, having the same distribution, with mean μ and variance σ^2 . Their sum is S_n and their mean is $\bar{X}_n = S_n/n$. We have already seen that $E(\bar{X}_n) = \mu$ and $\text{var}(\bar{X}_n) = \sigma^2/n$. If we define $Z_n = \sqrt{n}(\bar{X}_n - \mu)/\sigma$, we therefore have $E(Z_n) = 0$ and $\text{var}(Z_n) = 1$ for all n . This lack of dependence upon n suggests that we might ask whether or not Z_n has a limiting distribution as $n \rightarrow \infty$.

We will do this by examining the MGF of Z_n . We have

$$\begin{aligned} M_{Z_n}(t) &= E\left(\exp\left[t\sqrt{n}(\bar{X}_n - \mu)/\sigma\right]\right) = \exp\left[\frac{-t\mu\sqrt{n}}{\sigma}\right] E\left(\exp\left[t\frac{\bar{X}_n\sqrt{n}}{\sigma}\right]\right) \\ &= \exp\left[\frac{-t\mu\sqrt{n}}{\sigma}\right] E\left(\exp\left[\frac{t}{\sigma\sqrt{n}}S_n\right]\right) = \exp\left[\frac{-t\mu\sqrt{n}}{\sigma}\right] M_{S_n}\left(\frac{t}{\sigma\sqrt{n}}\right) \\ &= \exp\left[\frac{-t\mu\sqrt{n}}{\sigma}\right] \left[M_X\left(\frac{t}{\sigma\sqrt{n}}\right)\right]^n, \end{aligned}$$

where $M_X(\cdot)$ is the common MGF of all the X s. For convenience, we work with logarithms of both sides:

$$\ln M_{Z_n}(t) = \frac{-t\mu\sqrt{n}}{\sigma} + n \ln M_X\left(\frac{t}{\sigma\sqrt{n}}\right).$$

But

$$\begin{aligned} M_X\left(\frac{t}{\sigma\sqrt{n}}\right) &= E\left(\exp\left[\frac{t}{\sigma\sqrt{n}}X\right]\right) = E\left(1 + \frac{tX}{\sigma\sqrt{n}} + \frac{t^2X^2}{2\sigma^2n} + o(n^{-1})\right) \\ &= 1 + \frac{tE(X)}{\sigma\sqrt{n}} + \frac{t^2E(X^2)}{2\sigma^2n} + o(n^{-1}) = 1 + \frac{t\mu}{\sigma\sqrt{n}} + \frac{t^2(\mu^2 + \sigma^2)}{2\sigma^2n} + o(n^{-1}). \end{aligned}$$

This is of the form $1 + x$, where $|x| < 1$ providing n is large enough. Therefore we use the expansion $\ln(1 + x) = x - x^2/2 + x^3/3 - \dots$ to write, as $n \rightarrow \infty$,

$$\ln M_X\left(\frac{t}{\sigma\sqrt{n}}\right) = \frac{t\mu}{\sigma\sqrt{n}} + \frac{t^2(\mu^2 + \sigma^2)}{2\sigma^2n} - \frac{t^2\mu^2}{2\sigma^2n} + o(n^{-1}) = \frac{t\mu}{\sigma\sqrt{n}} + \frac{t^2\sigma^2}{2\sigma^2n} + o(n^{-1}).$$

We therefore have, as $n \rightarrow \infty$,

$$\ln M_{Z_n}(t) = \frac{-t\mu\sqrt{n}}{\sigma} + n\left(\frac{t\mu}{\sigma\sqrt{n}} + \frac{t^2}{2n} + o(n^{-1})\right) = \frac{t^2}{2} + o(1),$$

which tends to $t^2/2$. Therefore $M_{Z_n}(t) \rightarrow e^{t^2/2}$, which is the MGF of a standard normal variable. So the distribution of Z_n tends to the standard normal distribution, and we have

Result (the CENTRAL LIMIT THEOREM): if $X_1, X_2, \dots$ are independent random variables having a common distribution with means μ and variance $\sigma^2 < \infty$, then as $n \rightarrow \infty$

$$P(\mu + \sigma a/\sqrt{n} \leq \bar{X}_n \leq \mu + \sigma b/\sqrt{n}) \rightarrow \Phi(b) - \Phi(a).$$

Informally, you can think “for large n , $\bar{X}_n$ has approximately a normal distribution with mean μ and variance σ^2/n ”. The requirement $\sigma^2 < \infty$ guarantees that the standardised variable $Z_n \neq 0$. There are many generalisations of the theorem where the assumptions of independence, common distribution or finite variance of the X_i are relaxed.

5.4 Normal approximation to the Binomial

The Central Limit Theorem underpins many methods used in statistics (since we are often interested in studying averages of large numbers of observations). However, another application is to the approximation of other distributions such as the Binomial. Recall that if $X_1, \dots, X_n$ are independent Bernoulli random variables, each with success probability p , then their sum $S_n = \sum_{i=1}^n X_i$ is distributed as $\text{Bin}(n, p)$. We have seen that if n is large and np is moderate, the Binomial distribution can be approximated using the Poisson. The Central Limit Theorem now tells us that, if n is large, the distribution of S_n is approximately normal and therefore we can approximate the Binomial with the normal. Specifically, we know that the mean and variance of S_n are np and $np(1-p)$ respectively; hence for large enough n we can approximate $\text{Bin}(n, p)$ using $N(np, np(1-p))$.

There is a catch, however. The Binomial is a *discrete* distribution, whereas the normal is continuous. If we were to use an exact binomial calculation we would find, for example, that $P(S_n \leq k) = P(S_n < k+1)$. This is not true for the corresponding normal distribution. To get round this we use a CONTINUITY CORRECTION. In this particular example, we would calculate the normal probability corresponding to $k + 1/2$, as a sensible compromise between the values k and $k+1$.

Example 5.2: 10 green bottles accidentally fall from the top of a wall. The probability that any bottle is broken is 0.75, independently of all the others. Find the probability that at least 3 bottles remain intact (i) exactly:

(ii) using the Poisson approximation to the binomial:

(iii) using the normal approximation to the binomial:



6 Introduction to Statistics

Having studied probability, doubtless in far more detail than you would have liked, we are in a position to move on to the second main part of this course i.e. *Statistics*. As mentioned at the beginning of the course, Statistics can be regarded as a subject which deals with the interpretation of information (i.e. DATA) in the face of uncertainty. Most ‘interesting’ statistical methods are therefore underpinned by some application of probability theory. Before we discuss these, however, it is necessary to think a little about what data actually *are*, and it is also helpful briefly to discuss some simple methods for presenting data so as to convey information clearly and unambiguously. Next we will review some of the ideas that are necessary if we are to use probability to interpret data. In subsequent chapters, we will look at a few specific statistical techniques.

6.1 Data: types, and representation

Data usually arise as measurements or observations. Observations may be QUALITATIVE (i.e. non-numerical) or QUANTITATIVE (numerical).

Qualitative data can be CATEGORICAL, where the observations fall into non-ordered groups (e.g. sex, hair colour, method of transport), or ORDINAL, where the observations have a natural ordering (e.g. disease severity, level of satisfaction with this course).

Quantitative data can be DISCRETE, taking values in a countable set (e.g. number of railway accidents this year) or CONTINUOUS (e.g. body weight, concentration of pollutants in river water). In practice, continuous variables can only ever be measured to a certain degree of accuracy, so that a purist may argue that *observations* should always be treated as discrete. Thus the distinction is not always clear cut.

6.1.1 Presentation of qualitative data

Qualitative observations can be completely summarised by stating the number, or proportion, of times that each value or category occurs. The following methods may be used:

1. A FREQUENCY TABLE. List the categories and record the numbers, or proportions RELATIVE FREQUENCIES, of observations in each category.

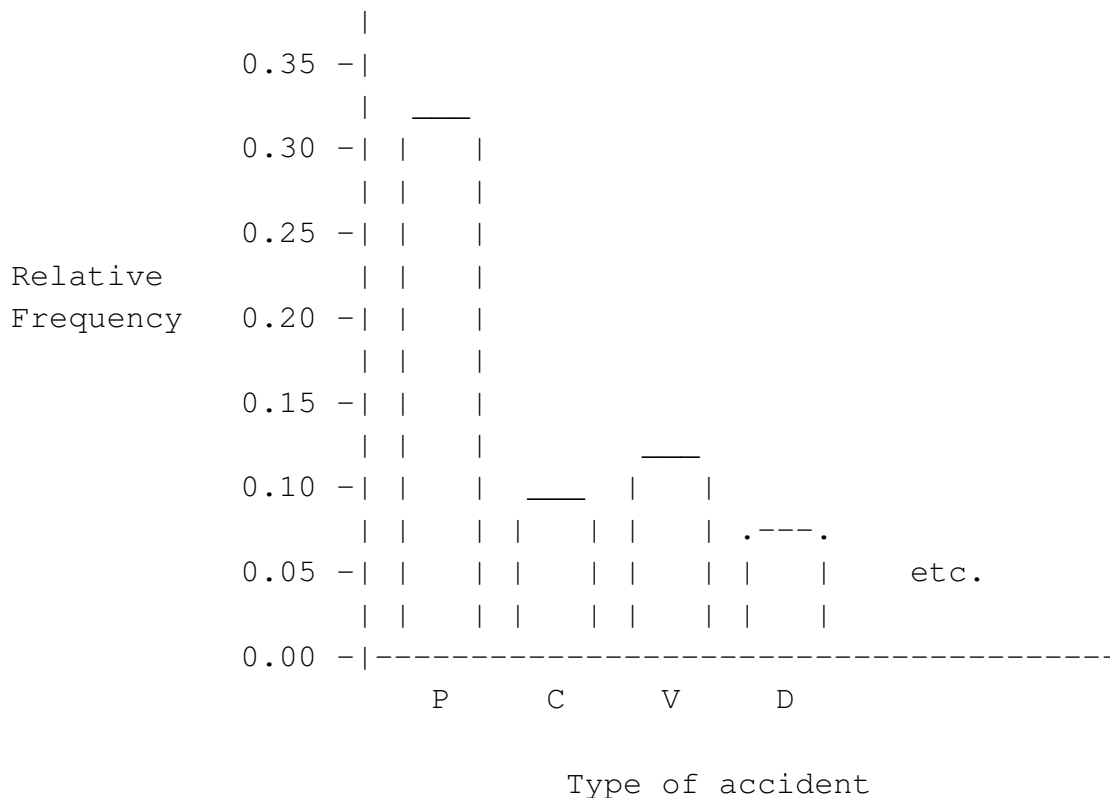
Example 6.1: Types of fatal accident in children under 15 years in the UK in 1987 were as follows:

Type of accident	Coding	No. of children	Relative frequency
Road — pedestrian	P	200	0.309
Road — cyclist	C	73	0.087
Road — vehicle occupant	V	96	0.114
Drowning	D	63	0.075
⋮	⋮	⋮	⋮
TOTAL		842	1

It is common, as here, to introduce meaningful codes to represent the values of categorical variables.

2. A BAR CHART. The categories are marked off along the abscissa (horizontal axis), and vertical lines or BARS are drawn to represent the numbers of observations or relative frequencies. The bars don't touch — this emphasizes the categorical nature of the observations. For ordinal data, the bars should be ordered to correspond with the ordering of categories (lowest on the left, highest on the right).

Example 6.1 (contd.):



3. A **PIE CHART** — relative frequencies are represented as ‘slices’ of a circle.

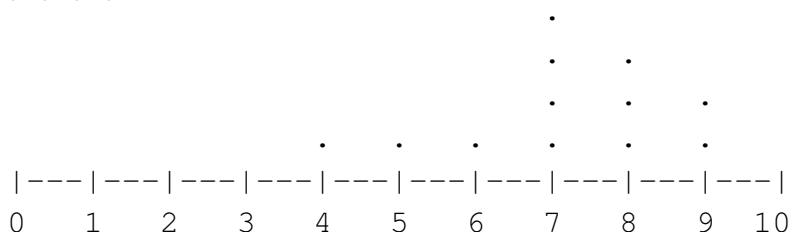
4. A PICTOGRAM. Used by the media to try and make Statistics seem interesting, and by politicians (and the media) to mislead. Intentionally or otherwise.

6.1.2 Presentation of discrete quantitative data

Again, we can present discrete qualitative data by specifying the relative frequency of each value. There are some other methods as well, reflecting the numeric nature of the data.

1. A FREQUENCY TABLE, as before.
2. A barchart of relative frequencies. NB again make sure the bars don't touch — this emphasises the discrete nature of the data. Such a bar chart is a bit like the probability mass function of a discrete random variable.
3. A DOT DIAGRAM (for fairly small amounts of data).

Example 6.2: The marks of 12 students for a piece of coursework were (out of 10): 5, 8, 7, 7, 4, 9, 8, 7, 8, 6, 7, 9:



4. A STEM-AND-LEAF DISPLAY (round data first if necessary)

Example 6.3: 11 systolic blood pressure measurements (in mm Hg) were 152, 105, 123, 131, 99, 115, 149, 137, 126, 124 and 128:

	Stem	Leaf
	9	9
	10	5
	11	5
Leaf unit = 1 mm Hg	12	3 4 6 8
	13	1 7
	14	9
	15	2

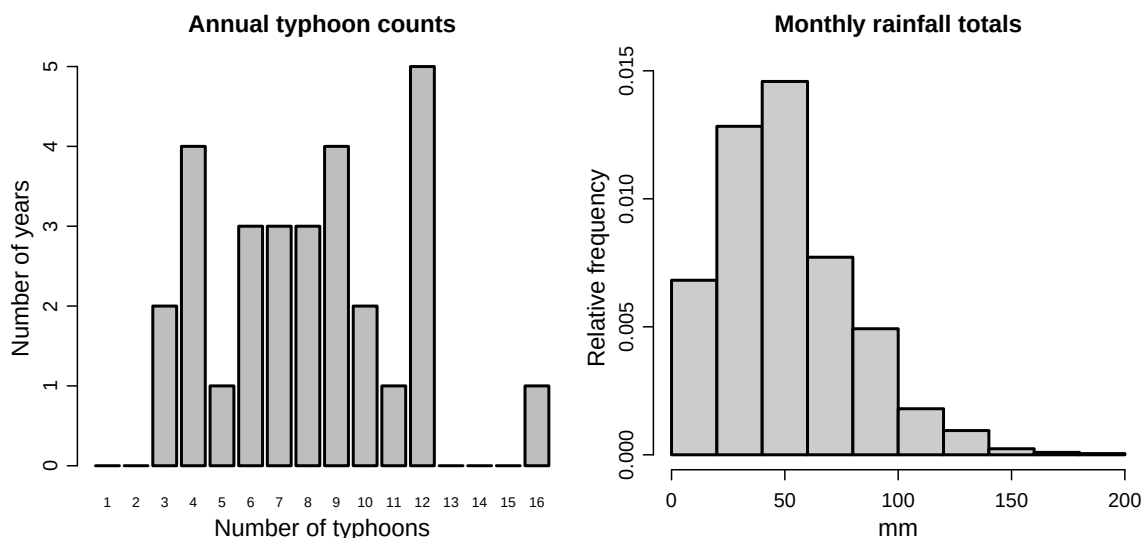
The stem-and-leaf plot allows some grouping of data, without losing information on the actual values.

6.1.3 Presentation of continuous data

Continuous data (or large amounts of discrete data) are usually summarised by a RELATIVE FREQUENCY HISTOGRAM obtained as follows:

1. Calculate the RANGE (= largest value - smallest value) of the data.
2. Divide the range into equal intervals (as a rough guide, 5–15 depending on the number of observations). It may be necessary to extend the range in order to get ‘sensible’ intervals e.g. if you have 50 observations ranging from 22 to 67, you might choose intervals $[20, 25)$, $[25, 30)$, $\dots$, $[65, 70)$. Choice of interval size is a compromise between ease of presentation and preservation of detailed information.
3. Calculate frequency in each interval (use tally marks if doing it by hand).
4. Calculate relative frequencies.
5. If necessary (e.g. if adjacent intervals contain very few observations), pool intervals.
6. Draw the histogram: the x -axis represents the values of the variable, and the frequencies are represented by the *areas* (NB not heights!) of rectangles.

Example 6.4: To illustrate the difference between a barchart and a histogram, the plots below show the frequency distribution of annual counts of intense typhoons in the north-west Pacific from 1972–2000; and the relative frequency distribution of monthly rainfall totals at Kew Gardens from 1900 to 1987. Typhoon counts are discrete and hence displayed using a barchart; rainfall totals are continuous and hence displayed using a histogram:



6.2 Numerical summaries of data

The previous section dealt with the presentation of data. It is often useful to condense large quantities of numeric data into a few numbers that preserve the main features of a dataset. Any function of the data is called a STATISTIC. Some important statistics are:

6.2.1 Measures of location

Just as the expected value of a random variable is a measure of location, so we can define similar measures for samples. They stand in the same relation to the histogram (barchart) of the data as do measures of location for random variables to the probability density (mass) function.

Suppose we have n data values $x_1, \dots, x_n$. If we arrange them into ascending order we get the ordered sample $x_{(1)}, \dots, x_{(n)}$. The following measures are often used:

1. The SAMPLE (ARITHMETIC) MEAN: $\bar{x} = (\sum_{i=1}^n x_i) / n$.
2. The SAMPLE MEDIAN, which is the value such that the same number of values lie above as below it. If n is odd and equal to $2m + 1$, the median is $x_{(m+1)}$. If n is even, the median is $\frac{1}{2} (x_{(m)} + x_{(m+1)})$.

The upper and lower QUARTILES of the data divide each half of the data in half again. If n is divisible by 4 ($n = 4k$ say), the lower quartile is $\frac{1}{2} (x_{(k)} + x_{(k+1)})$. Otherwise, it is $x_{(n/4+1)}$. Similarly for the upper quartile.

The median and quartiles are examples of PERCENTILES, which divide a dataset into 100 equal parts. The median is the 50th percentile; the quartiles are the 25th and 75th respectively. Rather than try and remember the formulae for calculating percentiles (including the median), think of the following graphical procedure: plot a graph that shows, for each value of x , the proportion of observations less than or equal to x . This graph will be a step function whose values increase from 0 to 1. To find the percentiles, draw horizontal lines across at the appropriate level. If the level corresponds to the horizontal part of a 'step', take the mid-point as the appropriate percentile. This graph may be called an EMPIRICAL (CUMULATIVE) DISTRIBUTION FUNCTION or an OGIVE.

The mean and median are close when the histogram (barchart) of the data is approximately symmetric (we say the data have an approximately symmetric DISTRIBUTION). The median is less sensitive to OUTLIERS (i.e. isolated large or small values which are not representative of the remainder of the dataset). If the data consist of many small values and a few large ones, the distribution is said to be RIGHT SKEWED or POSITIVELY SKEWED, and the mean will be greater than the median. Similarly, for LEFT (or NEGATIVELY) SKEWED data, the mean is less than the median:

6.2.2 Measures of dispersion

You've probably realised by now that we're defining statistics (recap: functions of the data) that correspond to the same ideas that we've already used in the study of random variables. So for measures of dispersion we have

1. The SAMPLE VARIANCE: $s^2 = (\sum_{i=1}^n (x_i - \bar{x})^2) / (n-1)$, or equivalently $[\sum_{i=1}^n x_i^2 - n\bar{x}^2] / (n-1)$. This may be greatly affected by outliers, and is most appropriate as a descriptive measure when the data have an approximately symmetric distribution. Note the use of the divisor $n - 1$. The reason for this will become clear later.
2. The SAMPLE STANDARD DEVIATION, s , is the square root of the sample variance.

Other measures are

1. The RANGE: $x_{(n)} - x_{(1)}$. Not very useful — typically it gets bigger as n increases!
2. The INTERQUARTILE RANGE i.e. difference between the upper and lower quartiles. It is less sensitive to outliers than the range.

6.3 Probability models

Often an experiment is repeated n times and we wish to analyse the results in a meaningful way. In Statistics, we regard the observations $x_1, \dots, x_n$ as the realised values of random variables $X_1, \dots, X_n$ (e.g. crop yields on n different plants, times to failure of n pieces of electronic equipment). If the conditions of the experiment are the same each time, we may be prepared to assume that the sample space and probability function are also the same, and therefore that $X_1, \dots, X_n$

all have the same probability distribution. If, in addition, the results of each experiment are not influenced in any way by the others, then $X_1, \dots, X_n$ can be assumed to be mutually independent in which case they are iid (see page 85).

Often we will not know the probability distribution of the X s. However, there may be good reasons (e.g. knowing the kind of process which generated the data, or from physical theory) to assume a particular form. In this latter case, typically we will not know the parameters of the distribution. For example, we may assume that the X s have a geometric distribution with unknown parameter p , or a normal distribution with unknown mean μ and variance σ^2 . Such assumptions constitute a PROBABILITY MODEL for the data (which may or may not be reasonable!).

Given the observations $X_1 = x_1, \dots, X_n = x_n$, we may want to make inferences or predictions. For example, what is the crop yield that we can expect for similar plots; what is the chance that a (similar) piece of equipment will fail within a prespecified guarantee period? In other words, we want to deduce properties of the distribution of the X s.

In a more complicated situation we might want to know if one variety of crop gives a higher average yield than another, with model

Variety 1		Variety 2
Yields: $X_1, \dots, X_n$		Yields: $Y_1, \dots, Y_n$
IID	$\leftarrow$ independent $\rightarrow$	IID
Expectation μ_X		Expectation μ_Y

In terms of the properties of the distributions, the question here is: is $\mu_X > \mu_Y$?

Alternatively, we might want to know how the application of fertiliser affects crop yield, with model

	Plot 1	...	Plot n
Fertiliser quantity	d_1	...	d_n
Yield	X_1	...	X_n
Expected yield is	$\mu_X(d) = \alpha + \beta d + \gamma d^2$		

where $\beta > 0$, $\gamma < 0$. What value of d gives the maximum expected yield? (If we know β and γ , this is straightforward — it's $-\beta/2\gamma$).

To answer such questions, we may want to estimate parameters (or a function of parameters) of a model, or to test hypotheses about them. In the above examples, we want to estimate μ_X , to test the hypothesis $\mu_X = \mu_Y$ and to estimate $-\beta/2\gamma$. In general, the methods and the results will depend on the model that is assumed. The more specific and detailed the model, the more accurate the results will be. But if the assumed model is incorrect the results may be disastrously misleading. Therefore there is a trade-off between the complexity of the model and its robustness to mis-specification.

6.4 Estimation

If $X_1, \dots, X_n$ are IID with unknown expectation $E(X) = \mu$, then an obvious estimate of μ is the sample mean $\bar{x}$. The random variable $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ is an ESTIMATOR of μ ; its observed value $\bar{x}$ is an ESTIMATE. We haven't had a definition in a box for a while:

Definition: A statistic (recall: function of the data) $T_n = T_n(X_1, \dots, X_n)$ is an ESTIMATOR of a parameter θ if its value $t_n = T_n(x_1, \dots, x_n)$ is used as an estimate of θ .

Since the X s are random variables, so is any statistic. Thus any estimator T_n has a probability distribution. The properties of this distribution determine whether or not T_n is a 'good' estimator.

Definition: If $E(T_n) = \theta$ then T_n is an UNBIASED estimator of θ . The difference $b(T_n, \theta) = E(T_n) - \theta$ is the BIAS of T_n as an estimator of θ .

An unbiased estimator will give us the right value 'on average' i.e. in a long series of experiments. Of course, this isn't much use if we can only do an experiment once! We would like to be fairly sure that an estimator produces an estimate which is close to the actual value of θ . This will happen if T_n is unbiased and has a small variance, or equivalently a small standard deviation. This suggests that we might wish to study the standard deviation of T_n at some stage:

Definition: The distribution of an estimator is called its SAMPLING DISTRIBUTION. The STANDARD ERROR of an estimator is the standard deviation of this sampling distribution.

This sounds a bit pedantic! The term *standard error* makes sense, however, in the context of estimation: any deviation of the estimator from the true value of θ is an error, and the terminology serves to emphasise this interpretation.

Definition: If T_n is unbiased for θ and $\text{Var}(T_n) \rightarrow 0$ as $n \rightarrow \infty$, then for any $\epsilon > 0$, $P(|T_n - \theta| < \epsilon) \rightarrow 1$ as $n \rightarrow \infty$. Such an estimator is CONSISTENT for θ .

Example 6.5: $X_1, \dots, X_n$ are iid variables with mean μ and variance $\sigma^2 < \infty$. Show that the 'obvious' estimator $\bar{X}_n$ is unbiased and consistent for μ .



You might be forgiven for thinking that unbiased estimators always exist. Surprisingly, they don't. It can be shown that, if the X s are iid Bernoulli variables with parameter p , and we set $\theta = 1/p$, no unbiased estimator of θ exists (despite that fact that $\bar{X}_n$ is consistent for p , according to example 6.5 above). The reason for this is that both the bias and the variance of T_n may depend on θ , and so strictly should be regarded as functions of θ . An unbiased estimator must have zero bias no matter what the value of θ is. The lack of an unbiased estimator simply means that no function of the data has an expectation that varies linearly with θ .

Unbiased estimators may not be unique — for example, if the data come from a normal distribution with mean μ then both the sample mean and sample median are unbiased estimators of μ (proof for the median omitted — we already know that the result holds for the mean). So, given two unbiased estimators, how do we choose between them? It seems reasonable to say 'we want the estimator that minimizes our chances of obtaining an estimate far from the true value'. One way of achieving this is via the criterion of mean square error:

Definition: The MEAN SQUARE ERROR of an estimator T for a parameter θ is defined as

$$\text{MSE}(T) = E[(T - \theta)^2]$$

In this definition, T may be biased. Suppose it has expectation $E(T)$. Then

$$\begin{aligned} \text{MSE}(T) &= E[(T - E(T) + E(T) - \theta)^2] \\ &= E[(T - E(T))^2 + 2(T - E(T))(E(T) - \theta) + (E(T) - \theta)^2] \\ &= \text{Var}(T) + 2(E(T) - \theta)E(T - E(T)) + (E(T) - \theta)^2 = \text{Var}(T) + b^2(T, \theta). \end{aligned}$$

It is seen that there is a tradeoff, in terms of MSE, between bias and variance. If we concentrate on unbiased estimators then the smallest possible MSE is achieved by the estimator with minimum variance. However, by considering estimators with a small amount of bias, we may find something that has a much smaller variance than the best unbiased estimator, so that we can reduce the MSE. In this case we would prefer to use the biased estimator, since we are more certain that the resulting estimate will be close to the true value of θ .

There are various ways of finding estimators with good properties — and of determining whether any particular estimator can be improved upon. These are covered in advanced Statistics courses.

So far, our discussion of estimation has only considered the case where we give a single number as our estimate of θ . Of course, since the estimate is a realised value of a random variable it is unlikely to be exactly equal to θ (in fact, in most cases the probability of our being right is zero — **Exercise:** why?). In this case, we might wish to give a *range* of θ values as our estimate. Such a range is called an INTERVAL ESTIMATE, as opposed to the POINT ESTIMATES we have been considering up to now. Typically, if the variance of our estimator is small then we will be reasonably sure that our point estimate is close to θ , and will therefore our interval estimate will be rather short. On the other hand, if the variance of the estimator is large then our interval estimate will be longer. Thus the length of the interval tells us something about the precision of the estimator. Viewed in another way, it tells us about the *uncertainty* surrounding our conclusions (recall, from the beginning of the course, that Statistics can be viewed as the interpretation of information in the face of uncertainty).

In some ways related to the idea of interval estimation is that of HYPOTHESIS TESTING. Suppose we wish to test whether the yields of two crop varieties differ, as in the example on page 98. In this case, we might proceed by saying ‘I wish to test the hypothesis that $\mu_X - \mu_Y = 0$ ’ (i.e. that there’s no difference between the yields). A sensible way to go about this would be to set $\theta = \mu_X - \mu_Y$ and construct an interval estimate of θ that accurately reflects our uncertainty. If the value zero lies within our interval, we must conclude that it is a plausible value of θ , and therefore we cannot argue convincingly that there is any difference between the varieties.

In the rest of the course, we have time to consider only some very specific (but widely used) statistical methods of estimation and hypothesis testing, based on normally distributed variables.

7 Normal probability models

In the last chapter, we discussed some general issues of estimation in statistics. We now illustrate some of the concepts in the special, but very important, case when the data values are assumed to come from one or more normal distributions. Before we get started however, it's useful to collect together a few useful results and definitions.

7.1 Distributions associated with the normal

7.1.1 The chi-squared distribution

Definition: If $Z_1, \dots, Z_n$ are independent $N(0, 1)$ random variables, the distribution of the sum of squares $\sum_{i=1}^n Z_i^2$ is called a CHI-SQUARED DISTRIBUTION WITH n DEGREES OF FREEDOM. We write $\sum_{i=1}^n Z_i^2 \sim \chi_n^2$.

In Example 3.8, we saw that if we square a single $N(0, 1)$ random variable, the result is distributed as $\Gamma(1/2, 1/2)$. So the χ_1^2 distribution is the same as $\Gamma(1/2, 1/2)$, and the χ_n^2 distribution is the same as that of a sum of n independent $\Gamma(1/2, 1/2)$ random variables. These each have MGF $(1 - 2t)^{-1/2}$, so their sum has MGF $(1 - 2t)^{-n/2}$, which is the same as the MGF of a $\Gamma(n/2, 1/2)$ distribution. So we have shown the following

Result: The χ_n^2 distribution is the same as the $\Gamma(n/2, 1/2)$ distribution. In particular, it has expectation n and variance $2n$.

A consequence of the definition above is that if $X_1, \dots, X_n$ are independent $N(\mu, \sigma^2)$ random variables then

$$\sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 \sim \chi_n^2.$$

7.1.2 The t distribution

Definition: If $Z \sim N(0, 1)$ and $U \sim \chi_n^2$, and Z and U are independent, then the distribution of the ratio $T = Z/\sqrt{U/n}$ is called the (STUDENT'S) t DISTRIBUTION WITH n DEGREES OF FREEDOM. We write $T \sim t_n$.

We will not derive any properties of the t distribution here. It is worth knowing the following, however:

- The density is an even function.
- For $n = 1$, the density is given by $f(t) = [\pi(1+t^2)]^{-1}$, $t \in \mathbb{R}$. This is the density of the CAUCHY DISTRIBUTION, which is one of the most awkward continuous distributions in probability and statistics. Its mean does not exist.
- For $n > 1$, the mean of the distribution exists and is zero.
- As $n \rightarrow \infty$ the t_n distribution tends to the standard normal. In fact, for $n > 30$ the t_n distribution can be approximated well using $N(0, 1)$.

To do probability calculations involving t -distributions, we use tables of their percentage points.

7.1.3 The F distribution

Definition: If U and V are independent random variables distributed as χ_m^2 and χ_n^2 respectively, the distribution of the ratio

$$W = \frac{U/m}{V/n}$$

is called the F -DISTRIBUTION WITH m AND n DEGREES OF FREEDOM. We write $W \sim F_{m,n}$.

Again, we will not derive any properties of this distribution. For $n > 2$, the mean of the distribution exists and is equal to $n/(n-2)$. Note that this does not depend on m .

As with the t -distribution, probability calculations involving F -distributions are usually done using tables. The people who compile tables have noticed that if $W \sim F_{m,n}$ then $1/W \sim F_{n,m}$ by definition — hence there is no need to tabulate *both* $F_{m,n}$ and $F_{n,m}$. This does mean that sometimes you have to translate a probability statement about W into a statement about $1/W$ before you can use the tables, however . . .

We are now in a position to return to problems of estimation.

7.2 One sample problem — known σ

Suppose we observe $x_1, \dots, x_n$, which are regarded as realised values of IID random values $X_1, \dots, X_n$ that are assumed to be normally distributed. This assumption can be checked in practice using informal graphical methods — e.g. examine the histogram of the data and see if it looks like a normal density, but be aware that there may be quite a lot of random fluctuation if n is small. Formal methods for checking normality exist, but will not be covered in this course.

Since all the X s are assumed to be IID, they all have the same mean and variance (μ and σ^2 say). Once these two parameters are known the common distribution of the X s is completely specified. Of course we don't know them, we only know the observed values $x_1, \dots, x_n$. The obvious (and, in fact, optimal from a variety of viewpoints) estimator of μ is $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$, with observed value $\bar{x}$. Now we know from Chapter 5 that a linear combination of normally distributed random variables is itself normally distributed, and that $\bar{X}_n$ has mean μ and variance σ^2/n . So $\bar{X}_n$ has sampling distribution $N(\mu, \sigma^2/n)$, and its standard error is $\sigma/\sqrt{n}$ (we will write $s.e.(\bar{X}_n) = \sigma/\sqrt{n}$). We see that it is consistent for μ .

If we assume that σ^2 is known (not very realistic, but let's keep things simple for the moment) then we know the sampling distribution of $\bar{X}_n$. We can use this to construct an interval estimate for μ , as discussed at the end of the last chapter, in a rather elegant way. The idea is to choose the interval in such a way that the probability of our interval excluding μ is equal to some rather small value, say α . Since the sampling distribution of $\bar{X}_n$ is symmetric, we have no reason for preferring one side of the estimate to the other and our calculated interval should also be symmetric. We therefore seek to define an interval $(\bar{X}_n - c, \bar{X}_n + c)$ such that

$$P(|\bar{X}_n - \mu| < c) = P\left(\left|\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}\right| < \frac{c}{\sigma/\sqrt{n}}\right) = P\left(|Z| < \frac{c}{\sigma/\sqrt{n}}\right) = 1 - \alpha,$$

where Z , as usual, denotes a standard normal random variable.

From the sketch, we see that $c\sqrt{n}/\sigma$ is the point at which $P(Z > c\sqrt{n}/\sigma) = \alpha/2$. Call this value $z_{\alpha/2}$ (it can be read from tables). Then $c = z_{\alpha/2}\sigma/\sqrt{n}$. Thus we can now give the following

Definition: The random interval

$$\left[\bar{X}_n - \frac{\sigma}{\sqrt{n}}z_{\alpha/2}, \bar{X}_n + \frac{\sigma}{\sqrt{n}}z_{\alpha/2}\right] = \bar{X}_n \pm (z_{\alpha/2} \times s.e.(\bar{X}_n))$$

is called a $100(1 - \alpha)\%$ CONFIDENCE INTERVAL for μ .

The interpretation is that, if we calculate many confidence intervals in this way from different sets of data drawn from the same distribution, in the long run we expect $100(1 - \alpha)\%$ of them to

contain μ . Notice that it is the *intervals* which vary between calculations, not μ . Notice also that, as n increases, the length of a confidence interval decreases for fixed α , reflecting the increased precision of the estimator for large sample sizes. The confidence interval approach is therefore a good way to convey information about uncertainty.

In practice, two commonly-used values of α are 0.05 (leading to 95% confidence intervals) and 0.01 (leading to 99% intervals). It may be useful to learn the corresponding $z_{\alpha/2}$ values, to save having to look them up in tables:

α	$z_{\alpha/2}$
0.05	1.96
0.01	2.57

7.2.1 Hypothesis testing

In some circumstances we will be less interested in estimating μ than in testing whether or not the data are consistent with a hypothesis that μ takes some prespecified value μ_0 . If we decide the data are *not* consistent with the value of μ being μ_0 , then we must accept that μ lies in some other set (e.g. that $\mu > \mu_0$, or $\mu \neq \mu_0$).

Definition: An assumption about a parameter, which we wish to test on the basis of available data, is called a NULL HYPOTHESIS and is denoted by H_0 . If the data are not deemed to support H_0 then we will conclude that an ALTERNATIVE HYPOTHESIS H_1 is supported.

If we know in advance (**NB** ‘in advance’ means *before seeing the data!*) that μ cannot possibly be less than (greater than) μ_0 then our alternative hypothesis will be $\mu > \mu_0$ ($\mu < \mu_0$). Occasions where we know things like this are rare in practice, however, and most of the time we will use the alternative $H_1 : \mu \neq \mu_0$.

A hypothesis is tested using a TEST STATISTIC, T say, chosen in such a way that

1. Its distribution under H_0 is known;
2. ‘Large’ values of $|T|$ indicate that the data are inconsistent with H_0 .

In the case where $X_1, \dots, X_n$ are iid $N(\mu, \sigma^2)$, σ is known and we wish to test

$$H_0 : \mu = \mu_0 \quad \text{against} \quad \begin{cases} \mu > \mu_0 \\ \mu \neq \mu_0 \\ \mu < \mu_0 \end{cases},$$

a suitable test statistic is

$$Z_n = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}}$$

with realised value z . Intuitively, this will tend to give ‘small’ values (i.e. values close to 0) when H_0 is true and ‘large’ values otherwise. Since $\bar{X}_n \sim N(\mu, \sigma^2/n)$, we find that

$$Z_n = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}} = \frac{\bar{X}_n - \mu + \mu - \mu_0}{\sigma/\sqrt{n}} = Z + \frac{\mu - \mu_0}{\sigma/\sqrt{n}} = Z + \sqrt{n} \left(\frac{\mu - \mu_0}{\sigma} \right),$$

where Z is a standard normal random variable. So, if H_0 is true, Z_n has a standard normal distribution. Therefore a natural way to see whether the data are consistent with H_0 is to calculate the probability that a standard normal random variable exceeds z .

Definition: The OBSERVED SIGNIFICANCE LEVEL of a test is the probability of obtaining a value of the test statistic at least as extreme as that observed, under H_0 . An alternative name for this is the p -VALUE of the test.

If the p -value is less than some prespecified ‘small’ value α (which we are free to choose) then we say the data are NOT CONSISTENT with the null hypothesis. How do we choose α ? Presumably in such a way that we don’t reject the null hypothesis too often when it’s actually true:

Definition: If H_0 is true and we reject it, we have made a TYPE 1 ERROR. If H_0 is false and we accept it, we have made a TYPE 2 ERROR.

From our construction of the test, we see that α is the probability of making a Type 1 error. Clearly we want to make this as small as possible. This is easy to do — we say ‘only reject H_0 if Z_n is infinite’, which gives us zero probability of making a Type 1 error! Of course, in this case the probability of making a Type 2 error is always 1, so this isn’t very helpful . . . we see that there must be a trade-off between the probabilities of the two types of errors. In practice, we often set $\alpha = 0.05$ or 0.01 .

Once we have chosen α , can we control the probability of making a Type 2 error (usually denoted by β)? It should be clear that this probability will depend on the true value of μ — the larger the difference $|\mu - \mu_0|$, the greater the probability that Z_n will fall within the REJECTION REGION of the test. If $|\mu - \mu_0|$ is relatively small, it may be that the Type 2 error rate is quite high.

However, from previous calculations we see that in general,

$$Z_n = Z + \sqrt{n} \left(\frac{\mu - \mu_0}{\sigma} \right) \Rightarrow Z_n \sim N \left(\sqrt{n} \left(\frac{\mu - \mu_0}{\sigma} \right), 1 \right).$$

It is clear from this that, whatever the value of $\mu \neq \mu_0$, the distribution of Z_n can be shifted as far away from zero as we like simply by increasing n (since the mean varies but the variance is constant). So we can control the Type 2 error rate by varying the sample size n .

Definition: The quantity $1 - \beta$ is called the **POWER** of a statistical test. It measures the test's ability to detect a departure from H_0 when it exists.

Since β depends on the true value of μ for any fixed n , so does the power. However, in applications we will usually have an idea of the magnitude of any deviation from μ_0 that is large enough to be of practical interest. This tells us how large a sample we need: we specify, in advance, the value of α , the magnitude of any difference that we consider to be practically important, and the desired probability of detecting a difference this large. Frequently this has to be balanced against cost considerations, but it gives you the general idea of how a statistician would go about designing an experiment.

Connection with confidence intervals: If you think about it, the **CRITICAL VALUE** of z in the test procedure above (i.e. the value below which we accept H_0 and above which we reject it) is $z_{\alpha/2}$, the quantity introduced during the calculation of confidence intervals. In fact, a little manipulation shows that if we carry out an α -level test for of $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$ for any value of μ_0 in the range

$$\left[\bar{x}_n - \frac{\sigma}{\sqrt{n}} z_{\alpha/2}, \bar{x}_n + \frac{\sigma}{\sqrt{n}} z_{\alpha/2} \right],$$

the null hypothesis is accepted; for any value of μ_0 outside this range, H_0 is rejected. This gives us an alternative interpretation of a $100(1 - \alpha)\%$ confidence interval: it is the set of values $\{\mu_0\}$ such that the null hypothesis $H_0 : \mu = \mu_0$ is accepted by an α -level test.

1-tailed tests: The discussion so far has focused exclusively on the commonest situation when the alternative hypothesis is $H_1 : \mu \neq \mu_0$. If we consider instead the **1-SIDED ALTERNATIVES** $H_1 : \mu > \mu_0$ then the appropriate significance level of the test will be $P(Z > z)$ rather than $P(|Z| > |z|)$, since only positive values of the test statistic would provide evidence against H_0 :

Any negative values, however large, can only have occurred by chance under this alternative hypothesis — if you're not entirely convinced that this is the case in any application, you should be using a 2-tailed test!

A similar modification is used for the other 1-tailed alternative $H_1 : \mu < \mu_0$.

Time for an example:

Example 7.1: The mean breaking strength of a certain type of cord has been established from long experience to be 18.3 oz, with a standard deviation of 1.2 oz. A new machine is introduced for manufacturing the cord. 100 lengths of the output from this new machine are tested to destruction, and found to have an average breaking strength of 17.9 oz. If it is reasonable to assume that the breaking strengths are normally distributed with the same standard deviation, is there any evidence that the new machine is fundamentally different from the old one? Calculate a 95% confidence interval for the true mean breaking strength of cord produced by the new machine.



7.3 One sample problem — σ unknown

The assumption, in the previous section, that σ is known is clearly unrealistic. In practice we can estimate σ^2 using the sample variance s^2 , which was introduced in Chapter 6:

$$\hat{\sigma}^2 = s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2 = \frac{1}{n-1} \left[\sum_{i=1}^n X_i^2 - n\bar{X}_n^2 \right].$$

The ‘hat’ symbol on $\hat{\sigma}^2$ is often used in Statistics to denote an estimate.

Can we say anything about the distribution of s^2 ? Well, recall from the beginning of this Chapter that

$$\sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 \sim \chi_n^2.$$

The left hand side here looks a bit like the definition of s^2 ; however, in the latter, we have $\bar{X}$ instead of μ . Let’s see what we can do with this. According to the definition, we have

$$\begin{aligned} (n-1) \frac{s^2}{\sigma^2} &= \sum_{i=1}^n \left(\frac{X_i - \bar{X}_n}{\sigma} \right)^2 = \sum_{i=1}^n \left(\frac{X_i - \mu + \mu - \bar{X}_n}{\sigma} \right)^2 \\ &= \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 - 2 \left(\frac{\bar{X} - \mu}{\sigma^2} \right) \sum_{i=1}^n (X_i - \mu) + \sum_{i=1}^n \left(\frac{\bar{X} - \mu}{\sigma} \right)^2 \end{aligned}$$

$$\begin{aligned}
&= \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 - 2n \left(\frac{\bar{X} - \mu}{\sigma} \right)^2 + n \left(\frac{\bar{X} - \mu}{\sigma} \right)^2 \\
&= \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 - \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \right)^2.
\end{aligned}$$

The first term here is distributed as χ_n^2 ; and the second as χ_1^2 (since we know that $\bar{X} \sim N(\mu, \sigma^2/n)$). Hence the expected value of $(n-1)s^2/\sigma^2$ is $n-1$. The expected value of s^2 is thus σ^2 , so it is unbiased as an estimator of σ^2 . This is why we use divisor $n-1$ instead of n .

Result: If $X_1, \dots, X_n$ are independent $N(\mu, \sigma^2)$ random variables,

$$\frac{(n-1)s^2}{\sigma^2} = \sum_{i=1}^n \left(\frac{X_i - \bar{X}_n}{\sigma} \right)^2 \sim \chi_{n-1}^2.$$

We will not prove this result here. Note from the above discussion, however, that it looks intuitively right — we showed that s_n^2 could be regarded as the difference between a χ_n^2 and a χ_1^2 . If we are prepared to accept that s^2 and $\bar{X}_n$ are independent, the result follows immediately (this is true, but not so easy to prove!).

If σ^2 is unknown, and is estimated by s^2 , we may be interested in testing hypotheses about either μ or σ^2 . We deal with each of these in turn.

7.3.1 Inference about μ

Suppose we wish to test some hypothesis $H_0 : \mu = \mu_0$, when σ^2 is unknown. The obvious test statistic to use is

$$T_n = \frac{\bar{X}_n - \mu_0}{s/\sqrt{n}} = \frac{(\bar{X}_n - \mu_0) / (\sigma/\sqrt{n})}{s/\sigma}$$

i.e. to proceed as if σ^2 were known, and just replace it by its estimate s^2 . To test hypotheses about μ , we need to know the distribution of T_n . Note now that, if H_0 is true, the numerator on the right-hand-side here is distributed as $N(0, 1)$, and that the denominator is of the form $\sqrt{U/(n-1)}$, where $U \sim \chi_{n-1}^2$ (we know that $(n-1)s^2/\sigma^2 \sim \chi_{n-1}^2$). If you compare this with the definition given at the beginning of the chapter, you will see that T_n has a t -distribution with $n-1$ degrees of freedom.

We can now test hypotheses about μ when σ^2 is unknown. For example, suppose we wish to test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$ (the form of the alternative hypothesis here tells us that a 2-tailed test is required). We proceed as follows:

1. Determine the form of the test statistic T_n (given above).
2. Determine the distribution of T_n under H_0 (i.e. $T_n \sim t_{n-1}$).

3. Determine the Type 1 error rate, α .
4. Find, from tables, the value, $t_{\alpha/2}$ say, such that under H_0 , $P(|T_n| > t_{\alpha/2}) = \alpha$ (this is the step that depends upon the alternative hypothesis, in exactly the same way as when σ^2 was known).
5. Calculate the actual value of T_n for the observed data, and accept or reject H_0 as appropriate.

Similarly, a $100(1 - \alpha)\%$ confidence interval for μ in this case is

$$\left[\bar{X}_n - \frac{s}{\sqrt{n}} t_{\alpha/2}, \bar{X}_n + \frac{s}{\sqrt{n}} t_{\alpha/2} \right] = \bar{X}_n \pm (t_{\alpha/2} \times s.e.(\bar{X}_n))$$

(recall that a hat $\hat{\cdot}$ denotes an estimate). If you compare tables of the t -distribution with those for $N(0, 1)$, you will find that $t_{\alpha/2} > z_{\alpha/2}$, so that confidence intervals using the t -distribution are wider than those using the normal. This reflects the extra uncertainty that arises from not knowing σ^2 .

Example 7.2: Cans of peaches have a nominal weight of 400g. A sample of 12 cans is taken, and each of the 12 cans is weighed. The sample mean weight is 403g, and the sample variance is 1.4. The weights of individual cans can be assumed to be iid and normally distributed. At the 95% level, is there any evidence that the production process is off target (i.e. that the true mean weight differs from 400g)? Calculate a 95% confidence interval for the true mean weight.



7.3.2 Inference about σ

Now suppose that we wish to test a hypothesis, not about μ but about σ^2 (this can be important in manufacturing for example, where it is usually not desirable for output quality to be extremely variable). It seems reasonable to base any test for σ^2 upon the sample variance s^2 , particularly since we know that $(n-1)s^2/\sigma^2 \sim \chi_{n-1}^2$. In fact, the procedure is pretty much identical to those we have followed so far. The only difference arises from the fact that the χ^2 distributions are not symmetric. So: to test $H_0 : \sigma^2 = \sigma_0^2$ against the alternative $H_1 : \sigma^2 \neq \sigma_0^2$, we go

1. Determine the form of the test statistic:

$$\Psi_n = \frac{(n-1)s^2}{\sigma_0^2} = \frac{\sum_{i=1}^n (X_i - \bar{X}_n)^2}{\sigma_0^2}.$$

2. Determine the distribution of Ψ_n under H_0 (i.e. $\Psi_n \sim \chi_{n-1}^2$).
3. Determine the Type 1 error rate, α .
4. Find, from tables, the two values, ψ_1 and ψ_2 say, such that under H_0 , $P(\Psi_n < \psi_1) = P(\Psi_n > \psi_2) = \alpha/2$. The need to consider two separate values arises from the asymmetry of the chi-squared distributions:

As usual, this step depends upon the alternative hypothesis. If the alternative were $H_1 : \sigma^2 > \sigma_0^2$ then we would only reject H_0 for unusually large values of Ψ_n , exceeding the point ψ'_2 such that $P(\Psi_n > \psi'_2) = \alpha$.

5. Calculate the actual value of Ψ_n for the observed data, and accept or reject H_0 as appropriate.

To obtain confidence intervals for σ^2 , use the fact that a $100(1 - \alpha)\%$ interval is the set of all values accepted by a 2-tailed α -level test. Any value σ_0^2 will be accepted providing

$$\psi_1 < \frac{(n-1)s^2}{\sigma_0^2} < \psi_2 ,$$

with ψ_1 and ψ_2 defined as above. Rearranging the left-hand inequality will give an upper limit for σ^2 ; the right-hand one will give a lower limit.

Example 7.2 (continued): The variability in the canning process is supposed to be such that the weights of cans of peaches have a standard deviation of 2g. Is there any evidence that the true variability differs from this? Calculate a 95% confidence interval for the true standard deviation.



In the unlikely event that μ is known, of course inference about σ can be based directly upon the fact that, under H_0 ,

$$\frac{\sum_{i=1}^n (X_i - \mu)^2}{\sigma_0^2} \sim \chi_n^2.$$

7.4 2-sample problems

Suppose now that we have 2 independent samples:

$$\begin{aligned} X_1, \dots, X_{n_1} & \text{ are iid } N(\mu_1, \sigma_1^2) \\ Y_1, \dots, Y_{n_2} & \text{ are iid } N(\mu_2, \sigma_2^2) . \end{aligned}$$

There are several obvious hypotheses we may wish to test:

1. $H_0 : \mu_1 = \mu_2$. We consider three possible cases here:

- (a) σ_1^2 and σ_2^2 are both known.
- (b) $\sigma_1^2 = \sigma_2^2$, but the value is not known.
- (c) $\sigma_1^2 \neq \sigma_2^2$ and both values are unknown.

2. $H_0 : \sigma_1^2 = \sigma_2^2$. Here, we consider two possible cases:

- (a) μ_1 and μ_2 are both known.
- (b) μ_1 and μ_2 are not known.

Obviously, we could take a long time going through each of these cases individually! However, the general procedure is the same each time. First we consider the easiest case.

7.4.1 Testing equality of means, variances known

The hypothesis $H_0 : \mu_1 = \mu_2$ is equivalent to $H_0 : \mu_1 - \mu_2 = 0$. It is natural to estimate $\mu_1 - \mu_2$ using the difference in sample means $\bar{X}_{n_1} - \bar{Y}_{n_2}$. Now $\bar{X}_{n_1} \sim N(\mu_1, \sigma_1^2/n_1)$, and $\bar{Y}_{n_2} \sim N(\mu_2, \sigma_2^2/n_2)$. Since the X s and Y s are independent, and sums (equivalently, differences) of normal random variables are themselves normal, we therefore have

$$\bar{X}_{n_1} - \bar{Y}_{n_2} \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right),$$

so that the standard error of $\bar{X}_{n_1} - \bar{Y}_{n_2}$ is

$$s.e.(\bar{X}_{n_1} - \bar{Y}_{n_2}) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}.$$

Given this result, it is natural to base inference around the test statistic

$$Z = \frac{\bar{X}_{n_1} - \bar{Y}_{n_2}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}},$$

which has a standard normal distribution under H_0 . Hypothesis tests for the difference $\mu_1 - \mu_2$ can now be performed, and confidence intervals calculated exactly as before: a $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is

$$\bar{X}_{n_1} - \bar{Y}_{n_2} \pm (z_{\alpha/2} \times s.e.(\bar{X}_{n_1} - \bar{Y}_{n_2})).$$

7.4.2 Testing equality of means, variances unknown but equal

We now suppose that $\sigma_1^2 = \sigma_2^2 = \sigma^2$, say, but that the actual value is unknown. As in the 1-sample case, it seems sensible to proceed as though σ^2 were known, but to replace σ^2 by some estimate of it wherever necessary. At this point, we have to decide how to estimate σ^2 . Clearly we could estimate it using

$$s_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (X_i - \bar{X}_{n_1})^2, \quad \text{or using} \quad s_2^2 = \frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (Y_i - \bar{Y}_{n_2})^2,$$

since these are both unbiased estimators of σ^2 . More generally, however, we might hope to obtain a more accurate estimator by combining s_1^2 and s_2^2 in a weighted average of the form $as_1^2 + bs_2^2$, for some weights a and b . How should we choose these weights? Well, we would like our estimator to (a) be unbiased (b) have as small a variance as possible. The expected value of the weighted average is

$$E(as_1^2 + bs_2^2) = aE(s_1^2) + bE(s_2^2) = (a + b)\sigma^2,$$

since we know that s_1^2 and s_2^2 are both unbiased for σ^2 . Hence, if our combined estimator is to be unbiased for σ^2 we require $a + b = 1$ so that $b = 1 - a$.

The value of a can now be chosen so as to minimise the variance of the combined estimator, which is

$$\text{Var}(as_1^2 + (1 - a)s_2^2) = a^2\text{Var}(s_1^2) + (1 - a)^2\text{Var}(s_2^2).$$

Now we already know that $(n_i - 1)s_i^2/\sigma^2 \sim \chi_{n_i-1}^2$ ($i = 1, 2$), and that the variance of a χ_ν^2 distribution is 2ν . It follows that $\text{Var}(s_i^2) = [\sigma^2/(n_i - 1)]^2 \times 2(n_i - 1) = 2\sigma^4/(n_i - 1)$. So the variance of our combined estimator is

$$2\sigma^4 \left(\frac{a^2}{n_1 - 1} + \frac{(1 - a)^2}{n_2 - 1} \right).$$

This can be minimised by differentiating with respect to a and setting to zero (it's quadratic in a , with a positive coefficient of a^2 , so any turning point is guaranteed to be a minimum). We find that

$$a = \frac{n_1 - 1}{n_1 + n_2 - 2}, \quad \text{whence} \quad b = 1 - a = \frac{n_2 - 1}{n_1 + n_2 - 2}.$$

We therefore have

Definition: Given two independent samples $X_1, \dots, X_{n_1}$ and $Y_1, \dots, Y_{n_2}$, from normal distributions with unknown means and unknown but equal variances, the POOLED ESTIMATOR of the unknown variance σ^2 is

$$s_P^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{\sum_{i=1}^{n_1} (X_i - \bar{X}_{n_1})^2 + \sum_{i=1}^{n_2} (Y_i - \bar{Y}_{n_2})^2}{n_1 + n_2 - 2}.$$

Using the χ^2 distributional results for s_1^2 and s_2^2 , together with the fact that the two estimators are independent (since the X s and Y s are), we find that

$$\frac{(n_1 + n_2 - 2) s_P^2}{\sigma^2} \sim \chi_{n_1 + n_2 - 2}^2.$$

We can now return to the problem of making inferences, or testing hypotheses, about $\mu_1 - \mu_2$. In the previous section such inferences were based on the test statistic

$$Z = \frac{\bar{X}_{n_1} - \bar{Y}_{n_2}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}.$$

If we replace the σ^2 s (which are now equal, but unknown) by their estimator s_P^2 , we get a test statistic

$$T = \frac{\bar{X}_{n_1} - \bar{Y}_{n_2}}{s_P \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{(\bar{X}_{n_1} - \bar{Y}_{n_2}) / \left(\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \right)}{s_P / \sigma}.$$

The numerator here has a standard normal distribution; and the denominator is of the form $\sqrt{U/\nu}$, where $\nu = n_1 + n_2 - 2$ and U is distributed as χ_ν^2 . Hence in this case the test statistic T has a t distribution with $(n_1 + n_2 - 2)$ degrees of freedom. This can now be used to carry out tests and calculate confidence intervals for $\mu_1 - \mu_2$. A $100(1 - \alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is now

$$\bar{X}_{n_1} - \bar{Y}_{n_2} \pm \left(t_{\alpha/2} \times s_P \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \right) = \bar{X}_{n_1} - \bar{Y}_{n_2} \pm (t_{\alpha/2} \times s.e.(\bar{X}_{n_1} - \bar{Y}_{n_2})).$$

7.4.3 Testing equality of means, variances unknown and unequal

In the case where $\sigma_1^2 \neq \sigma_2^2$, it might appear that things will actually be *more* straightforward than they were in the section above, because we don't have to worry about obtaining a pooled estimate of the common variance. The natural test statistic for testing hypotheses about $\mu_1 - \mu_2$ is

$$T = \frac{\bar{X}_{n_1} - \bar{Y}_{n_2}}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{(\bar{X}_{n_1} - \bar{Y}_{n_2}) / \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} / \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

(i.e. replace the σ^2 s in the Z -statistic by their estimates). However, the re-arrangement on the right-hand side here reveals a problem: the numerator is, as usual, a standard normal random variable, but the denominator is no longer the square root of a scaled chi-squared. Therefore this 'obvious' test statistic does not have a t distribution. In fact, the distribution of T here depends on both σ_1^2 and σ_2^2 (which are unknown), so that it is not a suitable test statistic.

To devise a test in this situation is a classical problem (the *Behrens-Fisher problem*) in Statistics. There are various resolutions, but not in this course. As a result, it would be good if the data were consistent with the hypothesis $\sigma_1^2 = \sigma_2^2$. We may wish to test this.

7.4.4 Testing equality of variances

Suppose we wish to test $H_0 : \sigma_1^2 = \sigma_2^2$, or equivalently $H_0 : \sigma_1^2/\sigma_2^2 = 1$. If μ_1 and μ_2 are unknown, we may base tests and confidence intervals around the ratio

$$F = \frac{s_1^2}{s_2^2} = \frac{\sigma_1^2 U / (n_1 - 1)}{\sigma_2^2 V / (n_2 - 1)}, \text{ say,}$$

where U and V are independent χ^2 random variables with degrees of freedom $n_1 - 1$ and $n_2 - 1$ respectively. Under H_0 , the σ^2 s cancel out and we have

$$F = \frac{U / (n_1 - 1)}{V / (n_2 - 1)} \sim F_{n_1 - 1, n_2 - 1},$$

from the definition of the F distribution at the beginning of this Chapter. Hence, if we can find the appropriate percentage points of this F distribution (from tables), we can decide whether the data are consistent with H_0 .

If μ_1 and μ_2 are known, our sample variance estimates will be

$$\tilde{s}_1^2 = \frac{1}{n_1} \sum_{i=1}^{n_1} (X_i - \mu_1)^2 \quad \text{and} \quad \tilde{s}_2^2 = \frac{1}{n_2} \sum_{i=1}^{n_2} (Y_i - \mu_2)^2,$$

which are proportional to $\chi_{n_1}^2$ and $\chi_{n_2}^2$ variables respectively: in this case, the ratio $\tilde{s}_1^2/\tilde{s}_2^2$ is distributed as F_{n_1, n_2} under H_0 .

Example 7.3: The following data come from an experiment in which 30 “Standard” brand sparklers were burned, and the burning time (in seconds) of each sparkler was measured. Fifteen of the sparklers were held vertically upwards while burning (group 1), and fifteen were held vertically downwards (group 2). The sample mean and variance of the burning times for the two groups were as follows:

	Group 1	Group 2
Sample mean:	70.00	65.13
Sample variance:	21.43	14.55

Use the model $X_{ij} \sim N(\mu_i, \sigma_i^2)$ independently for all i, j , where $X_{i,j}$ is the burning time of the j th sparkler in group i ($i = 1, 2; j = 1, 2, \dots, 15$) to answer the following questions:

2. Are the data consistent with a model in which the variance is the same in both groups (i.e. $\sigma_1^2 = \sigma_2^2$)?

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7.4.5 Testing equality of means, two paired samples

Occasionally, we encounter situations where we apparently have two groups of data as in the previous sections, but where the X s and Y s cannot be regarded as independent. An important example of this occurs when the X s and Y s represent different measurements made upon the same subject or, for example, responses of twins to different drugs. Data that arise in this way (where for each X there is a corresponding Y which was obtained in near identical conditions) are called **PAIRED DATA**. If we wish to test hypotheses about differences between groups in this case, we cannot use the methods of the previous sections since they assumed that the X s and Y s are independent. However, we can reduce paired data to a 1-sample problem, by studying the differences between each pair of observations. If μ_D is the true mean difference, then the null hypothesis $H_0 : \mu_1 = \mu_2$ corresponds to $H_0 : \mu_D = 0$, and can be tested in the usual 1-sample way using the sample mean difference $\bar{d}_n$ say, and (where necessary) the sample variance of the differences s_D^2 . Note that this eliminates the need to worry about whether or not $\sigma_1^2 = \sigma_2^2$. However, it can only be done if there is an obvious pairing in the data.

Example 7.4: In an experiment to compare two methods for finding the percentage of iron in a compound, ten different compounds were analysed by both methods. The results were as follows:

Compound number:	1	2	3	4	5	6	7	8	9	10
Method A:	13.3	17.6	4.1	17.2	10.1	3.7	5.1	7.9	8.7	11.6
Method B:	13.4	17.9	4.1	17.0	10.3	4.0	5.1	8.0	8.8	12.0

Is there any significant difference between the two measurement methods?



Note, in this last example, that the actual difference between the two measurement methods was far smaller than the natural variability between compounds. If we had not used a paired test, the between-compound variability would have swamped this small difference.

7.5 Extension to non-normal data

The procedures in this Chapter have been derived under the assumption that the individual data values were drawn from 1 or more normal distributions. If you look back, however, you'll see that basically all of the techniques rely upon two facts: (i) sample means are normally distributed (ii) sample variances are scaled chi-squared random variables. Now we already know, from the Central Limit Theorem, that the mean of a large sample is approximately normally distributed regardless of the underlying distribution. It can also be shown that the chi-squared approximation holds for the sample variance regardless of the underlying distribution, providing the sample size is big enough. Therefore, for large samples the methods in this Chapter can still be used to a good degree of approximation. In practice, 'large' usually means 'greater than 30'.

8 Linear Regression

The final topic in the Statistics part of this course deals with the situation where we have measurements on two variables, and wish to determine whether there is any relationship between them. Specifically, suppose we have n pairs of observations $\{(x_i, y_i) : i = 1, \dots, n\}$, and that a plot of these data shows that the points are more or less scattered about a straight line. This relationship could be used to predict y from x at some future stage, in a situation where only x data are available. Such a situation may arise in a variety of ways:

1. x values may be observable prior to y values in time. For example, rainfall amounts in West Africa in September are linked to hurricane formation in the Atlantic Ocean between July and November the following year. Hence we could use knowledge of West African rainfall to predict hurricane activity 9–12 months in advance (this is, in fact, essentially how long-range Atlantic hurricane forecasts are derived). Another example of this kind of situation is where we want to administer some dose of a drug to a patient. If x denotes the dose, and y denotes some response (such as a measure of disease remission, or a measure of adverse reaction), then we will want to say ‘if I give dose x , what will the value of y be?’.
2. x values may be cheaper to obtain than y values. For example, the fat content of minced meat can only be determined precisely by laboratory analysis. However, if light is passed through a meat sample then the proportion of light absorbed at certain wavelengths can be a useful proxy measure of fat content. If we can use a laboratory experiment to develop a CALIBRATION RELATIONSHIP between fat content and light absorption (i.e. a relationship that will predict fat content from light absorption), then we can estimate fat content quickly and cheaply by passing light through a meat sample. Such a procedure may be suitable for online quality monitoring in the meat industry (this idea, in fact, forms the basis of *near infra-red spectroscopy*, which is widely used in a variety of industrial applications).
3. y values may be unobservable outside a laboratory. An example of this is in radioactive dating, where the age of an object (the y value) may be estimated by studying the amount, x , of some radioactive isotope present.

We now ask the question: exactly how should we fit a straight line to data $\{(x_i, y_i) : i = 1, \dots, n\}$? Clearly we want to find the ‘best possible’ line, but it’s not obvious what ‘best possible’ means. It’s tempting to think that we mean ‘the line which is as close as possible to all of the data points’ — i.e. minimise the average perpendicular distance from each data point to the line. In fact however, if we’re interested in *prediction* this isn’t the best line, because if we’re interested in predicting y from x then our ultimate goal is to minimise some measure of prediction error, and this suggests that we focus on the *vertical* distances of points from the line:

According to this diagram, if we use the line $y = b_0 + b_1x$ to describe the observed pattern, our data points satisfy the n equations

$$y_i = b_0 + b_1x_i + e_i \quad (i = 1, \dots, n),$$

where e_i is the prediction error for the i th point. It seems reasonable, therefore, to choose the values of b_0 and b_1 that minimise the sum of squared prediction errors, $\sum_{i=1}^n e_i^2$ (you may think it would make more sense to minimise the sum of *absolute* prediction errors $\sum_{i=1}^n |e_i|$, but we quickly run into mathematical difficulties if we try that!).

This procedure seems sensible, if a little *ad hoc*. We can make things a little more rigorous if we try to embed the procedure within the framework of a probability model. In most situations, if we have two x values that are the same the associated y values will differ (because of experimental error, other factors affecting the y s or whatever). Thus y_i can be regarded as the observed value of a random variable Y_i , whose distribution depends on x_i . In this situation, x is called the EXPLANATORY (or INDEPENDENT) VARIABLE and Y is called the RESPONSE (or DEPENDENT) VARIABLE. Bearing in mind the form of the relationship above, an appropriate model for such a situation might be

$$Y_i = \beta_0 + \beta_1x_i + \varepsilon_i,$$

where the $\{\varepsilon_i\}$ are independent random variables with zero mean and common variance σ^2 . An equivalent way of expressing this (which is the key to generalising the ideas in more complicated situations) is to write $E(Y_i) = \beta_0 + \beta_1x_i = \mu(x_i)$, say. If we also regard the x s as realised values of random variables $\{X_i : i = 1, n\}$, we are effectively modelling the *conditional distribution* of Y given $X = x$. The curve $\mu(x)$ as a function of x is the REGRESSION CURVE of Y on x .

Definition: A LINEAR (STATISTICAL) MODEL is one in which the regression curve is a linear function of the parameters in the model.

NB: the regression curve does *not* have to be a linear function of x in this model (although many people mistakenly think that it does). For example, the model $E(Y_i|x_i) = \beta_0 + \beta_1 \ln(x_i)$ is a linear model, because the regression curve is linear in β_0 and β_1 .

8.1 The Least Squares approach to model fitting

We now have a probability model for the Y 's, and the fitting of a straight line to data is seen to be a problem of estimating the parameters (β_0, β_1 and σ^2) of this model. We'll deal with β_0 and β_1 first, then see how we can estimate σ^2 .

As indicated above, a sensible way of estimating β_0 and β_1 is to choose values that minimise the sum of squared errors

$$S(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 .$$

This estimation method is called the LEAST SQUARES METHOD of model fitting. It is appropriate for data that are

- HOMOSCEDASTIC, i.e. the variation in Y is the same for all x (in terms of our model, σ^2 is constant);
- independent.

If the $\{\varepsilon_i\}$ are normally distributed random variables, Least Squares fitting is, from various viewpoints, optimal in terms of the inferences we can make about β_0 and β_1 .

To minimise $S(\beta_0, \beta_1)$ we differentiate and set to zero:

$$\begin{aligned} \frac{\partial S}{\partial \beta_0} &= -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = -2n(\bar{y} - \beta_0 - \beta_1 \bar{x}) \\ \text{and } \frac{\partial S}{\partial \beta_1} &= -2 \sum_{i=1}^n x_i (y_i - \beta_0 - \beta_1 x_i) , \end{aligned}$$

where $\bar{x}$ and $\bar{y}$ denote the sample means of the x s and the y s respectively. The least squares estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ therefore satisfy the NORMAL EQUATIONS:

$$\bar{y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{x} = 0 \quad \text{and} \quad \sum_{i=1}^n x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0 .$$

From the first of these, we see that $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$. Substituting this into the second equation yields $\hat{\beta}_1 = [\sum_{i=1}^n x_i (y_i - \bar{y})] / [\sum_{i=1}^n x_i (x_i - \bar{x})]$. In fact, however, the formula for $\hat{\beta}_1$ is not usually

given in this form (we'll see why shortly). Noting that $\sum_{i=1}^n \bar{x}(x_i - \bar{x}) = \sum_{i=1}^n \bar{x}(y_i - \bar{y}) = 0$, the least squares estimates are usually given in the following form:

Result: the least squares estimates $\hat{\beta}_1$ and $\hat{\beta}_0$ are

$$\hat{\beta}_1 = \frac{C_{xy}}{C_{xx}} \quad \text{and} \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x},$$

where

$$C_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = \left(\sum_{i=1}^n x_i^2 \right) - n\bar{x}^2$$

and

$$C_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \left(\sum_{i=1}^n x_i y_i \right) - n\bar{x}\bar{y}.$$

The minimum sum of squared errors is therefore

$$\begin{aligned} S(\hat{\beta}_0, \hat{\beta}_1) &= \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = \sum_{i=1}^n \left(y_i - \bar{y} + \frac{C_{xy}}{C_{xx}} \bar{x} - \frac{C_{xy}}{C_{xx}} x_i \right)^2 \\ &= \sum_{i=1}^n (y_i - \bar{y})^2 - 2 \frac{C_{xy}}{C_{xx}} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) + \left(\frac{C_{xy}}{C_{xx}} \right)^2 \sum_{i=1}^n (x_i - \bar{x})^2 \\ &= C_{yy} - 2 \frac{C_{xy}}{C_{xx}} C_{xy} + \left(\frac{C_{xy}}{C_{xx}} \right)^2 C_{xx} = C_{yy} - \frac{C_{xy}^2}{C_{xx}}, \end{aligned}$$

where $C_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2 = (\sum_{i=1}^n y_i^2) - n\bar{y}^2$. $S(\hat{\beta}_0, \hat{\beta}_1)$ is called the RESIDUAL SUM OF SQUARES (RSS).

8.2 Properties of Least Squares estimators

Now we will try to find some of the properties of $\hat{\beta}_0$ and $\hat{\beta}_1$. So far, all we have done is to derive formulae for the calculation of $\hat{\beta}_0$ and $\hat{\beta}_1$ from data. In order to study their properties, we need to remember that the calculated values are regarded as the realised values of random variables. The properties all depend upon the observed values of the x s, which are fixed. We will use slightly sloppy notation, since $\hat{\beta}_0$ and $\hat{\beta}_1$ were estimates before and now we're treating them as estimators.

First of all, we study $\hat{\beta}_1$: we have $E(\hat{\beta}_1) = (C_{xx})^{-1} E(C_{xY})$ and $\text{Var}(\hat{\beta}_1) = (C_{xx})^{-2} \text{Var}(C_{xY})$. We write C_{xY} rather than C_{xy} to emphasise the fact that we are now treating the Y s as random variables. C_{xx} can be treated as a constant, however, since the x s are fixed. To find the properties of

$\hat{\beta}_1$ therefore, we need to know about C_{xY} :

$$C_{xY} = \sum_{i=1}^n (Y_i - \bar{Y})(x_i - \bar{x}) = \sum_{i=1}^n Y_i(x_i - \bar{x}) - \bar{Y} \sum_{i=1}^n (x_i - \bar{x}) = \sum_{i=1}^n Y_i(x_i - \bar{x}) ,$$

since $\sum_{i=1}^n (x_i - \bar{x}) = 0$. Now we know that $E(Y_i) = \beta_0 + \beta_1 x_i$, and that $\text{Var}(Y_i) = E(\varepsilon_i^2) = \sigma^2$, so we have

$$\begin{aligned} E(C_{xY}) &= E \left[\sum_{i=1}^n Y_i(x_i - \bar{x}) \right] = \sum_{i=1}^n (x_i - \bar{x}) E(Y_i) = \sum_{i=1}^n (x_i - \bar{x})(\beta_0 + \beta_1 x_i) \\ &= \beta_0 \sum_{i=1}^n (x_i - \bar{x}) + \beta_1 \sum_{i=1}^n x_i(x_i - \bar{x}) = \beta_1 \sum_{i=1}^n (x_i - \bar{x})^2 . \end{aligned}$$

Similarly, we have

$$\text{Var}(C_{xY}) = \sum_{i=1}^n (x_i - \bar{x})^2 \text{Var}(Y_i) = \sigma^2 \sum_{i=1}^n (x_i - \bar{x})^2 = \sigma^2 C_{xx} , \text{ whence}$$

$$E(\hat{\beta}_1) = \frac{1}{C_{xx}} E(C_{xY}) = \beta_1 \quad \text{and} \quad \text{Var}(\hat{\beta}_1) = \frac{1}{C_{xx}^2} \text{Var}(C_{xY}) = \frac{\sigma^2}{C_{xx}} .$$

To find the properties of $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$, we need the properties of $\bar{Y} = n^{-1} \sum_{i=1}^n (\beta_0 + \beta_1 x_i + \varepsilon_i)$:

$$\begin{aligned} E(\bar{Y}) &= E(\beta_0 + \beta_1 \bar{x} + \bar{\varepsilon}) = \beta_0 + \beta_1 \bar{x} , \\ \text{and} \quad \text{Var}(\bar{Y}) &= \text{Var}(\beta_0 + \beta_1 \bar{x} + \bar{\varepsilon}) = \text{Var}(\bar{\varepsilon}) = \frac{\sigma^2}{n} . \end{aligned}$$

Hence

$$E(\hat{\beta}_0) = E(\bar{Y}) - \beta_1 \bar{x} = \beta_0 \quad \text{and} \quad \text{Var}(\hat{\beta}_0) = \text{Var}(\bar{Y}) + \bar{x}^2 \text{Var}(\hat{\beta}_1) = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{C_{xx}} \right] .$$

In fact, we cheated a bit on the variance argument there — it only works if $\bar{Y}$ and $\hat{\beta}_1$ are uncorrelated (this goes way back to Section 4.1.2!). It can be shown that they *are*, and therefore the argument is OK. We won't prove it here though.

These results show that $\hat{\beta}_0$ and $\hat{\beta}_1$ are both unbiased estimators. Providing their variances tend to zero as $n \rightarrow \infty$, they will also be consistent. This will occur if $C_{xx} \rightarrow \infty$ with n , which is guaranteed providing the x values are not all the same (recall that $C_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$).

Note also that in general, large values of C_{xx} will give rise to small variances and hence precise estimates of β_0 and β_1 . In some situations we might be able to control the values of C_{xx} — for example in a laboratory experiment, we may be free to choose our x values. Since C_{xx} is effectively a measure of the variability in the x s, this tells us that we should choose the x s to be as variable as possible (subject to the constraints of the experiment, of course!). You might therefore conclude that we should put half the observations at each end of the x range (since each of the terms $\{(x_i - \bar{x})^2 : i = 1, \dots, n\}$ will then be as large as possible). In general, this is a silly idea though, since if you do this you won't be able to check whether a linear regression curve looks reasonable (you can only check this by producing a plot of y against x , so it's helpful to have a plot with several different x values on it).

A further page or so of algebra, which is neither particularly difficult nor very instructive, allows us to show that $E\left(S\left(\hat{\beta}_0, \hat{\beta}_1\right)\right) = (n-2)\sigma^2$. Hence it is natural to estimate σ^2 using

$$\hat{\sigma}^2 = \frac{S\left(\hat{\beta}_0, \hat{\beta}_1\right)}{n-2},$$

which is unbiased. $\hat{\sigma}^2$ is uncorrelated with both $\bar{Y}$ and $\hat{\beta}_1$. Note, however, that $\hat{\beta}_0$ and $\hat{\beta}_1$ are *not* uncorrelated.

8.3 Normal errors

So far we have not assumed anything about the errors $\{\varepsilon_i\}$, except that they are independent with common mean zero and variance σ^2 . If we make the extra assumption that $\varepsilon_i \sim N(0, \sigma^2)$ then, since $\hat{\beta}_0$ and $\hat{\beta}_1$ are linear combinations of normal variables, they are also normally distributed. In this case, we can regard linear regression as predicting a probability distribution (which in this case is normal) for Y given a value of x — we are saying that $Y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$. This is a slightly more sophisticated presentation than ‘fitting a straight line’!

It can also be shown that, in the normal case,

$$\frac{(n-2)\hat{\sigma}^2}{\sigma^2} \sim \chi_{n-2}^2.$$

From this result, t -tests and confidence intervals for β_0 and β_1 can be derived, following exactly the same procedures as in the previous Chapter. For example, to test the hypothesis $H_0 : \beta_1 = \beta_1^{(0)}$ (often, $\beta_1^{(0)} = 0$, so that we are testing whether there is any linear dependence of Y upon x), we would calculate the test statistic

$$T = \frac{\hat{\beta}_1 - \beta_1^{(0)}}{\hat{\sigma} \sqrt{1/C_{xx}}} = \frac{(\hat{\beta}_1 - \beta_1^{(0)}) / (\sigma \sqrt{1/C_{xx}})}{\hat{\sigma} / \sigma},$$

which is distributed as t_{n-2} under H_0 , since $\hat{\beta}_1 \sim N(\beta_1^{(0)}, \sigma^2/C_{xx})$ in this case. This t distribution can also be used to calculate confidence intervals for β_1 . A similar procedure can be followed for β_0 .

8.3.1 Confidence intervals for the regression line

In addition to deriving confidence intervals for β_0 and β_1 individually, we may be interested in knowing the uncertainty associated with our fitted regression line. The y -value on this line at the point x is a random variable, $\hat{Y}$ say (i.e. ‘the estimated value of Y at x ’), defined as

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x = \bar{Y} + \hat{\beta}_1(x - \bar{x}).$$

(the representation in terms of $\bar{Y}$ and $\hat{\beta}_1$ is necessary for variance calculation, because $\hat{\beta}_0$ and $\hat{\beta}_1$ are not independent). $\hat{Y}$ (which is called the **FITTED VALUE** at x) is a linear combination of independent normal random variables, and hence has a normal distribution. We have

$$\begin{aligned} E(\hat{Y}) &= E(\hat{\beta}_0 + \hat{\beta}_1 x) = \beta_0 + \beta_1 x, \\ \text{and } \text{Var}(\hat{Y}) &= \text{Var}(\bar{Y}) + (x - \bar{x})^2 \text{Var}(\hat{\beta}_1) = \sigma^2 \left[\frac{1}{n} + \frac{(x - \bar{x})^2}{C_{xx}} \right]. \end{aligned}$$

We should be familiar with the general form of confidence intervals by now: we estimate σ^2 using $\hat{\sigma}^2$, and calculate a $100(1 - \alpha)\%$ confidence interval for $\beta_0 + \beta_1 x$ as

$$\hat{\beta}_0 + \hat{\beta}_1 x \pm \left(t_{\alpha/2} \times \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{C_{xx}}} \right),$$

where $t_{\alpha/2}$ is the appropriate percentage point of a t distribution with $n - 2$ degrees of freedom. Notice that the confidence interval is narrowest at the point $x = \bar{x}$. This means that our most precise estimates are in the centre of our existing observations.

8.3.2 Confidence intervals for unobserved Y values

Suppose now that we’re not interested so much in confidence intervals for the fitted line as in communicating our uncertainty about future Y values. The obvious way to do this is to use the fact that our regression model says ‘given a value of x , Y has a normal distribution with mean $\beta_0 + \beta_1 x$ and variance σ^2 ’ — so we could just use our estimates $\hat{\beta}_0, \hat{\beta}_1$ and $\hat{\sigma}^2$, together with tables of the normal distribution, to make probability statements about future values of Y at a given x (e.g. ‘The probability that Y will be in the range $(y_{\text{lower}}, y_{\text{upper}})$ is 0.95’). However, in practice such a procedure will under-represent the true uncertainty about future Y values, because it does not account for the error in estimating the model parameters.

To account for the estimation error and obtain a correct assessment of uncertainty, note that the fitted value at x is $\hat{Y}$, as above; and that the observed value will be $\beta_0 + \beta_1 x + \varepsilon$, where $\varepsilon \sim N(0, \sigma^2)$. If we say ‘the predicted value of Y is $\hat{Y}$ ’, our prediction error will be

$$Y - \hat{Y} = [Y - (\beta_0 + \beta_1 x)] + [(\beta_0 + \beta_1 x) - \hat{Y}].$$

The first term here is just ε , and the second is the error in the fitted value that we just studied while deriving confidence intervals for the regression line. ε is independent of anything in the original dataset, so we have a linear combination of normal random variables again. Our prediction error is distributed as

$$N\left(0, \sigma^2 \left[1 + \frac{1}{n} + \frac{(x - \bar{x})^2}{C_{xx}}\right]\right).$$

As usual, having estimated σ^2 using $\hat{\sigma}^2$, we use a t distribution to calculate a $100(1 - \alpha)\%$ interval for Y :

$$\hat{\beta}_0 + \hat{\beta}_1 x \pm \left(t_{\alpha/2} \times \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x - \bar{x})^2}{C_{xx}}} \right).$$

Example 8.1: Data on IQ test (x) and chemistry exam (y) scores for 12 US freshmen were as follows:

$$\sum_{i=1}^{12} x_i = 725 \quad \sum_{i=1}^{12} x_i^2 = 44475 \quad \sum_{i=1}^{12} y_i = 1011 \quad \sum_{i=1}^{12} y_i^2 = 85905 \quad \sum_{i=1}^{12} x_i y_i = 61685.$$

1. Find the least squares regression line for predicting chemistry scores from IQ. Provide an estimate of the error variance.

-

8.4 The correlation coefficient

The slope $\hat{\beta}_1 = C_{xy}/C_{xx}$ provides a measure of the degree of *linear association* between the x and y values. However, a drawback is that its value depends on the units in which these variables are measured. So we may consider

$$r_{xy} = C_{xy} / \sqrt{C_{xx}C_{yy}}$$

which is scale-invariant and is called the SAMPLE CORRELATION COEFFICIENT between $\{x_i\}$ and $\{y_i\}$. Using the Cauchy inequality

$$\left(\sum a_i b_i\right)^2 \leq \left(\sum a_i^2\right) \left(\sum b_i^2\right),$$

it can be shown that $0 \leq r_{xy}^2 \leq 1$, whence the correlation coefficient takes values between -1 and +1. $|r_{xy}| = 1$ when the data values all lie perfectly on a straight line; the sign of r_{xy} indicates the direction of the line's slope. If $r_{xy} = 0$, there is no linear relationship between the x s and y s (**NB**: this does not mean to say that there is no relationship — there may be a quadratic relationship, for example).

It is quite common to regard the study of correlation as a scientific objective in itself (for example, there are procedures for testing whether or not a sample correlation coefficient is significantly different from zero). As we have seen however, the correlation coefficient is only a by-product of a regression analysis, so we can't really learn anything new by studying it (for example, we can test for association by testing $H_0 : \beta_1 = 0$, which will give exactly the same results as a standard test on the correlation coefficient). Note, however, that the residual sum of squares from the regression (page 125) is

$$S(\hat{\beta}_0, \hat{\beta}_1) = C_{yy} - \frac{C_{xy}^2}{C_{xx}} = C_{yy} (1 - r_{xy}^2).$$

Now C_{yy} is the sum of squares of the y_i about their average value $\bar{y}$. This is a measure of the variability — or equivalently uncertainty — in the y s. $S(\hat{\beta}_0, \hat{\beta}_1)$ is the sum of squared errors after fitting the least squares line, which is a measure of the residual uncertainty after incorporating the x information and is never larger than C_{yy} . In some sense then, the fitting of a regression line has reduced our uncertainty about the y s, from C_{yy} to $S(\hat{\beta}_0, \hat{\beta}_1)$. The quantity $C_{yy} - S(\hat{\beta}_0, \hat{\beta}_1)$ may be thought of as variability that has been explained by the regression. This shows that r_{xy}^2 has an interpretation as the *proportion* of explained variation. It is sometimes called the COEFFICIENT OF DETERMINATION, and may be multiplied by 100 and quoted as a percentage (e.g. 'this relationship explains 72% of the variation in the response').

In Section 4 we defined the correlation coefficient between two random variables X and Y as $\rho = \text{Cov}(X, Y) / \sigma_X \sigma_Y$ in an obvious notation. We also saw that ρ could be considered as a measure of linear association in the bivariate normal distribution. There is a close connection with the sample correlation coefficient considered here: r_{xy} is in fact an estimator of ρ .